AQUATIMBANG: A PREDICTIVE ANALYTICS-DRIVEN FISH PRICE TRANSPARENCY SYSTEM FOR INFORMED MARKET DECISION-MAKING

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ABSTRACT

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To God be the Glory.

Cozy Trio

DEDICATION

This humble work is dedicated to the Almighty God, who has provided the researchers with the wisdom, strength, and guidance to complete this study successfully.

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CHAPTER 1
PROJECT CONTEXT

The Project Context and Research Design section provides an overview of the environment, background, and purpose of the study, as well as the methods used to achieve its objectives. It explains the problem being addressed, the setting in which the project is applied, and the rationale behind the chosen research approach. Additionally, this section outlines the design, tools, and procedures used to collect and analyze data, ensuring that the study is systematic, reliable, and aligned with its intended goals.

Introduction and Background of the Study

Unfair pricing and lack of pricing transparency are widespread in wet fish markets, affecting both vendors and consumers globally. Fish provides over 3.3 billion people with at least 20 percent of their average animal protein intake. Small-scale fisheries alone account for at least 40 percent of the global catch and provide employment for an estimated 60.2 million people—approximately 90 percent of total fisheries employment worldwide (FAO, Duke University & WorldFish, 2023). Despite this critical role in food security and poverty alleviation, market distortions such as inconsistent pricing and lack of reliable price information disproportionately harm small-scale vendors, who often operate without access to transparent market data, leading to inequitable profit distribution and unstable incomes.

In Southeast Asia, where fish is a dietary staple and a major economic commodity, similar problems prevail. Small-scale fisheries account for more than 80% of the region's total fish production, yet market intermediaries often capture the largest share of the final selling price (SEAFDEC, 2025). The Philippines, as an archipelagic nation with over 1.6 million registered fisherfolk and one of the world's top fish-producing countries, exemplifies these challenges. Despite the sector's importance, local fish markets continue to suffer from inconsistent pricing, lack of price transparency, and unfair valuation of fresh fish products. Many vendors rely heavily on middlemen or guesswork to set their selling prices, a practice that leads to unequal profit distribution and limits their ability to earn a stable income.

Recent studies across different contexts have documented these difficulties. Alhassan et al. (2025) noted that price volatility and limited access to reliable market data are significant barriers to growth for small-scale fisheries, preventing vendors from effectively planning when and where to sell their products. The traditional, manual methods of recording and sharing price data remain prevalent in many developing-world fish markets, leading to data lags and inaccuracies that directly harm vendor incomes. In the Philippine setting, Campos et al. (2023) documented significant economic losses among fishing-dependent communities due to climate variations, further highlighting the vulnerability of those who lack access to reliable price and market information. These findings collectively demonstrate a clear need for digital tools that can improve data accuracy, accessibility, and timeliness in local fish markets.

Technological advancements have introduced predictive analytics and data-driven systems capable of enhancing market transparency. For example, Ibarra et al. (2023) demonstrated that fish prices can be forecasted with reasonable accuracy using time-series models. A study by Rahman and Akter (2025) on Southeast Asian fish markets showed that machine learning models could predict weekly price fluctuations, potentially helping vendors optimize their selling strategies. While these studies confirm the potential of predictive analytics, such systems are often technically complex, require substantial data infrastructure, and are not readily accessible to local vendors. This creates a clear research gap: existing predictive models are largely designed for academic or large-scale commercial use, leaving a disconnect between available technology and its practical, user-friendly application at the community level.

In addition to pricing challenges, the lack of a centralized and organized fish market data system further complicates decision-making. The Bureau of Fisheries and Aquatic Resources (2021) has strengthened its price monitoring program through the development of the Electronic Fisheries Price Monitoring System (eFPMS), which aims to automate data generation and provide close-to-real-time reporting for fisheries commodities. Similarly, the Department of Information and Communications Technology (2026) has deployed online price monitoring platforms that allow users to upload and share observed

prices. These initiatives underscore the growing recognition of digital tools in enhancing market transparency, yet they remain largely focused on data collection and reporting rather than on predictive analytics or fair, weight-based pricing mechanisms that directly benefit vendors.

To address the identified gap, this study proposes AquaTimbang: A Predictive Analytics-Driven Fish Price Transparency System for Informed Market Decision-Making. The system is designed to help Local Government Units (LGUs) digitally record daily prices and to provide vendors with transparent market data, a fair price calculator, and predictive price forecasts. By offering features such as a price dashboard, digital data recording, fair price computation, and trend forecasting, the system aims to improve transparency and consistency in fish pricing. Through these features, AquaTimbang seeks to bridge the gap between advanced time-series analytics and local market needs, ultimately promoting fairness, efficiency, and trust within the Philippine fish market.

This study directly contributes to several United Nations Sustainable Development Goals (SDGs). Primarily, it supports SDG 1 (No Poverty) by helping vendors secure fairer, more stable incomes through transparent, data-driven pricing. It advances SDG 2 (Zero Hunger) by promoting efficient and equitable fish distribution within local markets. The system also aligns with SDG 8 (Decent Work and Economic Growth) by fostering a more organized, transparent, and trustworthy market environment that empowers small-scale economic actors. Furthermore, it supports SDG 9 (Industry, Innovation, and Infrastructure) by applying accessible predictive analytics to a traditional sector, and SDG 12 (Responsible Consumption and Production) by reducing information asymmetries that lead to market waste and inefficiency.

Purpose and Description

The purpose of this study is to design and develop AquaTimbang, a predictive analytics-driven fish price transparency system that addresses inconsistent pricing, lack of transparency, and inefficient data management in local fish markets. The system aims to provide fair, weight-based pricing and timely price forecasts to vendors and local

government units (LGUs), enabling more informed decision-making and ultimately enhancing fairness and efficiency in fish market transactions.

AquaTimbang is a web and mobile-based system that integrates price monitoring and trend forecasting. Key features include a municipal fish price dashboard, a daily price entry and validation module for authorized LGU personnel, a fair price calculator that computes prices based on weight per kilogram, and a trend analysis and forecasting module that generates short-term price predictions from historical data. AquaTimbang serves multiple user roles—Fish Vendors, LGU personnel, and system administrator—through role-based access, promoting transparency, collaboration, and practical usability in real-world local market settings.

Objectives of the Study

The study aims to design, develop, and evaluate AquaTimbang to improve pricing transparency, support fair valuation of wet fish, and enhance data-driven decision-making in local fish markets.

Specifically, this study aims to:

1. Analyze functional and non-functional requirements from fish vendors and LGU personnel to inform system features including price recording, fair price computation, trend analysis and forecasting, and reporting.
2. Design the system architecture, user interface, database schema, and role-based access control for AquaTimbang as a web and mobile-based platform.
3. Develop the system's core features as defined in the project scope: user authentication, municipal fish price dashboard, daily price entry and validation, fair price calculator, forecasting module, reports module, user management, and fish species management.
4. Test the functionality, usability, and reliability of AquaTimbang through unit testing, integration testing, and user acceptance testing with selected stakeholders.
5. Deploy the system in a controlled local market environment, including user training, data initialization, and deployment documentation.

6. Evaluate the system's effectiveness in improving pricing transparency, fairness, and user decision-making based on feedback and system logs collected during deployment, with particular attention to the accuracy and usefulness of generated forecasts.

Scope and Limitations

This study focuses on the design, development, and evaluation of AquaTimbang, a predictive analytics-driven fish price transparency system aimed at improving price transparency, promoting fair valuation, and supporting informed decision-making for wet fish products in Pasil Fish Market Phase II, Cebu City.

User Roles and Responsibilities

1. Fish Vendors**,** can view current and historical fish prices, compute fair prices using the price calculator (weight × LGU-validated price per kg), and view price trends and forecasts.
2. **LGU Personnel,** can input and validate daily fish price data, monitor pricing trends, generate reports (PDF/CSV), and interpret forecast outputs for market guidance.
3. **System Administrator,** manages user accounts (add, edit, delete, approve), manages fish species list, oversees system configuration, and ensures data integrity.

Functional Features

1. Recording and viewing of current and historical fish prices within a local market.
2. Daily price entry and validation module that allows LGU personnel to input and verify fish price data.
3. Fair Price Calculation based on fish weight and price per kilogram, using LGU-validated price (Weight × Price per kg).
4. Fish price dashboard displaying price trends, validated reference prices, and forecast visualizations across wet fish species.
5. Predictive Analytics Module that generates short-term price forecasts by analyzing historical fish price data along with external factors such as weather conditions, fuel

prices, and local holidays. The module captures non-linear patterns and market shocks more effectively than models relying solely on price history. Forecasts are presented with confidence intervals to communicate uncertainty, enabling vendors and LGUs to make informed decisions even during unexpected market disruptions.

6. Trend analysis and forecasting module that analyzes historical price data and generates short-term price forecasts with confidence intervals.
7. Reports module that allows LGU personnel to generate and export price reports (PDF/CSV), including forecast summaries.
8. Role-based access control that provides different users with appropriate permissions for data viewing, entry, validation, and system management.

Data Coverage

1. The study focuses on selected wet fish species commonly sold in Pasil Fish Market Phase II.
2. Price data is collected daily from the participating local market during the development and testing period.

The Limitations of the study includes:

1. **Geographic Coverage.** Designed and tested only in Pasil Fish Market Phase II, Cebu City. Findings may not be directly applicable to other markets without adaptation.
2. **Fish Coverage.** Limited to selected wet fish species; does not cover other seafood or processed products.
3. **Predictive Analytics Capability.** No external regressors or complex deep learning models are employed in the core implementation.
4. **Data Entry Method.** Price data is entered manually by LGU personnel; no automated data collection (e.g., sensors, APIs).
5. **Internet Connectivity.** Requires a stable internet connection for real-time access; functionality is limited in offline or poor connectivity areas.
6. **External Integration.** No integration with national price databases, payment systems, or other government information systems.

7. **User Training and Adoption.** No long-term assessment of training effectiveness or sustained adoption beyond the testing and deployment period.

Significance of the Study

This study is significant as it introduces AquaTimbang, a platform that enhances transparency, fairness, and efficiency in local fish markets through data-driven pricing, centralized information, and predictive analytics.

Specifically, it benefits the following stakeholders:

Fish Vendors. Provides access to updated, reliable fish price information and short-term price forecasts, enabling fair selling prices and reducing exploitation. The fair price calculator helps vendors compute prices instantly based on weight and LGU-validated rates, improving income stability and customer trust. Forecasts allow vendors to anticipate price changes and adjust their selling strategies accordingly.

Local Government Units (LGUs). Supports monitoring and management of fish price data, better policy-making, price regulation, and data-driven planning to ensure fair market practices. The reporting module allows easy generation of price reports for transparency and record-keeping. Forecasts can assist LGUs in anticipating price surges and planning interventions.

Fish Market Industry. Contributes to the modernization of traditional fish markets by introducing digital tools and accessible time-series analytics, promoting an organized, transparent, and competitive marketplace.

Researchers and Future Developers. Serves as a reference for future research on price transparency systems, fisheries management, and time-series forecasting applications, with opportunities for expansion using more advanced models or external data sources.

RESEARCH DESIGN

This section discusses the research methods used in the study, the flow of the study, environment, respondents involved, instruments utilized, the treatment of data, and the definition of terms to ensure clarity and understanding of concepts related to the proposed AquaTimbang System.

Methods

This study will employ the System Development Research method, which focuses on the design, development, and evaluation of an information system intended to address a specific real-world problem. In this study, the researchers will develop AquaTimbang: A Predictive Analytics-Driven Fish Price Transparency System for Informed Market Decision-Making, a web and mobile-based platform designed to improve fish price transparency and provide accessible market information and forecasts for vendors and Local Government Units (LGUs). The system aims to address common issues such as inconsistent pricing, lack of centralized data, and limited access to reliable market information in local fish markets.

The researchers will utilize the Agile Development Methodology as the framework for developing the AquaTimbang system. Agile methodology supports iterative and incremental development, allowing continuous refinement of the system based on feedback from users such as fish vendors and LGU personnel. This approach promotes flexibility, collaboration, and adaptability throughout the development process, ensuring that the system evolves according to user needs and real-world conditions. Through regular testing and feedback cycles, the researchers can improve system functionality and usability effectively.

The study will also employ descriptive research to gather and analyze information regarding current practices in fish price monitoring and market transactions. Data will be collected through interviews, surveys, and observations involving Fish Vendors and LGU staff. These methods will help identify key challenges such as price inconsistency, lack of transparency, and absence of data-driven decision-making tools. The collected data will

serve as the foundation for defining system requirements and designing the AquaTimbang platform.

The development of AquaTimbang will follow the Software Development Life Cycle (SDLC), which consists of the following phases:

Table 1

Software Development Life Cycle (SDLC)

Phase	Description
Planning	Identification of the problem, definition of system objectives, and determination of the project scope and limitations for the AquaTimbang system.
Requirement Analysis	Collection and analysis of system requirements through interviews, surveys, and observations with Fish Vendors, LGU personnel, and System Admin. Special attention to forecasting needs and ARIMA parameter selection.
System Design	Development of system architecture, database design, user interface, and data flow for the fish price transparency system, including integration of forecasting model computation.
Development	Implementation and coding of system modules including the fish price dashboard, price entry and validation module, fair price calculator, forecasting engine, and role-based access system.
Testing	Conducting functional testing, forecast accuracy evaluation (e.g., MAPE, RMSE), and user acceptance testing (UAT) to ensure the system operates correctly and meets the needs of its intended users.
Deployment	Implementation of the web and mobile-based system prototype and demonstration of the platform for intended users such as Fish Vendors, LGU personnel, and System Administrator.

Evaluation	Assessment of system performance, usability, forecast usefulness, and effectiveness through user feedback and testing results to identify areas for improvement.

Flow of the Study

The flow of the study presents the inputs, processes, and output involved in the development of AquaTimbang: A Predictive Analytics-Driven Fish Price Transparency System for Informed Market Decision-Making. The inputs include the existing practices in fish price monitoring and valuation, system requirements gathered from Fish Vendors and Local Government Unit (LGU) personnel, as well as the necessary design components for system development. The process involves data gathering, system design, development, testing, and evaluation of the proposed system. The output of the study will be the developed AquaTimbang system, which aims to provide reliable fish price information, fair price computation, and predictive insights to support better decision-making in local fish markets.

Input. The input phase consists of the foundational knowledge, data, and system requirements necessary for the development of the proposed AquaTimbang system. This includes documenting the existing practices in fish price monitoring and valuation, gathering system requirements through stakeholder consultations with Fish Vendors and LGU personnel, and identifying the essential design components needed to guide system development and functionality.

Process. The process phase involves the systematic procedures undertaken in the development of the proposed system. This includes data gathering, system design, system development, system testing, and system evaluation. Each stage ensures that the system is properly analyzed, designed, built, and assessed to meet the identified requirements, address existing gaps in price transparency, and fulfill the objectives of the study.

Output. The output phase represents the final product of the study, which is the developed AquaTimbang system. The proposed system provides a predictive analytics–driven solution for fish price transparency by delivering reliable fish price information, enabling fair price computation, and offering predictive insights to support informed market decision-making for Fish Vendors, LGUs, and other market stakeholders.

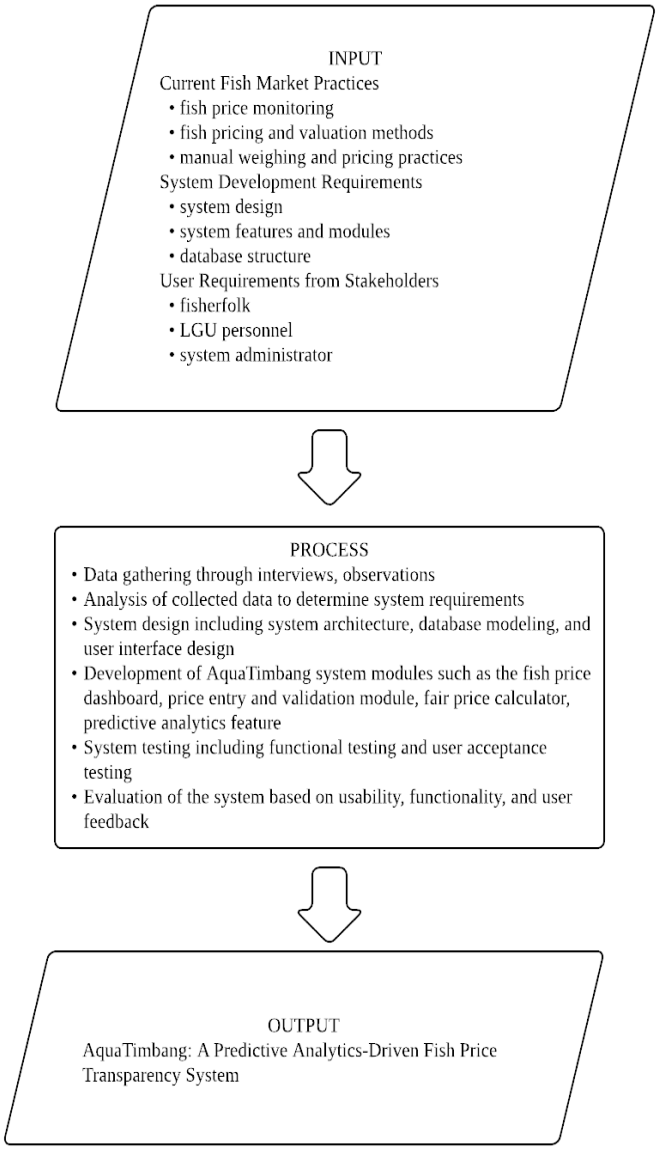


Figure 1

AquaTimbang's Flow of the Study

As shown in Figure 1, it illustrates the three-phase structure of the study—Input, Process, and Output—guiding the development of AquaTimbang. It begins with the identification of existing fish price monitoring practices and stakeholder requirements, followed by sequential stages of data gathering, system design, development, testing, and evaluation. The study culminates in a functional system that delivers transparent pricing, fair computation, and predictive insights for local fish market decision-making.

Environment

The study will be conducted in Cebu City, specifically in Pasil Fish Market Phase II, Cebu City, and the Department of Veterinary Medicine and Fisheries (DVMF). These locations were selected for several reasons. First, Pasil represents a realistic and active fish trading and landing environment where fishermen bring in their daily catch and vendors engage in price negotiation, providing authentic conditions for observing actual fish pricing practices. Second, this area reflects common challenges such as manual pricing, lack of standardized valuation, and limited access to reliable market data, making it an ideal setting for evaluating the effectiveness of the proposed system. Lastly, the Department of Veterinary Medicine and Fisheries (DVMF) serves as the administrative center where Local Government Unit (LGU) personnel oversee market-related activities and data monitoring.



Figure 2

Geographic Location of Pasil Fish Market PHASE II and DVMF

As shown in Figure 2, it illustrates the environment that allows the researchers to gather insights on existing price monitoring practices, data recording processes, and policy implementation. Additionally, the accessibility of these locations to the researchers enables efficient data gathering, close coordination with respondents, and timely collection of feedback during the system development and evaluation phases.

Respondents

The respondents of this study will be selected through purposive sampling, a non-probability sampling technique wherein participants are chosen based on their direct relevance to the subject of the study. This approach is deemed most appropriate given that the evaluation of the AquaTimbang system requires individuals who possess firsthand experience with fish pricing, market monitoring, daily fish trading operations, system management, and the use of forecast information. According to Etikan et al. (2016), purposive sampling is appropriate in research where participants are selected based on their ability to provide valuable insights related to the objectives of the study.

Table 2

Distribution of Respondents

Respondent category	Sample Size
Fish Vendors	25
LGU Personnel	5
LGU System Administrator	2
Total	**32**

The study will involve three groups of respondents from selected fish markets in Cebu City, particularly in Pasil and Cebu City Hall, with the total number of participants subject to the availability of qualified individuals at the time of the study.

Treatment of Data

All data collected from respondents will be handled with strict confidentiality in accordance with the Data Privacy Act of 2012. Any personal or identifiable information will not be disclosed in any part of the study. The data gathered from interviews, surveys, and system evaluation questionnaires will be carefully organized, tabulated, and analyzed. Quantitative data collected from the survey and evaluation instruments will be processed using descriptive statistics such as frequency distribution, percentage, and mean. These statistical tools will help determine the respondents' level of agreement and overall assessment of the AquaTimbang system in terms of functionality, usability, reliability, performance, and forecast quality. Forecast accuracy will also be assessed quantitatively using metrics such as Mean Absolute Percentage Error (MAPE) and Root Mean Square Error (RMSE) on hold-out test data.

Scoring Procedures

A five-point Likert scale was used to measure the responses of the respondents in evaluating the AquaTimbang system. The weighted mean was used to determine the result of each evaluation item, with the following formula:

$$\mathbf{WM = \Sigma(f \times x) / n}$$

Where:

- **WM** = Weighted Mean
- **f** = Frequency of each rating
- **x** = Scale value (1–5)
- **n** = Total number of respondents

Table 3

Interpretation of Weighted Mean Scores for System Evaluation

Weights	Scale Range	Interpretation	Parameters
5	4.50 – 5.00	Strongly Agree (SA)	The respondents believed that the AquaTimbang system is of excellent quality. The level of judgment was Excellent or 100% favorable.
4	3.50 – 4.49	Agree (A)	The respondents believed that the AquaTimbang system is of very good quality. The level of judgment was Very Good or 80% favorable.
3	2.50 – 3.49	Neither Agree or Disagree (N)	The respondents believed that the AquaTimbang system is of good quality. The level of judgment was Good or 60% favorable.
2	1.50 – 2.49	Disagree (D)	The respondents believed that the AquaTimbang system is of fair quality. The level of judgment was Fair or 40% favorable.
1	1.00 – 1.49	Strongly Disagree (SD)	The respondents believed that the AquaTimbang system is of poor quality. The level of judgment was Poor or 20% favorable.

Definition of Terms

AquaTimbang. A web-based predictive analytics-driven fish price transparency system designed to provide fair, weight-based pricing, centralized market data, and short-term price forecasts.

Daily Price Entry and Validation Module. A component allowing authorized LGU personnel to input, verify, and update fish price data.

Fair Price Calculator. A tool that computes fish price using: Weight × LGU-validated price per kg.

Fish Vendors. Individuals engaged in selling wet fish as a primary livelihood in local markets.

Local Government Unit (LGU). Government authorities responsible for monitoring, recording, and regulating fish price data and market activities.

Predictive Analytics. The use of historical time-series data and statistical or machine learning models to generate short-term future price estimates.

Role-Based Access Control (RBAC). A security mechanism restricting system access based on user roles (admin, LGU, vendor).

System Administrator. User with full system access responsible for managing accounts, fish species, and data integrity.

Trend Analysis and Forecasting Module. A module that analyzes historical price data using time-series methods and shows forecasted price trends to support decision-making.

CHAPTER II
REVIEW OF RELATED LITERATURE AND STUDIES

In this chapter, we examine various related studies that provide empirical evidence and insights that support the development of AquaTimbang, a predictive analytics-driven fish price transparency system for informed market decision-making in local fish markets. The studies discussed highlight several key components, including fair price computation, centralized data management, predictive analytics for price forecasting, and role-based access for multiple stakeholders. These studies establish a foundation for the system design, addressing gaps in current practices such as inconsistent pricing, lack of transparency, and inefficient data recording, while offering strategies for improvement.

Related Literature

The related literature reviewed in this study provides a comprehensive foundation for the development of AquaTimbang, a web and mobile-based system designed to address the challenges faced by fish vendors, and local government units in the Philippines. These studies focus on several key aspects that are integral to the success of the AquaTimbang system, including weight-based fair price calculation, centralized fish price databases, municipal dashboards for price monitoring, predictive analytics for identifying market trends, and role-based access control (RBAC) to ensure data integrity. Each of these components plays a crucial role in improving transparency, streamlining market operations, and supporting data-driven decision-making among stakeholders.

The economic valuation of fish is fundamentally linked to biological size distribution, creating a direct relationship between physical measurement and market pricing. Sjöberg (2015) demonstrated that fish markets exhibit high sensitivity to size composition, where even minor shifts toward smaller individuals can generate significant revenue losses for local fishers. This finding directly supports AquaTimbang's fair price calculator feature, which computes prices using the formula Weight × LGU-validated price per kg. The study was made possible by employing size-composition analysis of fish stocks, quantitative revenue modeling, and access to commercial landing data combined with biological sampling protocols that allowed researchers to correlate physical fish.

Yildiz et al. (2023) identified that size-focused market dynamics influence fishing practices, reinforcing the need for weight-based transaction systems. Their research was enabled through long-term ecological monitoring of marine food webs, economic impact assessment methodologies, and the use of multi-species fishery data with stock assessment models to trace the feedback loop between market incentives and fishing practices. Together, these studies provide empirical justification for integrating weight as a primary variable in fair price computation, ensuring that AquaTimbang's calculator addresses both economic fairness for vendors and accurate valuation of fish products based on measurable physical characteristics.

The transition from fragmented, manual price recording to centralized digital systems addresses the inconsistent pricing and inefficient data management documented in Philippine fish markets. Grantham et al. (2022) demonstrated that digital tracking tools such as NFC and RFID significantly reduce data gaps in small-scale fisheries, directly supporting AquaTimbang's daily price entry and validation module. This study was made possible by deploying NFC and RFID technologies in active fishing communities, supported by collaborative partnerships with local cooperatives, training programs for fisherfolk on device usage, and baseline data collection to measure the 44% underreporting gap. Martin et al. (2021) further established that curated, accessible databases are essential for regulating market activity and preventing economic losses, validating the need for AquaTimbang's municipal fish price dashboard. Their database curation research was enabled through interoperability framework design, stakeholder consultation processes that identified specific data needs for market regulation, data quality assessment protocols, and traceability standards adapted from commercial supply chains. Within the Philippine context, Yang and Pomeroy (2017) provided evidence that Community-Based Fisheries Management (CBFM) improves equity through localized control, directly aligning with AquaTimbang's role-based access control. Their study was feasible due to community-based participatory research methods, a longitudinal case study design across multiple fishing communities, local government partnerships, and household survey instruments validated for fisheries contexts. Chuenpagdee et al. (2019) reinforced this by demonstrating that centralized data platforms integrated with local knowledge systems

create greater market visibility. Their crowdsourced knowledge platform research was enabled through multi-stakeholder workshops, digital platform prototyping, qualitative analysis of local knowledge integration, and comparative case studies across Southeast Asian fisheries to identify common transparency barriers. Collectively, these studies show that while local communities provide sustainable practices, a centralized system like AquaTimbang provides the visibility necessary to transform local price data into actionable, data-driven decisions.

The application of predictive analytics to fisheries management enables stakeholders to anticipate price fluctuations and make proactive selling decisions. Tandi and Shrirao (2023) established that agile AI and data-driven solutions provide essential frameworks for processing raw market data into actionable trends, directly supporting AquaTimbang's trend analysis and forecasting module. Their agile AI research was made possible through real-time data processing architectures, machine learning pipeline development adapted for fisheries contexts, cloud computing infrastructure, and algorithm validation using historical market transaction data. In the Philippine setting, Bambao et al. (2024) demonstrated through the FISHDA system that time-series forecasting effectively mitigates risks of supply-demand imbalances, validating AquaTimbang's use of seasonal forecasting techniques that account for weekly patterns and external market drivers. The FISHDA system was enabled through time-series forecasting model development using seasonal ARIMA approaches (SARIMA/SARIMAX), supply-demand data collection from local markets, predictive accuracy testing against actual market outcomes, and partnerships with local government units for data access. Thasni et al. (2025) further showed that digital transformation in supply chains improves traceability and operational efficiency. Their study was conducted using end-to-end traceability system design, blockchain or distributed ledger integration, operational efficiency metrics comparing digital versus traditional methods, and pilot implementations in active market settings. Collectively, these studies confirm that predictive analytics incorporating seasonal patterns and exogenous factors (e.g., weather, fuel prices, holidays) – within AquaTimbang's defined scope of SARIMAX-based forecasting – can meaningfully support vendor decision-making by capturing real-world market dynamics beyond simple trend identification.

Effective fisheries management in the Philippine context requires a transition from top-down mandates toward an interactive governance model that accounts for decentralized market forces. Steenbergen et al. (2019) provide a critical framework illustrating that successful governance must function across multiple institutions—including the private sector and local cultural practices—rather than following a universal, rigid "ideal." This study was made possible through multi-institutional governance analysis, comparative case studies across Southeast Asian fisheries, stakeholder mapping that included private sector and cultural practices, and qualitative assessment of governance mechanisms outside formal regulatory structures. This concept is central to the design of the AquaTimbang Municipal Dashboard, which serves as a digital meeting point for various stakeholders, ensuring that price monitoring is not a localized government activity but a transparent, multi-institutional process that aligns with the study's objective of supporting LGUs with data-driven planning and price regulation.

The necessity for such an integrated system is further underscored by the findings of Simbre et al. (2024). Their study of urban fish farming in Metro Manila reveals a significant disconnect: despite formal government support, fishers continue to suffer from persistent socio-economic and environmental gaps, suggesting that "government assistance" alone is insufficient without a data-driven mechanism to identify and address real-time market challenges. This research was enabled through urban fisheries case study design in Metro Manila, mixed-methods data collection combining socio-economic surveys and environmental assessments, gap analysis comparing government support programs with actual fisher outcomes, and identification of persistent challenges despite formal assistance.

By integrating these perspectives, it becomes clear that for Philippine aquaculture and local fish markets to achieve true sustainability, governance must evolve into a multiscale system that monitors the diverse institutional drivers of the market chain. AquaTimbang addresses this by providing a centralized dashboard that translates raw data into "visibility" for local government units, allowing them to provide more targeted and effective support to fisherfolk alike. The enabling characteristics common to both

Steenbergen et al. (2019) and Simbre et al. (2024) include: multi-stakeholder governance frameworks, localized case study approaches, identification of gaps between policy and practice, and emphasis on data-driven mechanisms to bridge institutional disconnects. These factors directly inform AquaTimbang's municipal dashboard design, which integrates price monitoring, trend visualization, and report generation to support LGUs in moving beyond generic assistance toward targeted, evidence-based market interventions.

Effective fish market systems require differentiated access permissions to maintain data integrity while ensuring usability across diverse user groups. Kamal et al. (2026) demonstrated that process-centric approaches with embedded quality controls improve traceability and accountability in value chains, directly supporting AquaTimbang's role-based architecture. Their value chain resilience research was made possible through process-centric workflow mapping, quality control embedded design, stakeholder role analysis to define specific permissions for different user types, and traceability system implementation in active market environments. Zhang et al. (2015) provided the theoretical foundation for Group/Role-Based Access Control (RBAC), showing that scalable access models prevent unauthorized data modification while enabling appropriate stakeholder interaction. Their RBAC model development was enabled through formal access control modeling, scalability testing across different organizational sizes, security protocol design that prevents unauthorized field-level data access, prototype implementation, and vulnerability assessment. This aligns precisely with AquaTimbang's implementation of RBAC to ensure vendors cannot alter price records, LGU personnel cannot modify system settings, and administrators maintain overall data integrity—fostering trust and accountability within the local market environment.

Related Studies

This section reviews existing studies that are directly comparable to the features and goals of AquaTimbang: A Predictive Analytics-Driven Fish Price Transparency System for Informed Market Decision-Making. The studies are grouped according to their relevance to the following system features: fish price monitoring and market transparency systems, government price reporting and digital data platforms, predictive analytics and

price forecasting in fisheries, role-based digital platforms for market data management, and digital transformation in small-scale fisheries. Each study is discussed in terms of how its findings, design approaches, or technical contributions relate to specific components of AquaTimbang.

The University of Dar es Salaam (2026) developed the TanFishMarket Digital Marketplace Platform in partnership with the Food and Agriculture Organization (FAO) and national stakeholders to create a comprehensive digital ecosystem that connects fishers, traders, buyers, and regulators. The system integrates key functionalities such as digital trading, stock management, logistics coordination, and market analytics, enabling transparency and efficiency across the fisheries supply chain. Developed under the FISH4ACP programme funded by the European Union, the platform has trained more than 200 community members, with 62% women and 48% youth, demonstrating strong inclusivity and community engagement. This study was made possible through multi-stakeholder partnership development (FAO, national agencies, local communities), a participatory design process that included training programs for marginalized groups, integration of end-to-end digital ecosystem features (trading, stock management, logistics, analytics), and a funding mechanism from the European Union that enabled large-scale implementation. The platform highlights how centralized data systems and analytics can empower stakeholders by providing real-time access to relevant market information. This study is directly comparable to AquaTimbang's municipal fish price dashboard and stakeholder integration features. Both systems aim to unify multiple actors within a single digital platform, enhance transparency, and improve access to reliable market data. While TanFishMarket operates at a broader scale, AquaTimbang focuses on localized municipal implementation, making it more accessible and tailored to the needs of small-scale fisherfolk. The integration of analytics and user-centered design principles in TanFishMarket further supports AquaTimbang's goal of delivering simplified, actionable insights to users.

The Bureau of Fisheries and Aquatic Resources (2025), through its Fisheries Industry Development and Support Services Division (FIDSSD), conducted a Training-Workshop on Fisheries Price Monitoring Protocols and Procedures to standardize

data collection across Regions 9–13. This initiative supports the development of the Electronic Fisheries Price Monitoring System (eFPMS), which aims to automate price data generation and provide near real-time reporting of fisheries commodities. This initiative was enabled through region-wide training workshops covering five administrative regions, standardized data collection protocols developed with government experts, stakeholder capacity building that trained personnel on consistent reporting methodologies, government funding through BFAR, and inter-agency coordination between FIDSSD and regional offices. The program emphasized structured reporting, consistent methodologies, and improved accuracy in determining prevailing market prices. This initiative aligns closely with AquaTimbang's price entry and validation modules, which also focus on structured data collection and real-time reporting. By incorporating standardized input methods and validation processes, AquaTimbang reflects the same objectives of improving data accuracy and reliability while providing a more accessible interface for local government units and stakeholders.

The Department of Information and Communications Technology–Cordillera Administrative Region (2026), in partnership with the Baguio City government, launched an online consumer price reporting platform that allows users to upload and share observed prices of commodities, including fish. This system leverages citizen participation to establish prevailing market prices and identify price variations across different locations. This platform was made possible through citizen participation mechanisms that enabled crowdsourced price data, web-based system development with accessible user interfaces, cross-market price comparison features that allowed users to see variations across locations, pilot deployment in Baguio City with active community engagement, and user feedback collection that informed iterative improvements. The initiative demonstrates how digital tools can empower consumers to contribute to market transparency while enabling broader data collection through community engagement. Although AquaTimbang primarily assigns price data entry to authorized personnel rather than the general public, the concept of a web-based reporting system and accessible user interface strongly aligns with its design. The success of this platform supports AquaTimbang's implementation of a centralized dashboard that allows comparison of prices across markets and improves transparency.

The Federal Antimonopoly Service of Russia (2026) implemented a mandatory seafood price reporting policy requiring transactions exceeding 10 tons to be registered with the St. Petersburg Commodity Exchange. This policy was introduced to address price inflation and improve transparency across the seafood supply chain. The study revealed that products may pass through as many as six intermediaries, with cumulative markups reaching 50% or more, significantly increasing retail prices. The policy also introduced a price review mechanism requiring justification for price increases exceeding 5%, ensuring accountability among producers and distributors. This regulatory approach was enabled through legal and regulatory framework development that mandated transaction reporting, commodity exchange integration for centralized registration, enforcement mechanisms including a 5% price increase justification threshold, government mandate authority backed by the Federal Antimonopoly Service, and market monitoring infrastructure that tracked intermediary markups. This regulatory approach is highly relevant to AquaTimbang's price validation module and fair price calculator. Both aim to reduce inefficiencies and prevent excessive markups by ensuring that pricing data is transparent and verifiable. While the Russian model relies on government enforcement, AquaTimbang provides a system-based solution that empowers users with tools to evaluate fair pricing, thereby reducing information asymmetry within the market.

The Cooperative Development Authority (2021) reported that fisherfolk marketing cooperatives face persistent challenges due to the dominance of middlemen in price determination. The study highlighted the cycle of "selling cheap and buying high," which contributes to financial instability and limits income potential among fisherfolk. This study was made possible through primary data collection from active fisherfolk cooperatives, economic analysis of middlemen dominance using price comparison methods, documentation of income instability through financial records and cooperative ledgers, and qualitative interviews with cooperative members to understand the "selling cheap and buying high" cycle. These findings emphasize the need for systems that provide direct access to accurate pricing information and reduce dependency on intermediaries. This study strongly validates AquaTimbang's objective of empowering fisherfolk by providing reliable and transparent pricing data. By enabling users to access verified market prices,

AquaTimbang addresses the systemic inequities identified in the report and promotes more equitable participation in the fisheries market.

Ibarra et al. (2023) demonstrated that fish prices can be effectively forecasted using time-series models, providing empirical evidence that predictive analytics can support informed decision-making in fisheries markets. The study showed that historical price data, combined with appropriate seasonal adjustments and external drivers (e.g., weather, fuel costs), can be used to identify trends and predict future price movements. However, the researchers also noted that such systems are often complex and not easily accessible to local users, creating a gap between advanced analytical tools and practical application. This fish price forecasting research was enabled through historical price dataset compilation spanning multiple years, time-series model specification tailored to fisheries data (including seasonal components and exogenous regressors where applicable), forecast accuracy validation using statistical error measures (e.g., MAPE, RMSE), and explicit identification of accessibility limitations as a key factor for future work. Kuhl and Malthouse (2024) defined predictive analytics as a branch of data mining focused on forecasting future trends using multiple predictors and statistical models. They emphasized that computational tools are essential for handling large datasets and simplifying complex calculations. Similarly, iMerit (2025) explained that predictive analytics is a forward-looking approach that relies on structured and high-quality data to generate reliable forecasts, highlighting techniques such as regression, classification, and time-series modeling (including seasonal approaches that can incorporate external factors).

This explanatory study was enabled through data quality assessment methodologies that define structured data requirements, regression and classification technique documentation specifically for fisheries market applications, and practical examples linking data quality to forecast reliability. These studies collectively provide strong theoretical and methodological support for AquaTimbang's predictive analytics feature. By leveraging historical price data and external factors (fuel prices, weather, holidays) to forecast trends, AquaTimbang aligns with established practices in data-driven decision-making. At the same time, the identified challenges in accessibility reinforce

AquaTimbang's focus on simplifying predictive outputs, ensuring that insights are understandable and usable for non-technical users such as fish vendors.

Ferraiolo et al. (2001) established the theoretical foundation for Role-Based Access Control (RBAC), demonstrating that assigning permissions based on predefined roles ensures security, accountability, and efficient system management. The RBAC model enforces the principle of least privilege, ensuring that users can only access information relevant to their responsibilities. This foundational study was made possible through formal access control modeling using set theory and permission matrices, security analysis of unauthorized access risks, scalability testing across different organizational sizes, and the development of standardized RBAC frameworks (e.g., NIST model). Song et al. (2023) developed a blockchain-based fund management system that improves transparency and accountability by recording transactions on a tamper-evident ledger. This approach ensures that all transactions are traceable and cannot be altered without detection, addressing issues of trust and data integrity. Their study was enabled through blockchain or distributed ledger integration for tamper-evident record-keeping, traceability system design that logs all data modifications, prototype implementation in a multi-user financial context, and vulnerability assessment to detect unauthorized alteration attempts. These studies are directly applicable to AquaTimbang's system architecture, which incorporates role-based access control for different user groups, including fish vendors, LGU officers, and administrators. By ensuring that users have access only to relevant data and functions, AquaTimbang enhances security and maintains data integrity, fostering trust among stakeholders.

Stein (2023) examined the impact of mobile construction management applications and found that mobile tools significantly improve productivity, reporting accuracy, and decision-making speed. By enabling real-time data capture at the source, these systems reduce delays between data collection and analysis. This mobile application research was enabled through productivity measurement instruments (e.g., task completion time), a before-after comparison design with control groups, decision-making speed metrics captured through system logs and user surveys, and emphasis on intuitive interfaces as a

critical factor for user adoption. The study also emphasized the importance of intuitive interfaces in ensuring user adoption and efficiency. Nunez et al. (2018) developed a mobile cloud computing platform designed for small and medium enterprises, focusing on usability, accessibility, and adaptability. Their mobile cloud computing study was conducted using user-centered design methodology involving iterative prototyping and user testing, lightweight system architecture suitable for limited technical infrastructure (e.g., low bandwidth, basic smartphones), SME pilot testing in real business environments, and usability evaluation instruments such as the System Usability Scale (SUS). The study highlighted that user-centered design and lightweight system architecture are critical for successful implementation, particularly in environments with limited technical infrastructure. These findings strongly support AquaTimbang's mobile-first design and real-time data collection capabilities. By allowing users to input and access price data directly from market locations, AquaTimbang reduces reliance on manual processes and improves data accuracy. The emphasis on usability ensures that the system remains accessible to users with varying levels of technical expertise.

Alhassan et al. (2025) conducted a comprehensive study on digital transformation challenges in small-scale fisheries, identifying price volatility and limited access to reliable market data as major barriers to growth. The study found that fisherfolk often rely on informal and inconsistent sources of information, which limits their ability to make informed decisions. Additionally, traditional methods of recording and sharing price data were found to be manual, inefficient, and prone to errors. This comprehensive study was made possible through a mixed-methods research design combining surveys, interviews, and market observations across multiple fishing communities; price volatility metrics that quantified price fluctuations over time; digital readiness assessments that measured current technology adoption and barriers; and barrier identification frameworks that categorized challenges into data accessibility, accuracy, and timeliness. The study emphasized the need for digital systems that can provide accurate, timely, and accessible market data to improve decision-making and economic outcomes for fishing communities. These findings strongly reinforce the importance of AquaTimbang's centralized database and digital platform. By addressing the challenges of data inaccuracy, limited accessibility, and delayed reporting,

AquaTimbang provides a practical solution that enhances transparency and supports informed decision-making among fish vendors and local government units.

Related Systems

This section examines existing fish price monitoring and market analytics systems that share functional similarities with the proposed AquaTimbang system. Two major platforms, the BFAR Price Monitoring Service and the Sinay Fish Price App, were reviewed to identify their capabilities and relevance to this study.

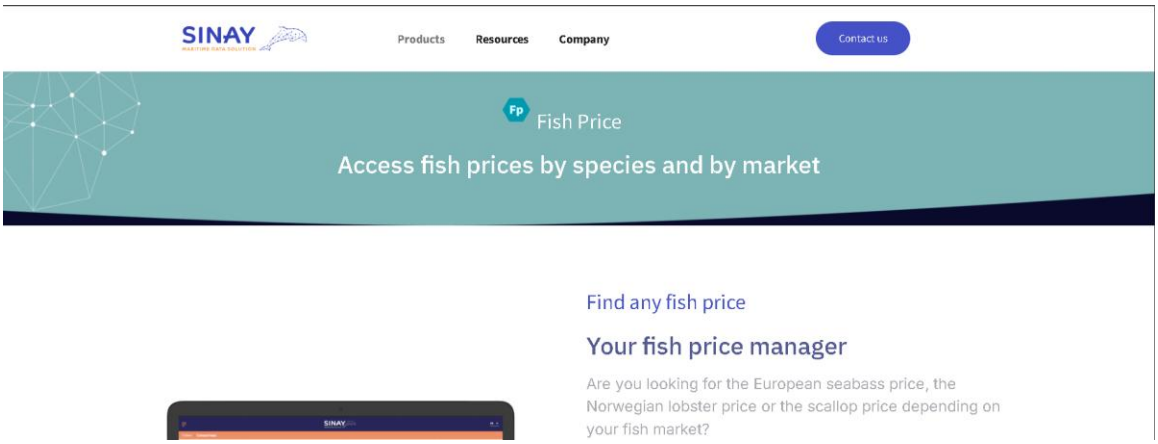


Figure 3

Sinay Dashboard

As shown in Figure 3, Sinay Fish Price App is a digital platform developed by Sinay, a technology company specializing in environmental and maritime data solutions. The application provides real-time and historical pricing data for various fish species across different markets. It is designed for professionals such as fishermen, shipowners, and fishmongers, offering features like market-specific price lists, average price calculations, and intake volume history to support informed commercial decisions (Sinay, 2026). The platform aggregates public data from sources such as France Agrimer and updates pricing information on a weekly basis. While it serves as a comprehensive tool for the fisheries sector, its focus is primarily on European markets and it does not address the specific needs of local Philippine fish markets, such as localized species, manual weighing practices, or integration with Local Government Unit (LGU) monitoring processes.

Despite its comprehensive data offerings, Sinay Fish Price App presents substantial limitations for local fish markets in the Philippines. The platform focuses primarily on European markets and does not cover Philippine fish species such as galunggong, tulingan, and bolinaw, which are essential to the daily transactions of local fisherfolk and vendors. Moreover, the system lacks a weight-based price calculator, a feature critical for fair and standardized transactions in local markets where fish are sold by weight.

AquaTimbang addresses these gaps by providing coverage of selected Philippine fish species, integrating a weight-based fair price calculator to ensure consistent and accurate transactions, and incorporating basic predictive analytics to help users identify price trends and make better buying and selling decisions. Unlike Sinay, which is designed for commercial users in European markets, AquaTimbang is purpose-built for fisherfolks and LGUs in the Philippine context, offering a more accessible and locally relevant platform.

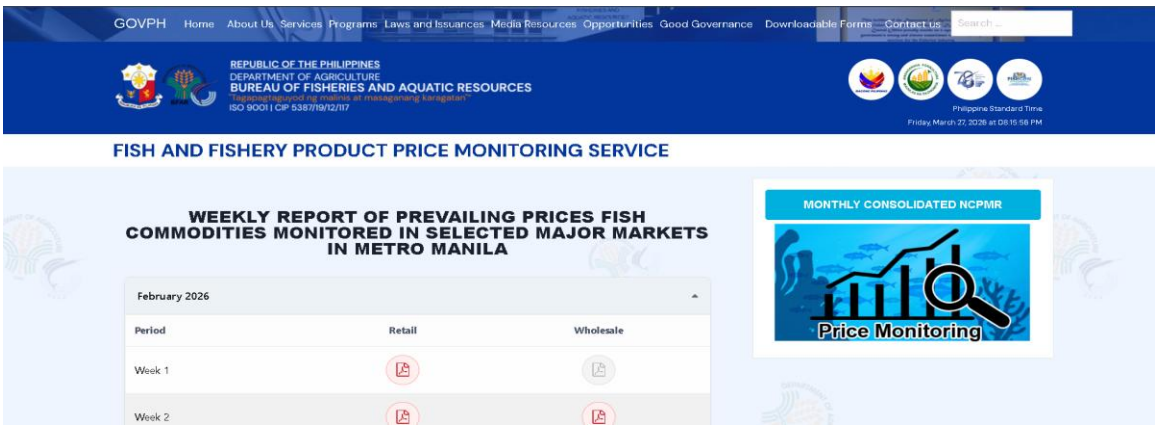


Figure 4

BFAR Dasboard

As shown in Figure 4, the BFAR Fish and Fishery Product Price Monitoring Service is a public service initiative by the Philippine Bureau of Fisheries and Aquatic Resources (BFAR). It aims to provide transparent and accessible price information on fish and fishery products in markets across the Philippines. This service supports the government's mandate to stabilize market prices, protect consumers, and provide fish vendors with up-to-date market data to improve their livelihood (BFAR, 2025). The program has been strengthened through training workshops for implementers and the

development of the Electronic Fisheries Price Monitoring System (eFPMS), which automates data generation and provides close-to-real-time reporting for fisheries commodities.

Despite its valuable role in government-led price monitoring, the BFAR system presents several limitations that hinder its effectiveness for end-users. The system is primarily designed for government monitoring purposes and lacks direct user-facing tools for fish vendors to actively access and utilize price information for their daily transactions. It does not include a weight-based fair price calculator, which is essential for ensuring consistency and fairness in actual market transactions. Furthermore, the system lacks predictive analytics capabilities, offering only historical data without insights into future price trends.

AquaTimbang addresses these gaps by providing a web and mobile-based platform that enables direct access for vendors, allowing them to actively participate in the price transparency process. The system integrates a weight-based fair price calculator to standardize transactions and incorporates basic predictive analytics to help users anticipate price fluctuations and make informed decisions AquaTimbang is purpose-built for multi-stakeholder use, offering role-based access for fish vendors and LGUs, and a mobile-optimized interface that is accessible even in market environments.

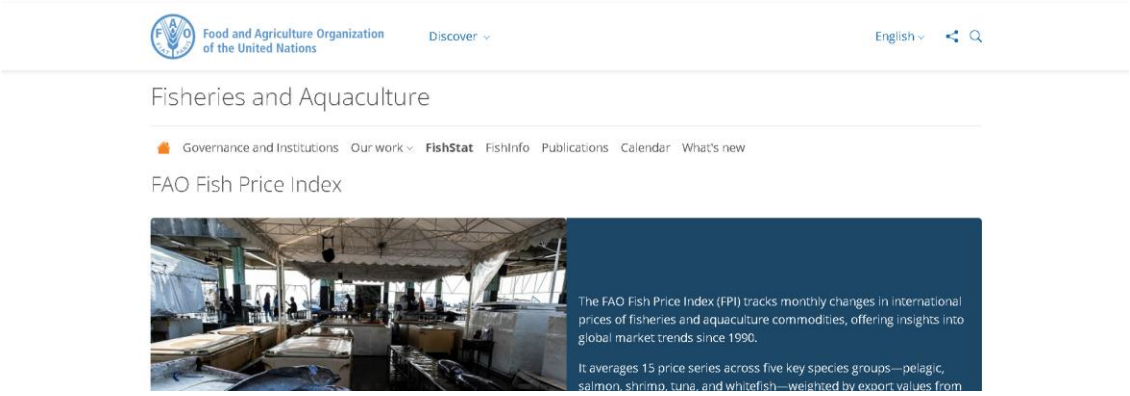


Figure 5

FAO Dashboard

As shown in Figure 5, the FAO Fish Price Index is a global barometer developed by the Food and Agriculture Organization of the United Nations that measures monthly changes in international fish prices. It covers 15 price series across five major species groups: shrimp, salmon, tuna, pelagic fish, and whitefish. Prices are sourced from European Market Observatory for Fisheries and Aquaculture (EUMOFA), Statistics Norway, and INFOFISH. As of February 2026, the index averaged 129.1 points, up 11.8 percent from February 2025 (FAO, 2026).

The FAO Fish Price Index operates at a global level and does not provide localized pricing for Philippine markets. It lacks coverage of Philippine wet fish species such as galunggong and tulingan, and does not include a weight-based price calculator, predictive analytics, or mobile accessibility for local fish vendors.

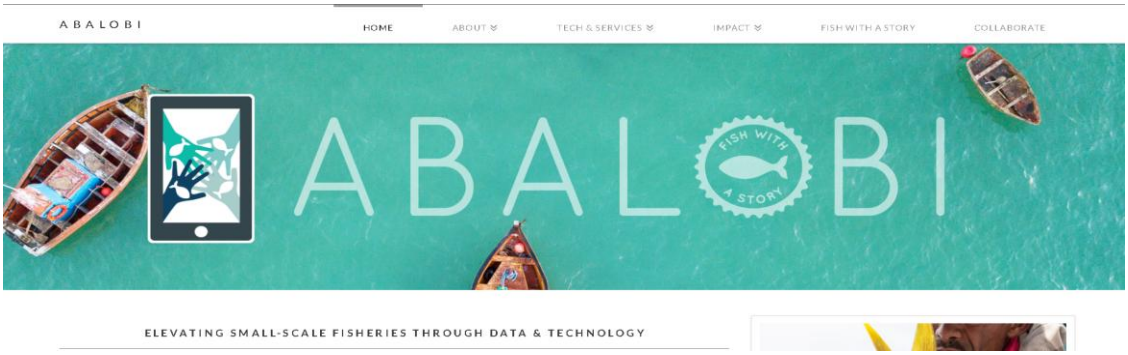


Figure 6

ABALOBI Dashboard

As shown in Figure 6, the ABALOBI is a mobile and web-based platform that enables small-scale fishers to capture, own, visualize, and use fisheries catch landing data. Recognized by MIT, the platform supports fisher groups, cooperatives, which connects fishing communities with fair local and global markets and linked financial services. Features include multilingual, offline-compatible catch landing data entry, real-time data dashboards, a web-based administration console, and customizable lists of gear, species, and vessels.

ABALOBI focuses primarily on catch documentation rather than price transparency and predictive analytics. It does not provide a weight-based price calculator for fish sold by weight, price trend forecasting, or features for local government units to input and validate market prices. The platform is designed for South African and East African contexts and does not address the specific needs of Philippine fish markets, including local fish species and integration with Philippine government monitoring systems.

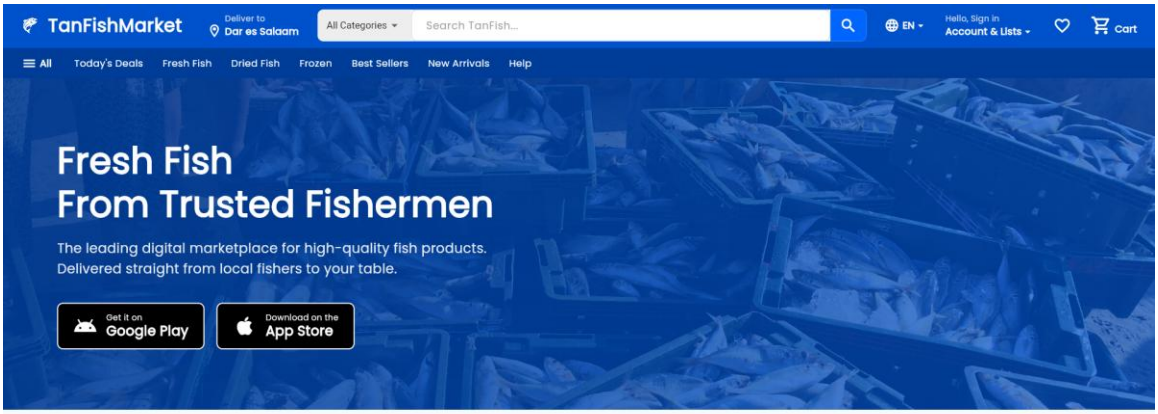


Figure 7

TanFish Market Dashboard

As shown in Figure 7, the TanFish Market is a digital marketplace launched in February 2026 designed to connect fishers from Lake Tanganyika with buyers locally and internationally. The platform allows fishers to list daily catches and processed fish products with detailed descriptions and photos, manage inventory, and receive orders digitally. Buyers can browse a diverse catalog of fresh, verified fish products, place orders securely, and track deliveries in real-time. The platform integrates secure payment gateways and logistics coordination to reduce post-harvest losses and improve market access for fishing communities. The initiative was developed in partnership with the University of Dar es Salaam, FAO, and the European Union's FISH4ACP programme (TanFish Market, 2026).

TanFish Market facilitates direct sales but lacks structured price monitoring systems, predictive analytics, and tools for government price validation. AquaTimbang addresses these gaps by focusing on price transparency and market monitoring, integrating

weight-based calculation, predictive analytics, and LGU validation features tailored to Philippine fish markets.



Figure 8

University of Dar es Salaam's Dashboard

As shown in Figure 8, The University of Dar es Salaam's College of Information and Communication Technologies (CoICT) developed a digital marketplace platform for fishing communities along Lake Tanganyika. Announced in April 2025, the platform digitally links fishers, processors, traders, and end consumers through mobile phones or computers, enabling real-time inventory and price monitoring, AI-powered messaging, secure payment channels (from mobile money to bank transfers), and logistics coordination. The platform also provides interactive maps for location-based services and hosts educational content that encourages sustainable fishing. The project includes a training program targeting at least 200 local users, with 60 percent women and 40 percent youth, to foster community ownership and long-term viability. Stakeholders involved include the Ministry of Livestock and Fisheries, Tanzania Fisheries Research Institute (TAFIRI), e-Government Agency (e-GA), NMB Bank PLC, and the Tanzania Revenue Authority (TRA) (UDSM, 2025).

The UDSM platform focuses on marketplace functions and lacks weight-based price calculation and predictive analytics features. It is designed for Tanzanian fishing communities and does not address the specific context of Philippine fish markets. While the platform includes price monitoring capabilities, it does not provide structured price

comparison tools across multiple markets or historical trend analysis. The platform is still in the development and pilot testing phase, with full implementation pending.

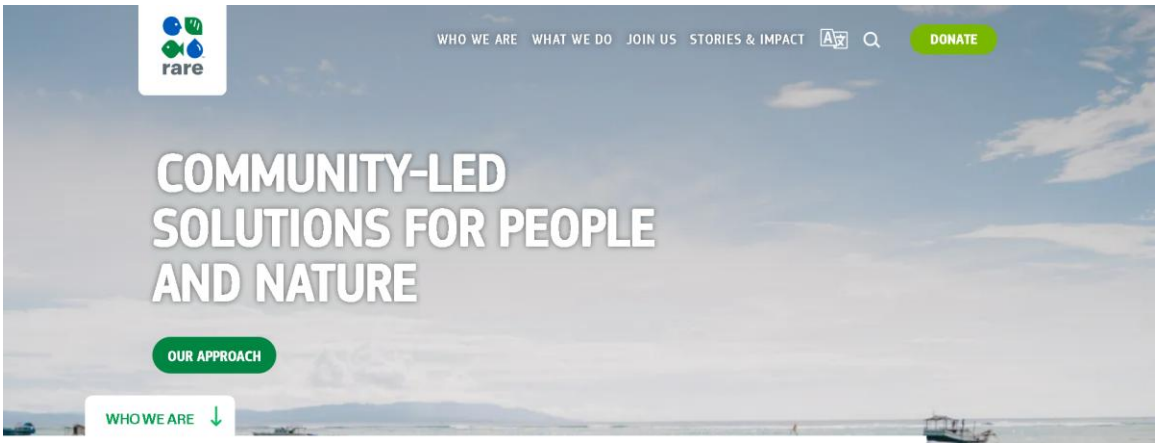


Figure 9

RARE Dashboard

As shown in Figure 9, RARE is an international conservation organization that invests in people and nature to solve conservation and development challenges by partnering with local leaders and communities to co-manage, protect, and restore natural resources against threats like climate change (RARE, 2025). RARE Philippines developed the OurFish App to support sustainable fisheries management. The mobile application is designed for fish vendors and fish buyers to record their daily catch and transaction data, enabling better data collection for fisheries management. The app allows users to log their catches, record sales transactions, and gain data insights to improve their business operations. RARE Philippines has conducted training sessions with local government units (LGUs) to reintroduce the app to coastal communities, including in San Carlos City, where it was initially implemented in 2017 before being paused during the COVID-19 pandemic (San Carlos City Government, 2024). The app supports LGUs in formulating appropriate interventions, rules, and programs for maintaining fisheries resources.

OurFish App focuses primarily on catch documentation and transaction recording rather than price transparency and predictive analytics. It does not provide a weight-based fair price calculator, price trend analysis, or predictive forecasting features. The app is

designed for general fisheries data collection and does not offer specialized features for comparing prices across different markets or predicting future price fluctuations. Additionally, the app's availability and active use have been inconsistent, with usage declining during the pandemic, indicating challenges in sustained adoption. AquaTimbang addresses these gaps by focusing specifically on price transparency with a weight-based fair price calculator, integrating basic predictive analytics to help users anticipate price trends, and providing municipal-level price dashboards with LGU validation features tailored to Philippine fish market.

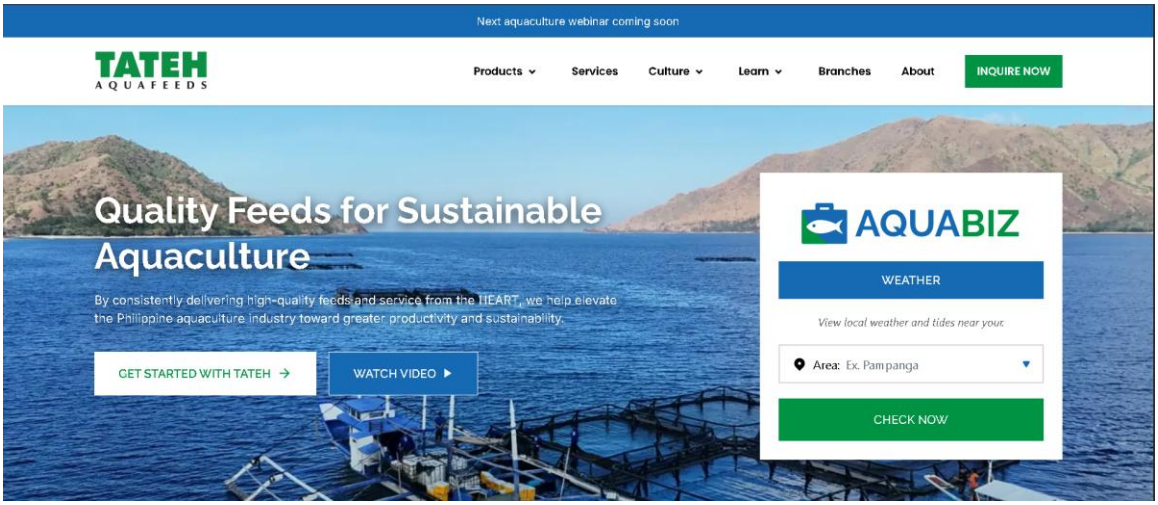


Figure 10

TATEH Dasboard

As shown in Figure 10, Tateh AquaBiz is a mobile application developed by Santeh Feeds Corporation, a Philippine-based company serving the aquaculture industry. The app provides current wholesale fish prices, species search, feed lookup, hatchery directories, disease identification, and growth calculators (Santeh Feeds Corporation, 2025).

AquaBiz focuses on aquaculture production rather than market price transparency. It lacks historical price data, predictive analytics, a weight-based price calculator, and LGU validation features. AquaTimbang addresses these gaps by providing weight-based price calculation, predictive analytics, and municipal-level price dashboards with LGU validation tailored to Philippine fish markets.

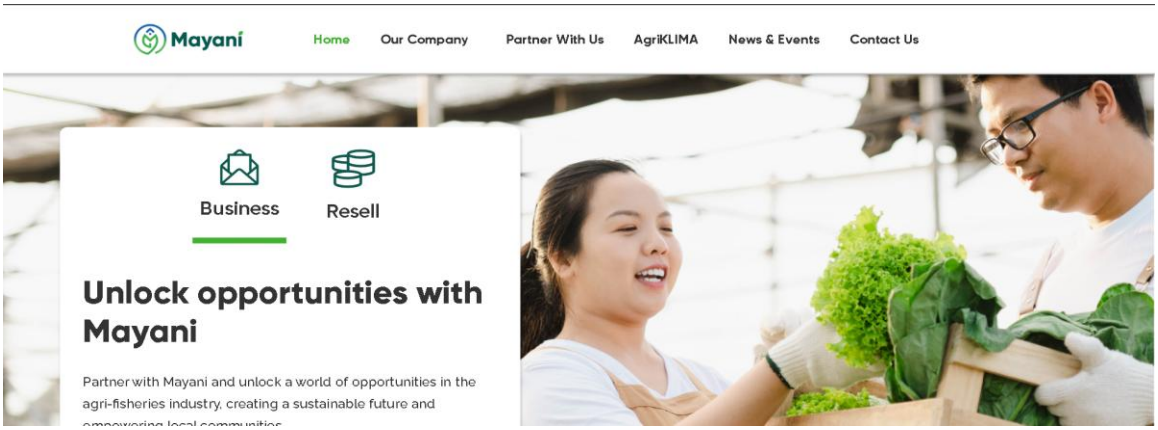


Figure 11

Mayani Dasboard

As shown in Figure 11, Mayani is a Philippine-based social enterprise that connects smallholder farmers and fisherfolk directly to consumers and businesses through a farm-to-table digital platform. The platform enables users to purchase fresh produce online, supports rural livelihoods by providing sustainable market access, and promotes local agriculture and fisheries through technology-driven supply chains (Mayani, 2023).

Mayani focuses on e-commerce and direct-to-consumer sales rather than price transparency and market monitoring. It does not provide a weight-based price calculator, predictive analytics for price trends, role-based access for different stakeholders, or features for LGU price validation. AquaTimbang addresses these gaps by providing weight-based price calculation, predictive analytics, and municipal-level price dashboards with LGU validation features tailored to Philippine fish markets.



Figure 12

Bantay Presyo Dashboard

As shown in Figure 12, Bantay Presyo is a price monitoring platform developed by the Philippine Department of Agriculture (DA) that provides daily price updates of various agricultural and basic commodities across major markets nationwide. The system aims to promote market transparency by making current price information accessible to consumers, traders, and government agencies.

Bantay Presyo focuses primarily on agricultural commodities and does not provide specialized features for fish market price transparency. It lacks a weight-based price calculator, predictive analytics for price trends, role-based access for different stakeholders, and mobile-friendly interface for fish vendors. AquaTimbang addresses these gaps by focusing specifically on fish price transparency with weight-based calculation, predictive analytics, and municipal-level dashboards with LGU validation features tailored to Philippine fish markets.

Comparative Matrix

Table 4 presents a comparative overview of AquaTimbang against the two related systems reviewed in this section. The table highlights key differences in core features, target users, and platform capabilities, demonstrating how AquaTimbang addresses the gaps found in existing systems.

Table 4

COMPARATIVE MATRIX

System	Type	Price Transparency	Weight-Based Calculator	Predictive Analytics	LGU Validation	Mobile Access	LocalPH Focus	% Score
AquaTimbang	Integrated Platform	✓	✓	✓	✓	✓	✓	100%
Sinay Fish Price App	International Commercial	✓	✗	✗	✗	✓	✗	33%
BFAR eFPMS	Government Monitoring	✓	✗	✗	✓	✗	✓	50%
FAO Fish Price Index	International Index	✗	✗	✗	✗	✗	✗	17%
ABALOBI	Catch Documentation	✗	✗	✗	✗	✓	✗	17%
TanFish Market	Digital MarketPlace	✗	✗	✗	✗	✓	✗	17%
UDSM Platform	Digital MarketPlace	✓	✗	✗	✗	✓	✗	33%
OurFish App (RARE)	Catch Documentation	v	✗	✗	✓	✓	✓	67%
Tateh AquaBiz	Aquaculture Support	✓	✗	✗	✗	✓	✓	50%
Mayani	E-Commerce	✗	✗	✗	✗	✓	✓	33%
Bantay Presyo (DA)	Price Monitoring	✓	✗	✗	✓	✗	✓	50%

The comparative matrix of selected fish price monitoring systems based on key criteria. Existing systems primarily focus on price monitoring, catch documentation, or digital marketplaces, but most lack weight-based calculation, predictive analytics, or LGU validation. In contrast, the proposed AquaTimbang integrates price transparency, weight-based calculator, predictive analytics (SARIMAX), LGU validation, and mobile access. Designed specifically for Philippine local fish markets, AquaTimbang addresses inconsistent pricing and lack of centralized data, empowering vendors and enabling data-driven LGU regulation.

TECHNICAL BACKGROUND

The development of AquaTimbang: A Predictive Analytics-Driven Fish Price Transparency System for Informed Market Decision-Making is grounded in modern data management and analytical technologies that integrate price monitoring, fair computation, and basic predictive insights to address the specific challenges of inconsistent pricing, lack of transparency, inefficient data management, and information asymmetry in local fish markets. Traditional fish market operations rely heavily on manual processes such as verbal price negotiations, handwritten records, and inconsistent weighing practices. While these methods have been used for generations, they lack transparency, standardization, and efficiency, directly harming fish vendors who earn unstable incomes and local government personnel who lack reliable data for regulation and planning.

Introduction to the Technology

AquaTimbang is specifically designed for deployment in local fish markets, where fish vendors, local government personnel, and system administrators require accurate, timely, and verifiable pricing data to facilitate fair transactions and informed decision-making. To address existing limitations, the system adopts a data-driven architecture that enhances market transparency through structured data collection (daily price entry and validation), standardized price computation (weight-based fair price calculator), a centralized municipal fish price dashboard, and basic trend analysis (e.g., moving averages). This integration ensures that pricing decisions are no longer based on subjective or fragmented information, thereby improving market fairness, accountability, and monitoring capability. By providing accessible and reliable market intelligence, AquaTimbang empowers vendors to secure fairer incomes and enables local governments to implement data-driven price regulation.

Theoretical Concepts

The development of AquaTimbang is grounded in several theoretical concepts that form the foundation for its design, functionality, and intended impact. These concepts address the identified challenges of inconsistent pricing, lack of transparency, inefficient

data management, and limited access to predictive information in local fish markets. Each theory is operationalized through specific system features.

Price transparency theory holds that when market participants have access to clear, consistent pricing information, market efficiency improves and information asymmetry decreases. Van Campenhout (2024) examined how ICT-based price information services reduce search costs and increase bargaining power for farmers in developing countries. A 2026 randomized controlled trial on mobile money services demonstrated that transparent pricing increases trust among counterparties and improves market efficiency. In AquaTimbang, this theory is operationalized through the Municipal Fish Price Dashboard, which displays current and historical validated fish prices across selected species. By making pricing data equally accessible to fish vendors, local government personnel, and consumers, the system directly reduces information asymmetry that currently disadvantages vendors.

Information asymmetry occurs when one party in a transaction possesses superior information, leading to market inefficiencies and exploitation. Van Campenhout (2024) noted that incomplete and asymmetric information contributes to inefficiencies in food supply chains, with smallholders lacking bargaining power against traders. Ritonga et al. (2026) studied the rice industry in Indonesia and found that intermediaries control price information, resulting in asymmetric market power that disadvantages farmers. AquaTimbang directly addresses this through its Fair Price Calculator (Weight × LGU-validated price per kg) and centralized database. By ensuring that all stakeholders access verified pricing information, the system reduces the exploitative practices documented in agricultural value chains, promoting fair valuation and reducing vendor dependency on middlemen.

Predictive analytics uses historical data and statistical models to forecast future trends. Warthon and Valenzuela (2026) developed PredictaMAR, an integrated decision-support platform for artisanal fisheries that reduced operational uncertainty and fuel consumption by 20-25%. Zhou et al. (2026) applied LSTM neural networks for fish

habitat prediction, achieving high accuracy. Halvorsen et al. (2026) examined ecological forecasting for fisheries management, demonstrating the importance of model transferability. AquaTimbang integrates these principles through its Predictive Analytics Module, which applies the Seasonal Autoregressive Integrated Moving Average with Exogenous Regressors (SARIMAX) model to historical daily price data combined with external factors such as fuel prices, weather conditions, and holiday indicators. SARIMAX extends the classical ARIMA framework by explicitly modeling weekly and annual seasonality (using seasonal parameters P, D, Q, s) and incorporating exogenous variables that influence fish prices. The system uses pmdarima.auto_arima with seasonal=True and exogenous regressors to automatically select optimal non-seasonal (p, d, q) and seasonal (P, D, Q, s) parameters based on AIC, then fits a SARIMAX model using statsmodels.tsa.statespace.sarimax.SARIMAX. While SARIMAX is more sophisticated than simple moving averages or non-seasonal models, it remains interpretable and suitable for local market contexts without requiring deep learning infrastructure, while capturing realistic market drivers. This approach provides actionable short-term price forecasts to vendors and local government personnel for informed market decision-making, including the effects of holidays, fuel cost fluctuations, and weather disruptions.

DDDM theory posits that organizations and individuals achieve better outcomes when decisions are based on data analysis rather than intuition alone. Warthon and Valenzuela (2026) demonstrated that satellite-derived oceanographic data enables risk-aware decision-making for small-scale fishers, reducing exploratory navigation time and fuel consumption. Ritonga et al. (2026) advocated for digital price transparency platforms as a pathway toward equitable market reform. AquaTimbang operationalizes DDDM by providing stakeholders with accessible market intelligence through the Municipal Fish Price Dashboard, ARIMA-generated price forecasts, and Reports Module (PDF/CSV export). This enables evidence-based planning, allowing vendors to decide when to sell and LGUs to anticipate price fluctuations.

RBAC is a security paradigm where permissions are assigned to organizational roles rather than individuals, ensuring users access only information relevant to their

responsibilities. Ortiz-Crespo et al. (2024) developed Ushauri, a digital advisory service in Tanzania using a co-design methodology that distinguished between farmer-users and extension agent-users with different access levels. In AquaTimbang, RBAC differentiates permissions among Fish Vendors (view prices, use calculator, monitor trends), LGU Personnel (input and validate daily prices, generate reports), and System Administrator (manage accounts, fish species, data integrity). This theory ensures data integrity and appropriate access, fostering trust among all stakeholders.

UCD is a design philosophy that places end-users at the center of development, with iterative testing to ensure usability and accessibility. Ortiz-Crespo et al. (2024) used a co-design methodology to elicit feedback from farmers and extension agents, creating a system that addressed actual user needs. Oxfam (2026) emphasized that interventions must be designed with local community participation to ensure effectiveness. AquaTimbang follows UCD principles through its Agile Development Methodology, enabling continuous refinement based on feedback from fish vendors, local government personnel, and administrators collected during interviews, surveys, and user acceptance testing.

CBFM theory emphasizes that fisheries management is most effective when local communities are empowered to participate in decision-making and monitoring. Oxfam (2026) documented how civil society action in the Philippines led to community-based coastal resource management that ensures better access and influence for poor fishing communities. Van Campenhout (2024) noted that ICT applications must be designed with local context in mind. AquaTimbang embodies CBFM principles by enabling multi-stakeholder participation where fish vendors, local government personnel, and administrators each have designated roles in market monitoring and decision-making.

Digital transformation involves integrating digital technologies to improve data management, market access, transparency, and operational efficiency. Warthon and Valenzuela (2026) demonstrated that the PredictaMAR platform reduced fuel consumption and stabilized fishers' incomes. Oxfam (2026) noted that sustained digital innovation is required to protect fisher livelihoods. Halvorsen et al. (2026) emphasized data-driven

forecasting models for effective fisheries management. AquaTimbang applies digital transformation theory by replacing fragmented, manual processes (verbal negotiations, handwritten records, inconsistent weighing) with a centralized digital platform that improves data accuracy, accessibility, and timeliness.

Related Technologies

The implementation of the AquaTimbang System involves several interconnected technologies that work together to ensure accurate price monitoring, fair price computation, and reliable price forecasting. These technologies support the system's data collection, predictive analytics, and user access management processes, forming an integrated digital infrastructure for price transparency within local fish market environments.

Django Framework

Django is a high-level Python web framework that follows the Model-View-Template (MVT) architectural pattern. It provides built-in tools including an Object-Relational Mapper (ORM), authentication system, admin interface, and security features (Django Software Foundation, 2025). In AquaTimbang, Django serves as the backend foundation. The ORM manages database interactions with MySQL (storing price records, user accounts, fish species, external factor data, and SARIMAX forecast outputs). Django's authentication system is extended with custom user groups (Fish Vendor, LGU Personnel, System Administrator) to enforce role-based access control. The Django REST Framework (DRF) builds RESTful APIs that the React frontend consumes.

React.js

React.js is a JavaScript library for building user interfaces, using a component-based architecture and a virtual DOM for efficient updates (React Community, 2024). In AquaTimbang, React.js powers the Municipal Fish Price Dashboard, Daily Price Entry Form, Fair Price Calculator, and SARIMAX Forecast Visualizer. State management uses React hooks (useState, useEffect) and Context API. Data fetching uses Axios to communicate with the Django REST API. Responsive design (CSS3, Flexbox, Grid) ensures the system is accessible on desktop computers (used by LGU personnel) and tablets

(used by vendors in market stalls), following a mobile-first design principle. Forecast charts (line graphs with confidence intervals) are rendered using Chart.js or Recharts for clarity.

MySQL Database

MySQL is an open-source relational database management system (RDBMS) known for reliability, performance, and ACID compliance (Oracle Corporation, 2025). In AquaTimbang, MySQL stores all structured data: fish species catalogs, historical price records (by date, species, and validated price per kg), external factor data (e.g., daily fuel prices, weather conditions, holiday flags), user accounts with assigned roles, fair price calculation logs, and SARIMAX forecast results including predicted values, confidence intervals, and accuracy metrics (MAPE, RMSE). The schema is normalized to avoid redundancy, with indexes on frequently queried columns (date, species, and external factor IDs) to keep dashboard queries fast—essential for real-time price viewing by vendors.

SARIMAX Forecasting

Pandas provides high-performance data structures (DataFrame) for data manipulation (The Pandas Development Team, 2025). NumPy enables numerical computing with multi-dimensional arrays (Harris et al., 2020). Statsmodels supplies the SARIMAX class for seasonal model fitting, exogenous regressor integration, and forecasting (Seabold & Perktold, 2010). Pmdarima provides auto_arima for automatic selection of (p, d, q) and seasonal (P, D, Q, s) parameters based on AIC (Smith et al., 2017).

In AquaTimbang, these libraries implement the Predictive Analytics Module. Historical price records, together with external factor data (e.g., fuel prices, weather conditions, holiday indicators), are retrieved from MySQL into Pandas DataFrames, indexed by date. The system calls pmdarima.auto_arima with seasonal=True and exogenous variables included to determine the optimal SARIMAX model order, then fits a SARIMAX model using statsmodels.tsa.statespace.sarimax.SARIMAX. Short-term forecasts (e.g., next 7 days) are generated along with confidence intervals, incorporating both the time-series dynamics and the influence of future external factors. Accuracy is

evaluated using Mean Absolute Percentage Error (MAPE) and Root Mean Square Error (RMSE) on hold-out test data. Results are presented as interactive charts and textual insights on the dashboard, keeping the output interpretable for non-technical users while leveraging a robust seasonal forecasting method that accounts for real-world market drivers.

Cloud Hosting (AWS / GCP)

Cloud hosting platforms like Amazon Web Services (AWS) and Google Cloud Platform (GCP) provide on-demand virtual servers, storage, and managed services (Amazon Web Services, 2024; Google Cloud, 2025). For AquaTimbang, the system is deployed on a virtual private server (e.g., AWS EC2 or GCP Compute Engine) running a Linux distribution (Ubuntu Server). The Django application is served via Gunicorn with Nginx as a reverse proxy. MySQL runs on the same or separate instance. Static and media files are optionally served from cloud object storage (AWS S3 or GCP Cloud Storage). Daily automated backups of the MySQL database are scheduled, and HTTPS is enforced using Let's Encrypt certificates. This cloud deployment ensures that the prototype is accessible to stakeholders (fish vendors, local government personnel, system administrators) for testing and evaluation during the deployment phase, while simulating a real-world hosting scenario. SARIMAX model retraining can be triggered on a schedule (e.g., nightly via cron job) using the same cloud infrastructure, incorporating the latest price and external factor data.

CHAPTER III
RESEARCH METHODOLOGY

This Chapter describes the overall approach used in conducting the study. It outlines the research methods, procedures, and techniques applied to gather data, analyze information, and develop the project. The purpose is to ensure that the process followed is organized, consistent, and appropriate for achieving the study's objectives.

PLANNING PHASE

The planning phase is the initial stage in the system development process, where the objectives, scope, and overall direction of the project are clearly defined. During this phase, key requirements are identified, and the feasibility of developing the proposed system is assessed. It involves understanding the current problems, determining system goals, and outlining the resources needed for implementation.

Requirement Specifications: Feasibility Study

Requirement Specification defines the functional and operational needs of the proposed AquaTimbang: A Predictive Analytics–Driven Fish Price Transparency System. It identifies the essential features and system capabilities required to support fair price computation, historical data tracking, predictive trend analysis, and administrative monitoring within the local fish market environment. To ensure that these requirements are grounded in practical viability, the system is examined through technical, economic, and organizational feasibility analyses, which assess development risks, financial sustainability, and stakeholder alignment. Together, these analyses provide a structured foundation for defining the system's functional and operational specification.

Technical Feasibility

This technical feasibility analysis evaluates the proposed AquaTimbang: A Predictive Analytics-Driven Fish Price Transparency System for Local Government Units (LGUs). It identifies potential risks in system development and adoption, and outlines mitigation strategies to ensure successful implementation.

Table 5

Technical Feasibility

Criteria	**Description**	**Risk Level**	**Mitigation Strategy**
Familiarity with the Application	LGU Officers have limited experience with digital platforms for fish price monitoring. Manual processes dominate, creating a learning curve for users.	Moderately High	Conduct training sessions, design a user-friendly interface, and provide helpdesk support to ensure smooth adoption.
Familiarity with the Technology	Required technologies (MySQL, Pandas/NumPy for analytics, AWS/GCP hosting, RBAC) are widely available and supported. Development teams are familiar with these tools.	Low	Use established frameworks, follow secure coding practices, and leverage cloud services for scalability and reliability.
Project Size	The system includes multiple modules: price	Medium	Prioritize core modules (price entry, fair price

	entry/validation, fair price calculator, municipal dashboard, predictive analytics, and user management. Managing all within the capstone timeline may be challenging.		calculator, dashboard). Advanced features (trend forecasting, extended reporting) can be phased in later.
Compatibility	LGUs already use basic IT infrastructure (computers, internet, spreadsheets). AquaTimbang is compatible with existing workflows and requires only moderate upgrades.	Low	Ensure cross-device/browser testing, adopt lightweight system architecture, and provide offline fallback options where internet connectivity is weak.

Economic Feasibility

Economic feasibility evaluates whether the expected benefits of the system outweigh the estimated costs of development, implementation, operation, and maintenance. It analyzes the financial viability of the system by assessing development expenses, operational costs, potential savings, and long-term benefits. Economic feasibility also determines whether the system can provide cost-effective improvements in efficiency,

security, and overall operational performance while ensuring that available resources are used efficiently.

		2026	2027	2028	2029	Total
Benefits						
	Increased Sales	₱30,000	₱31,800	₱33,708	₱35,730	₱131,238
	Reduced Pricing Errors	₱8,000	₱8,320	₱8,653	₱8,999	₱33,972
	Optimized LGU Staff Time	₱5,000	₱5,200	₱5,408	₱5,624	₱21,232
Total Benefits		₱43,000	₱45,320	₱47,769	₱50,353	₱186,442
Present Value of Total Benefits (10%)		₱39,091	₱37,455	₱35,891	₱34,396.00	₱146,833
Development Costs						
	Labor: Analysis & Design	₱10,000	₱0	₱0	₱0	₱10,000
	Labor: Implementation	₱15,000	₱0	₱0	₱0	₱15,000
	Development Training	₱2,000	₱0	₱0	₱0	₱2,000
	Office Space & Equipment	₱1,000	₱0	₱0	₱0	₱1,000
Total Development Costs		₱28,000	₱0	₱0	₱0	₱28,000
Operational Costs						
	Cloud Hosting & Domain	₱4,000	₱4,160	₱4,326	₱4,499	₱16,985
	Internet & Communication	₱2,000	₱2,080	₱2,163	₱2,250	₱8,493
	LGU Personnel Time	₱5,200	₱5,408	₱5,624	₱5,849	₱22,081
Total Operational Costs		₱11,200	₱11,648	₱12,113	₱12,598	₱47,559
Total Costs		₱39,200	₱11,648	₱12,113	₱12,598	₱47,559
Present Value of Total Costs (10%)		₱39,200	₱10,589	₱10,011	₱9,466	₱69,266
Total Benefits - Total Costs		₱3,800	₱33,672	₱35,656	₱37,755	₱110,883
Cumulative Net Cash Flow		₱3,800	₱37,472	₱73,128	₱110,883	
Present Value of Net Benefits		-₱109	₱26,866	₱25,880	₱24,930	₱77,567
Intangible Benefits						
	Enhanced public trust	Transparent fish pricing increases community confidence in LGU market governance.				
	Better decision-making	LGU officers gain access to real-time, verified price data for policy formulation.				
	Improved LGU image	Adoption of digital innovation positions the LGU as modern and responsive.				

Figure 13

Economic Feasibility

Figure 13 presents the cost-benefit analysis of the AquaTimbang System by comparing the projected development and operational costs with the expected benefits over several years. It highlights the financial gains from reduced manual processing, improved security efficiency, and faster entry and exit verification. The analysis also shows the system's return on investment (ROI), break-even point, and net present value (NPV), indicating that the project is financially feasible and beneficial for long-term campus security operations.

Organizational Feasibility

Organizational feasibility examines whether the proposed system can be successfully adopted and sustained within the existing structure, culture, and resources of the organization. It evaluates the alignment of AquaTimbang with local government priorities, the roles and responsibilities of stakeholders, and the capacity of organizational management to support implementation. This analysis ensures that the system is not only technically viable but also organizationally practical, with clear leadership, stakeholder participation, and structured management processes to guarantee long-term success.

AquaTimbang is organizationally feasible because the LGU has a legal responsibility to monitor and validate fish prices, fisherfolk have a clear need for transparent pricing, and a designated system administrator can manage the backend. Potential challenges—such as low fisherfolk participation or limited technical literacy—can be addressed through community orientation, a simple mobile interface, and phased implementation.

AquaTimbang is strategically aligned with the Local Government Unit's (LGU) mandate to ensure fair trade, protect fisherfolk from exploitative middlemen, and modernize market governance. By digitizing daily price entry and validation, the system supports the Department of Veterinary Medicine and Fisheries (DVMF) in Cebu City, which oversees market-related activities and data monitoring. The project also complements national initiatives such as the Electronic Fisheries Price Monitoring System (eFPMS) and the Department of Information and Communications Technology's online price monitoring platforms by extending their focus from data collection to predictive analytics and fair, weight-based pricing.

At a broader level, AquaTimbang contributes to the United Nations Sustainable Development Goals (SDGs), specifically SDG 1 (No Poverty) by enabling vendors to secure fairer and more stable incomes, SDG 2 (Zero Hunger) by promoting equitable fish distribution, SDG 8 (Decent Work and Economic Growth) by fostering transparent and trustworthy market environments, SDG 9 (Industry, Innovation, and Infrastructure) by applying accessible predictive analytics to traditional markets, and SDG 12 (Responsible Consumption and Production) by reducing inefficiencies caused by information asymmetry. By aligning with both local governance priorities and global development goals, AquaTimbang ensures organizational support and long-term relevance.

The stakeholders of AquaTimbang play a vital role in ensuring the success of the project. Fish vendors are the primary beneficiaries, as they will use the fair price calculator, access historical and current prices, and monitor trends. Their participation is crucial because it ensures data accuracy and adoption of the system. Local Government Unit

(LGU) personnel are responsible for inputting and validating daily fish price data, generating reports, and overseeing market transparency, while the system administrator manages accounts, species lists, and system configuration to guarantee data integrity and security.

The LGU Fisheries Officer or Market Supervisor serves as the project champion, with the authority to mandate system use, allocate staff for validation, and encourage vendor participation. Their leadership ensures that AquaTimbang is embedded into official market operations, while the system administrator, either an LGU-assigned IT personnel or a dedicated role, champions the technical side by maintaining system uptime, safeguarding data security, and ensuring accurate computation of average prices.

Fish vendors, as end users, are central to the system's success. Their active participation in submitting and validating prices ensures reliable computation of fair prices. To encourage adoption, AquaTimbang offers a mobile-friendly interface, offline-capable data entry for areas with poor connectivity, and visual confirmation of submitted prices to build trust. The LGU may also conduct orientation sessions and provide small incentives, such as priority stall assignments, to boost initial participation.

Organizational management within the LGU operates through a structured hierarchy involving the Market Office, Agriculture Office, and IT Office. Responsibilities are clearly distributed, with market inspectors validating prices, IT personnel managing backend operations, and administrators overseeing accounts and system integrity. Management is prepared to issue a memorandum mandating AquaTimbang use for daily price validation, designate a dedicated system administrator, and schedule training sessions for LGU staff and vendors. A phased rollout, beginning with selected fish species and gradually expanding, will help manage workload and refine processes incrementally, ensuring smooth adoption and sustainability of the system.

Table 6

Stakeholder Analysis

Stakeholder	**Role in the System**	**Contribution to Feasibility**
Stakeholder Champion	LGU Fisheries Officer or Market Supervisor with authority to mandate system use	Provide leadership, enforce adoption, allocate staff, and embed AquaTimbang into official operations
LGU Personnel	Market regulators responsible for inputting and validating daily fish price data	Guarantee transparency and accountability by generating reports and overseeing market practices
System Administrator	Technical manager overseeing accounts, species lists, and system configuration	Maintain data integrity, ensure system uptime, and safeguard security for reliable operations
Fish Vendors	End users and primary beneficiaries who use the fair price calculator and monitor trends	Ensure data accuracy and adoption through active participation; benefit from fair and transparent pricing
Organizational Management	LGU hierarchy including Market Office, Agriculture Office, and IT Office	Supervise operations, schedule training, enforce system use, and align processes with organizational goals

System Context Diagram (DFD Level 0)

The System Context Diagram, also referred to as a Data Flow Diagram Level 0 (DFD Level 0), presents the highest-level view of RecyConnect as a single system process surrounded by all external entities that interact with it. This diagram identifies the system boundary and shows all data flows entering and leaving the system without revealing any internal processes. It serves as the starting point for the more detailed functional decomposition that follows. AquaTimbang interacts with three external entities—Fish Vendors, LGU Personnel, and System Administrator—and the central system process.

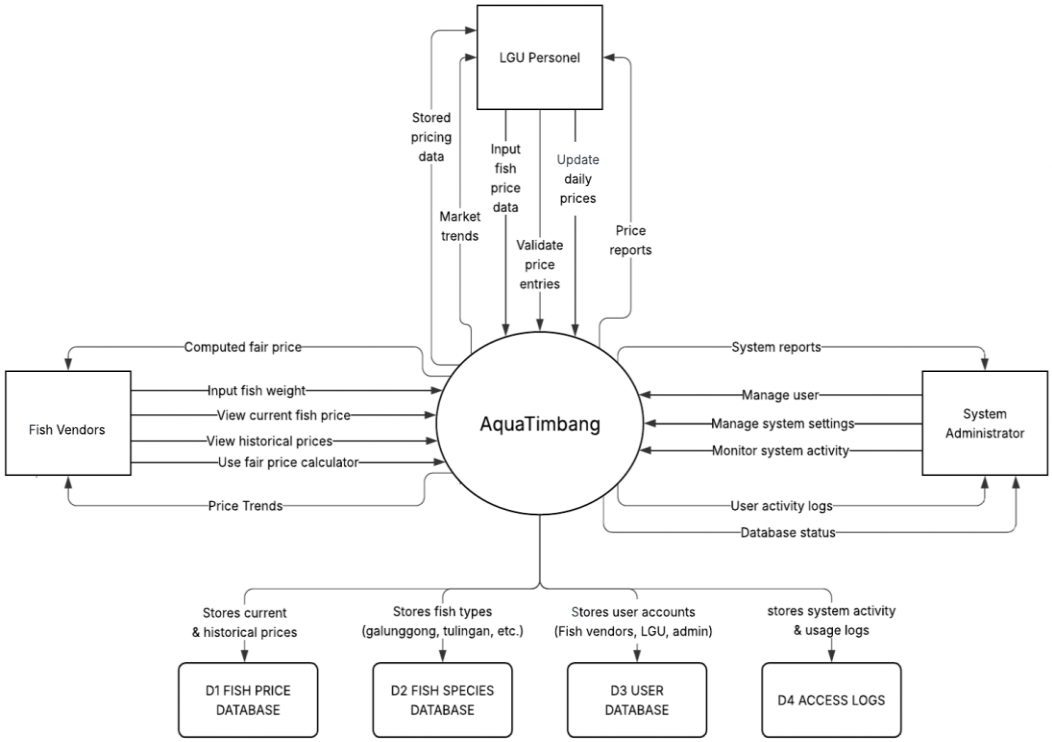


Figure 14

Data Flow Diagram Level 0

Figure 14 presents the Data Flow Diagram (DFD) Level 0 of the AquaTimbang system Fish Vendors input fish weight and access the fair price calculator, enabling them to view computed fair prices, current LGU-validated reference prices, historical price data, and price trends. LGU Personnel input fish price data, update daily prices, validate entries, and in return receive stored pricing data, market trends, and price reports. The System Administrator manages user accounts, configures system settings, and monitors system activity, while receiving system reports, user activity logs, and database status. All data are stored in four data stores: D1 Fish Price Database (current and historical prices), D2 Fish Species Database (fish types such as galunggong and tulingan), D3 User Database (accounts for vendors, LGU, and admin), and D4 Access Logs (system activity and usage logs). These stores enable centralized data management, support participatory pricing, and ensure transparent and efficient fish price monitoring.

User Flow Diagram

The User Flow Diagram shows how different users interact with the system by illustrating the step-by-step process of accessing and using the AquaTimbang System, including authentication, and system operations.

Fish Vendor

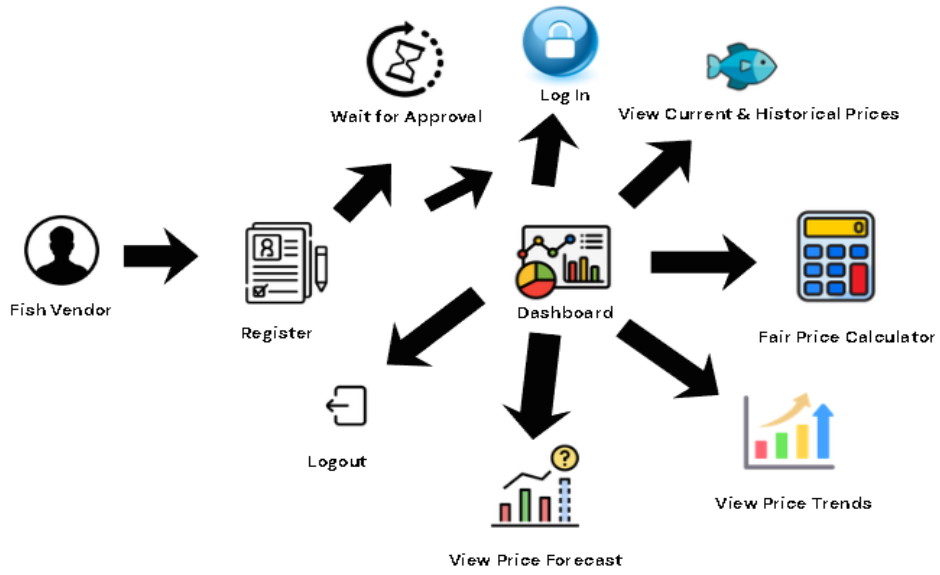


Figure 15

Fish Vendor Program Workflow

Figure 15 shows the Fish Vendor workflow in AquaTimbang, after registering an account, the fish vendor must wait for approval from either an LGU officer or the System Administrator. Once approved, the fish vendor can log in and access a dashboard that provides a quick overview of current & historical fish prices.The fish vendor is responsible for viewing current and historical prices, computing fair prices using the weight-based calculator, and checking price trends & forecasts to decide when to sell. This layout emphasizes the fish vendors's role in actively participating in price setting and accessing transparent pricing information for informed market decisions.

LGU Officer

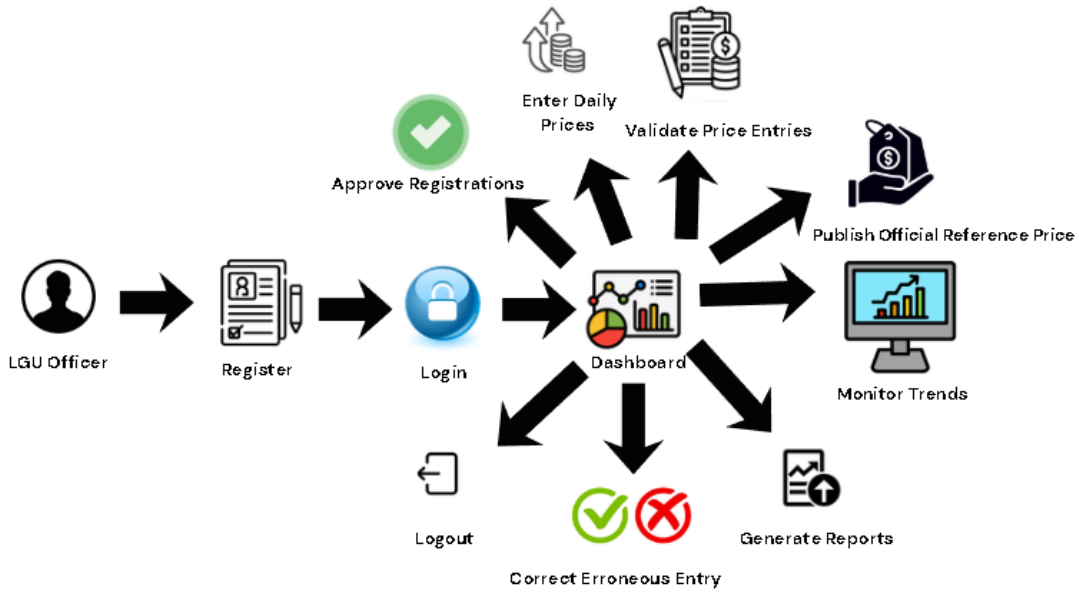


Figure 16

LGU Officer Program Workflow

Figure 16 shows the LGU Officer Program workflow in AquaTimbang, upon logging in, the LGU Officer can access a dashboard that provides a comprehensive view of fish price data within their municipality. The LGU personnel is responsible for reviewing and approving pending fish vendor registrations, entering daily prices, validating and publishing the official reference price, monitoring pricing trends, generating reports, and correcting erroneous entries while preserving an audit trail. This layout emphasizes the LGU personnel's crucial role in maintaining data integrity, ensuring fair pricing through validation, and supporting evidence-based policy decisions.

System Administrator

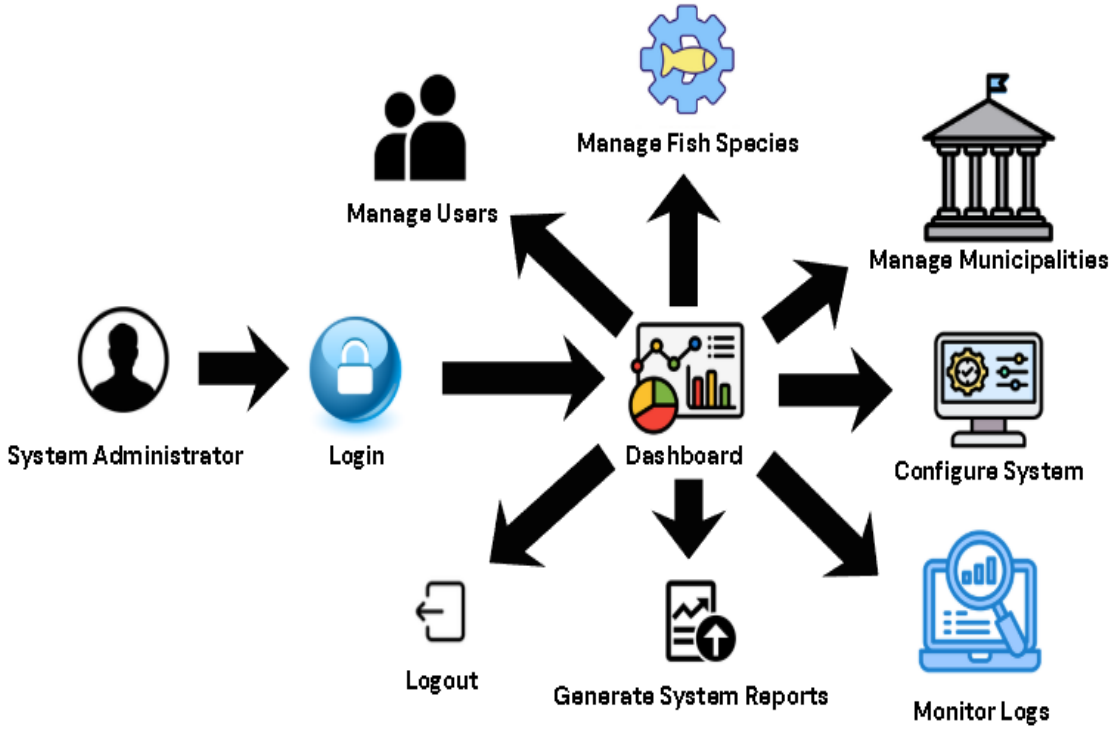


Figure 17

System Administrator Program Workflow

Figure 17 shows the System Administrator workflow in AquaTimbang, after gaining access, the System Administrator can access a dashboard that provides an overview of system operations. The System Administrator is responsible for managing user accounts and role-based access control (RBAC), approving registrations, assigning roles, managing fish species and municipalities, configuring system parameters, monitoring logs, and generating system reports. This layout emphasizes the System Administrator's role in maintaining system security, data accuracy, and user management across all stakeholder groups.

Functional Decomposition Diagram

Functional Decomposition Diagram breaks down the complex system into its component parts. A function in this context refers to a specific task within a larger process that is decomposed into smaller, more manageable sub-functions. Decomposition allows the research team to assign responsibilities clearly, trace requirements to specific features, and ensure that all requirements are addressed in the system design.

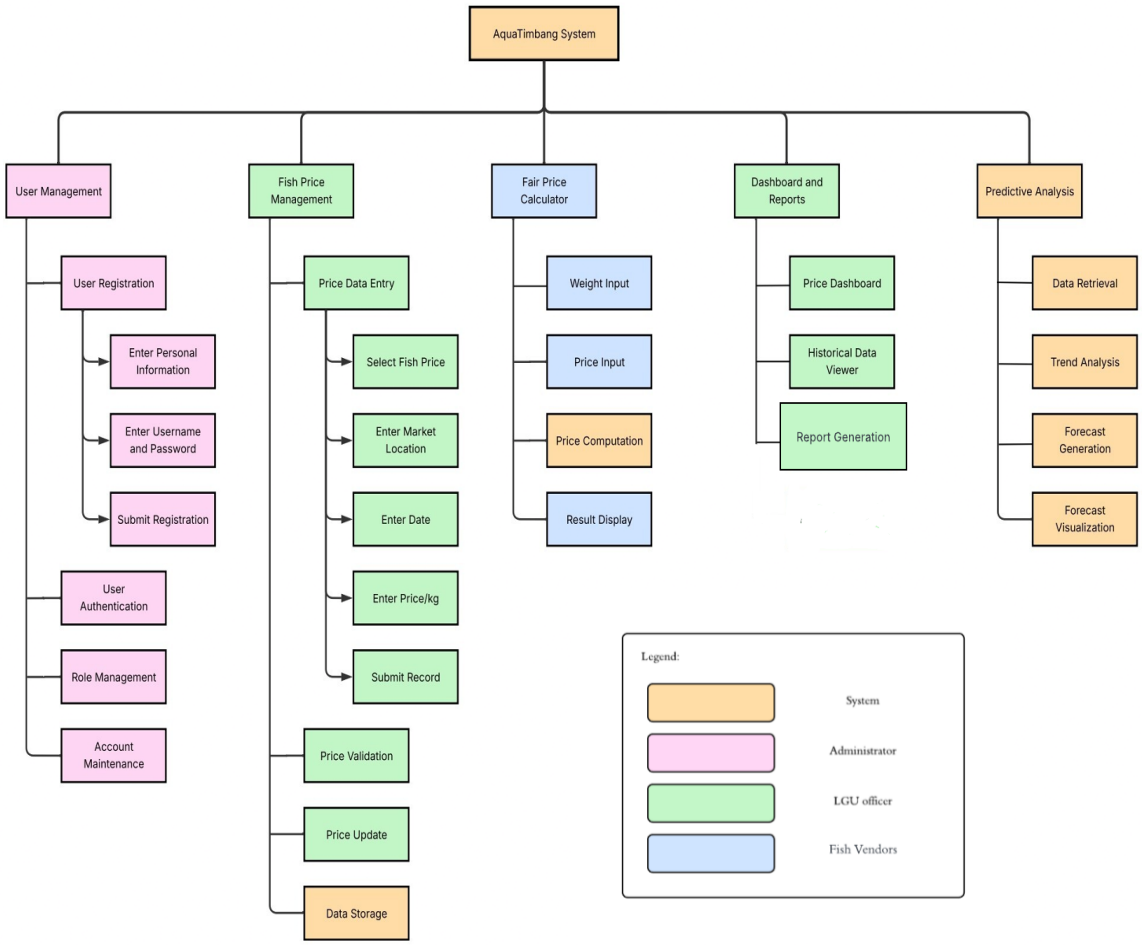
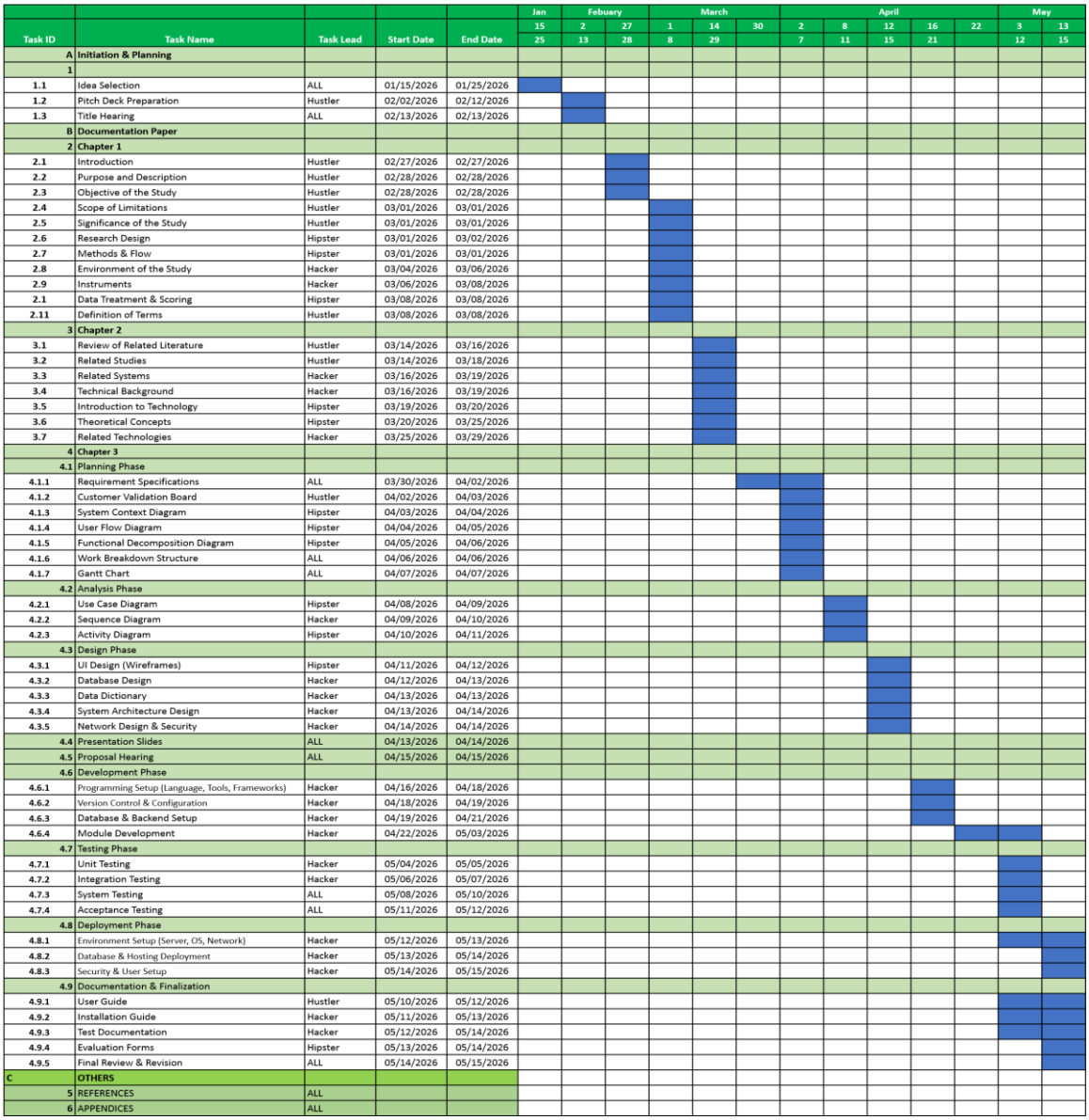


Figure 18

Functional Decomposition Diagram

Figure 18 shows the AquaTimbang Functional Decomposition Diagram, showing the six top-level modules and their sub-functions in a hierarchical tree structure. The legend distinguishes primary modules from sub-function components.

Work Breakdown Structure (With Gantt Chart)



Task ID	Task Name	Task Lead	Start Date	End Date
A	**Initiation & Planning**			
1				
1.1	Idea Selection	ALL	01/15/2026	01/25/2026
1.2	Pitch Deck Preparation	Hustler	02/02/2026	02/12/2026
1.3	Title Hearing	ALL	02/13/2026	02/13/2026
B	**Documentation Paper**			
2	**Chapter 1**			
2.1	Introduction	Hustler	02/27/2026	02/27/2026
2.2	Purpose and Description	Hustler	02/28/2026	02/28/2026
2.3	Objective of the Study	Hustler	02/28/2026	02/28/2026
2.4	Scope of Limitations	Hustler	03/01/2026	03/01/2026
2.5	Significance of the Study	Hustler	03/01/2026	03/01/2026
2.6	Research Design	Hipster	03/01/2026	03/02/2026
2.7	Methods & Flow	Hipster	03/01/2026	03/01/2026
2.8	Environment of the Study	Hacker	03/04/2026	03/06/2026
2.9	Instruments	Hacker	03/06/2026	03/08/2026
2.1	Data Treatment & Scoring	Hipster	03/08/2026	03/08/2026
2.11	Definition of Terms	Hustler	03/08/2026	03/08/2026
3	**Chapter 2**			
3.1	Review of Related Literature	Hustler	03/14/2026	03/16/2026
3.2	Related Studies	Hustler	03/14/2026	03/18/2026
3.3	Related Systems	Hacker	03/16/2026	03/19/2026
3.4	Technical Background	Hacker	03/16/2026	03/19/2026
3.5	Introduction to Technology	Hipster	03/19/2026	03/20/2026
3.6	Theoretical Concepts	Hipster	03/20/2026	03/25/2026
3.7	Related Technologies	Hacker	03/25/2026	03/29/2026
4	**Chapter 3**			
4.1	Planning Phase			
4.1.1	Requirement Specifications	ALL	03/30/2026	04/02/2026
4.1.2	Customer Validation Board	Hustler	04/02/2026	04/03/2026
4.1.3	System Context Diagram	Hipster	04/03/2026	04/04/2026
4.1.4	User Flow Diagram	Hipster	04/04/2026	04/05/2026
4.1.5	Functional Decomposition Diagram	Hipster	04/05/2026	04/06/2026
4.1.6	Work Breakdown Structure	ALL	04/06/2026	04/06/2026
4.1.7	Gantt Chart	ALL	04/07/2026	04/07/2026
4.2	Analysis Phase			
4.2.1	Use Case Diagram	Hipster	04/08/2026	04/09/2026
4.2.2	Sequence Diagram	Hacker	04/09/2026	04/10/2026
4.2.3	Activity Diagram	Hipster	04/10/2026	04/11/2026
4.3	Design Phase			
4.3.1	UI Design (Wireframes)	Hipster	04/11/2026	04/12/2026
4.3.2	Database Design	Hacker	04/12/2026	04/13/2026
4.3.3	Data Dictionary	Hacker	04/13/2026	04/13/2026
4.3.4	System Architecture Design	Hacker	04/13/2026	04/14/2026
4.3.5	Network Design & Security	Hacker	04/14/2026	04/14/2026
4.4	Presentation Slides	ALL	04/13/2026	04/14/2026
4.5	Proposal Hearing	ALL	04/15/2026	04/15/2026
4.6	Development Phase			
4.6.1	Programming Setup (Language, Tools, Frameworks)	Hacker	04/16/2026	04/18/2026
4.6.2	Version Control & Configuration	Hacker	04/18/2026	04/19/2026
4.6.3	Database & Backend Setup	Hacker	04/19/2026	04/21/2026
4.6.4	Module Development	Hacker	04/22/2026	05/03/2026
4.7	Testing Phase			
4.7.1	Unit Testing	Hacker	05/04/2026	05/05/2026
4.7.2	Integration Testing	Hacker	05/06/2026	05/07/2026
4.7.3	System Testing	ALL	05/08/2026	05/10/2026
4.7.4	Acceptance Testing	ALL	05/11/2026	05/12/2026
4.8	Deployment Phase			
4.8.1	Environment Setup (Server, OS, Network)	Hacker	05/12/2026	05/13/2026
4.8.2	Database & Hosting Deployment	Hacker	05/13/2026	05/14/2026
4.8.3	Security & User Setup	Hacker	05/14/2026	05/15/2026
4.9	Documentation & Finalization			
4.9.1	User Guide	Hustler	05/10/2026	05/12/2026
4.9.2	Installation Guide	Hacker	05/11/2026	05/13/2026
4.9.3	Test Documentation	Hacker	05/12/2026	05/14/2026
4.9.4	Evaluation Forms	Hipster	05/13/2026	05/14/2026
4.9.5	Final Review & Revision	ALL	05/14/2026	05/15/2026
C	**OTHERS**			
5	REFERENCES	ALL		
6	APPENDICES	ALL		

Figure 19

AquaTimbang's Work Breakdown Structure with Gantt Chart

Figure 19 presents the timeframe and a representation of the schedules and length of time involved in developing AquaTimbang. The Work Breakdown Structure (WBS) decomposes the project into manageable, assignable work packages organized hierarchically by the development phase. Each work package identifies the responsible team member.

ANALYSIS PHASE

The analysis phase focuses on understanding the system requirements and defining how the proposed system will function. During this stage, the system's processes, user interactions, and data flow are examined to ensure that all functional needs are clearly identified.

For the AquaTimbang, this phase involves analyzing user roles, system behavior, and interaction scenarios through diagrams such as use case, sequence, and activity diagrams. These models help establish a clear representation of how the system operates and serve as a foundation for the design and development phases.

Use Case Diagram

The Use Case Diagram presents all system use cases and their relationships to the three primary actors of Aquatimbang: Fishvendor, LGU officer and System Administrator. The diagram uses UML notation, including include and extend relationships, to show functional dependencies between use cases. Four role-specific diagrams are presented followed by seven fully dressed use case descriptions for the most critical system interactions.

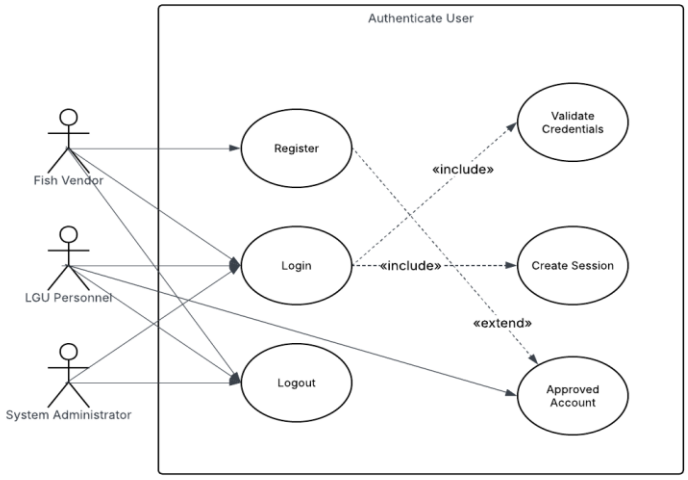


Figure 20

Authenticate User Use Case Diagram

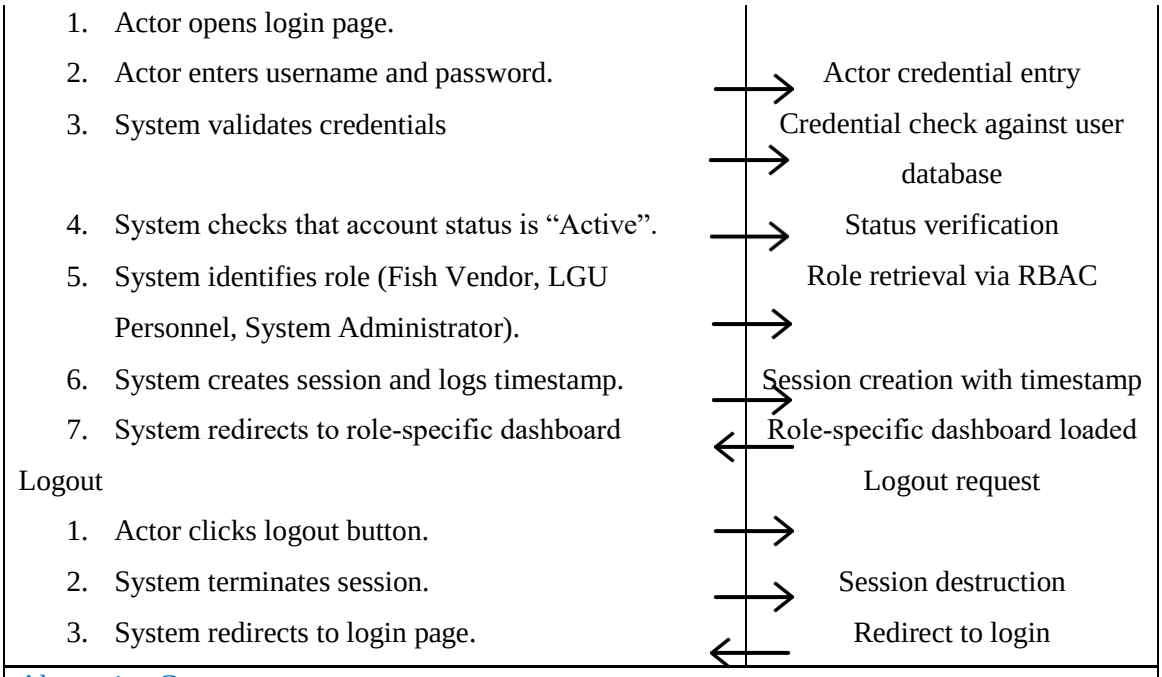
Figure 20 shows the Authenticate User use case. In this flow, the Fish Vendor, LGU Personnel, and System Administrator interact with the system to manage their access. Each

stakeholder can register for an account, log in to gain entry, and log out when finished. During login, the system validates credentials, creates a session, and may extend to approve the account if required.

Table 7

Authentication User

Use Case Name: Authenticate User (Register, Login, Logout)	ID: UC-1	Priority: High
Actor: Fish Vendor, LGU Personnel, System Administrator		
Description: Fish Vendor self-registers, then wait for LGU Personnel approval. LGU Personnel accounts are created by System Administrator. System Administrator accounts are created manually. After approval, any actor logs in with credentials; system validates, identifies role, creates session, and redirects to role-specific dashboard. Actor may also log out.		
Trigger: Actor navigates to login or registration page. Type: ☑ External ☐ Temporal		
Preconditions: System is running. For login: account exists and is active (approved).		
Normal Course:		Information for Steps:
Fisherfolk Registration	→	Registration link/button
1. Fish Vendor selects “Register”.		
2. Fish Vendor enters personal information (name, contact, municipality, etc.) and chooses username/password.	→	Registration form
3. System validates that username is unique and required fields are filled.	→	Validation check
4. System creates account with status “Pending Approval” and role “Fish Vendor”.	→	Account creation in database
5. System displays: “Registration successful. Please wait for LGU approval.”	←	Confirmation message
6. (Later) LGU Personnel approves or rejects the account (see UC-8)	→	Approval interface
Login for approved user	→	Login page interface



Alternative Course:

3a. Invalid credentials (login): system displays error, prompt re-entry.

3b. Account not yet approved (login): display “Your account is pending approval. Please wait for LGU confirmation.”

3c. Account rejected (login): display “Your registration was rejected. Contact your LGU office.”

3d. Account lockout after 5 failed attempts: lock 15 minutes, notify email.

3e. Forgot password: generate reset link (valid 1 hour), user sets new password.

3f. LGU Personnel or System Admin login: no registration step; accounts pre-created by admin.

Postconditions:

Registration: pending account created. Login: active session, dashboard. Logout: session destroyed.

Exceptions:

E1. Database connection lost: display error, retry.

E2. Account disabled by admin: display “Account disabled. Contact administrator.”

Summary Inputs	Source	Outputs	Destination

Registration data (name, contact, municipality, username, password)	Fish Vendor via registration form	Pending account record	User database
Username, password	Actor via login screen	Session ID	Session datastore
Approval/rejection decision	LGU Personnel / System Administrator	Account status update	User database
Logout request	Actor	Session termination	Session datastore
Error / status messages	System	Display to actor	Actor screen

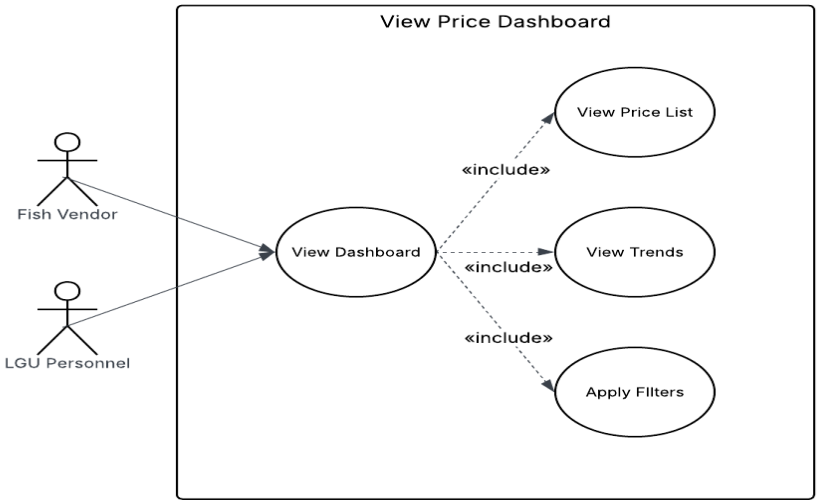


Figure 21

View Price Dashboard Use Case Diagram

Figure 21 shows the View Price Dashboard use case. In this flow, both Fish Vendor and LGU Personnel interact with the system to access market information. By using the dashboard, they can view the price list, analyze price trends, and apply filters to refine the displayed data.

Table 8

View Price Dashboard

Use Case Name: View Price Dashboard	ID: UC-2	Priority: High
Actor: Fish Vendor, LGU Personnel		
Description: The actor views current and historical fish prices. The system displays default current prices (today, all species, actor's municipality) and allows filters (species, date range, municipality). The system updates the display based on selected filters. No vendor suggestions are shown.		
Trigger: Actor selects the price dashboard from the main menu. Type: ☑ External ☐ Temporal		
Preconditions: Actor is logged in. At least some price data exists.		
Normal Course: 1. The actor opens the price dashboard. 2. The system displays current prices per fish species (default view: today, all species, actor's municipality). 3. The actor applies filters (species, date range, municipality). 4. The system retrieves and displays filtered price data. 5. The system shows a trend visualization (line/bar chart) based on the filtered data.		Information for Steps: Dashboard request → ← Default current price list Filter selections (species, date range, municipality) → ← Filtered price records ← Trend chart data
Alternative Course: 4.1 No data available (branch at step 2 or 4): system displays "No data available for the selected filters."		
Postconditions: Actor views updated price data, average suggestions, and trends.		
Exceptions: E1. Data retrieval timeout: display "Unable to load price data. Please check your connection."		
Summary Inputs	Source	Outputs / Destination

Dashboard request	Actor	Current price list	Actor display
Filter selections	Actor	Filtered price data	Actor display
Trend chart data	System	Visualization	Actor display
No data message	System	Alert display	Actor

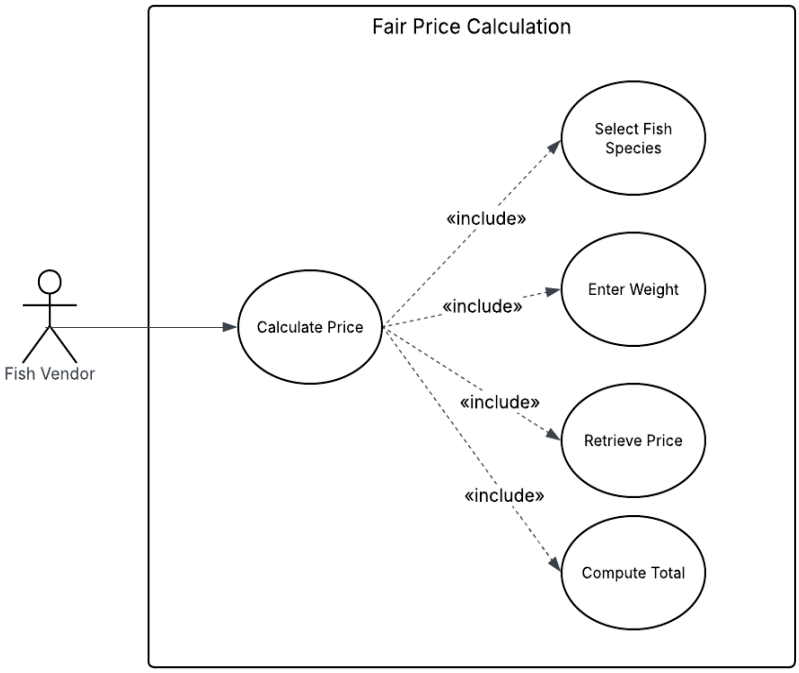


Figure 22

Fair Price Calculation Use Case Diagram

Figure 22 shows the Fair Price Calculation use case. In this flow, the Fish Vendor uses the system to determine the fair price of fish. The vendor begins by selecting the fish species, then entering the weight of the catch. The system proceeds to retrieve the current price data and finally computes the total amount.

Table 9

Fair Price Calculation

Use Case Name: Fair Price Calculation	ID: UC-3	Priority: High
Actor: Fish Vendor		
Description: The fish vendor calculates the total price of fish based on weight and the LGU-validated reference price. The actor selects a fish species and enters weight in kilograms. The system retrieves the latest reference price per kg, computes the total, and displays the result.		
Trigger: The actor accesses the price calculator feature. Type: ☑ External ☐ Temporal		
Preconditions: Actor is logged in. Current reference price exists for at least one fish species.		
Normal Course: 1. The actor opens the calculator. 2. The actor selects a fish species from a dropdown list. 3. The actor enter weighs in kilograms. 4. The system retrieves the latest price per kg for the selected species. 5. The system calculates total price = weight × price per kg. 6. The system displays the result (total price).	Information for Steps: → Calculator interface → Fish species selection → Weight entry (kg) → Price per kg retrieval → Total value calculation ← Total price display	
Alternative Course: 4.1 No price available: system displays “No current price found for this species. Please select another.”		
Postconditions: Total price is displayed.		
Exceptions: E1. Weight is zero or negative: system displays “Weight must be greater than zero.”		
Summary Inputs	Source	Outputs	Destination

Fish species	Actor	Price per kg	Calculation engine
Weight (kg)	Actor	Total value	Actor display
Error message	System	Alert display	Actor

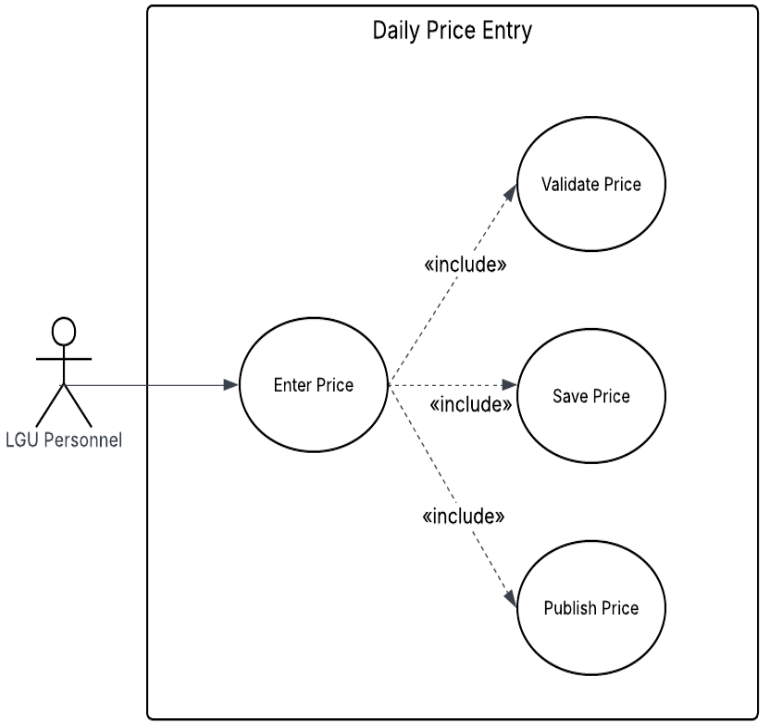


Figure 23

Daily Price Entry and Validation Use Case Diagram

Figure 23 shows the Daily Price Entry and Validation use case. In this flow, the LGU Personnel interacts with the system to record official market prices. The process begins when the personnel enter the price, after which the system automatically validates the entry, saves the data, and finally publishes the price for public access.

Table 10

Daily Price Entry and Validation (LGU Personnel)

Use Case Name: Daily Price Entry and Validation	ID: UC-4	Priority: High
Actor: LGU Personnel		
Description: LGU personnel enters and validates daily fish prices for each species. The actor selects a fish species, enters the official price per kilogram, and the system records it with a timestamp and LGU officer ID. The price becomes immediately available on the vendor dashboard and calculator.		
Trigger: LGU personnel accesses the daily price entry module (or scheduled daily task). Type: ☑ External ☐ Temporal		
Preconditions: Actor is logged in. LGU personnel accesses the daily price entry module (or scheduled daily task).		
Normal Course: 1. Actor opens the daily price entry module. 2. System displays a list of active fish species. 3. For each species, actor enters the official price per kg (or updates existing price). 4. System validates that the price is non-negative and within reasonable bounds (optional configurable min/max). 5. System saves the validated price with LGU officer ID and timestamp. 6. System publishes price to dashboard and price calculator. 7. System validates that the price is non-negative and within reasonable bounds (optional configurable min/max).		Information for Steps: Price entry interface Species list Price Entry Validation Check Price record storage Publication Confirmation message
Alternative Course: 4.1 Invalid price (negative or out of bounds): system displays error, prompts re-entry. 4.2 Actor may edit previously entered prices for the same date before finalizing.		
Postconditions: Daily prices stored and visible to all actors. Audit log records LGU action.		

Exceptions: E1. Database write error: display error, retry.			
Summary Inputs	Source	Outputs	Destination
Fish Price Validated	Fish species, price per kg – LGU Personnel	Validated price record	Price database
Approval/adjustment/rejection	LGU Personnel Error Message	Published price Notification	Dashboard, calculator Actor

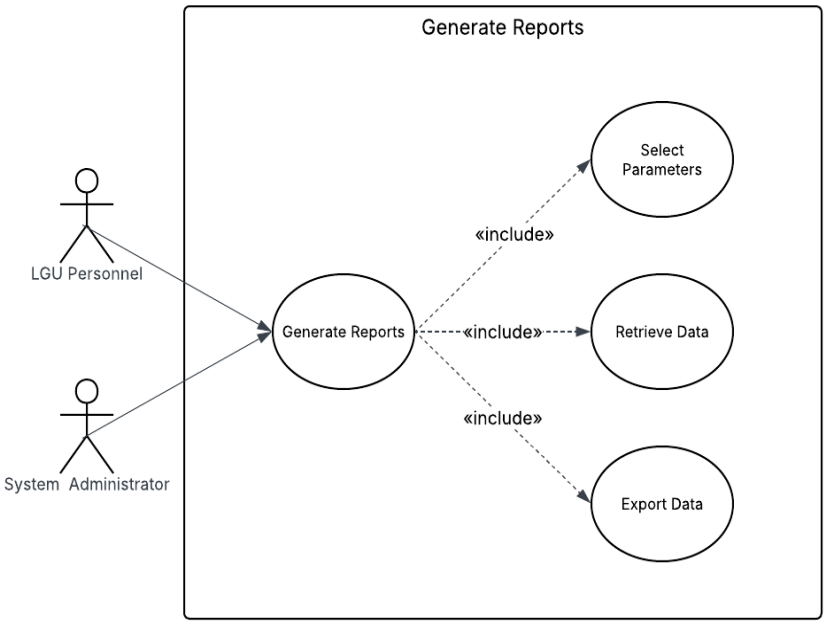


Figure 24

Generate Reports Use Case Diagram

Figure 24 shows the Generate Reports use case. In this flow, both LGU Personnel and System Administrator interact with the system to produce structured reports. The process begins when the user selects parameters, followed by the system retrieving the necessary data, and finally allowing the user to export the report.

Table 11

Generate Reports

Use Case Name: Generate Reports	ID: UC-5	Priority: Medium
Actor: LGU Personnel, System Administrator		
Description: The actor generates reports containing fish price history and validation logs. The actor selects parameters (date range, species, report type), the system retrieves data and generates a report, and the actor exports as PDF or CSV.		
Trigger: Actor is logged in. Data exists for selected parameters. Type: ☑ External ☐ Temporal		
Preconditions: Actor is logged in. Data exists for selected parameters.		
Normal Course: 1. Actor opens the reports module. 2. Actor selects report parameters (date range, species, report type: price history or validation log). 3. System retrieves data matching the parameters. 4. System generates the report (summary tables, charts). 5. Actor exports the report (PDF or CSV). 6. System downloads the file.	Information for Steps: → Reports interface → Parameter selection → Filtered data retrieval → Report object → Export format selection ← Downloadable file	
Alternative Course: 3.1 No data available: system displays “No records found for the selected parameters.”		
Postconditions: Report generated and available for download.		
Exceptions: E1. Export fails: display “Export failed. Please try again.”		

Summary Inputs	Source	Outputs	Destination
Report parameters (date, species, type)	Actor	Filtered price data	Report engine
Export format (PDF/CSV)	Actor	PDF/CSV file	Actor’s device

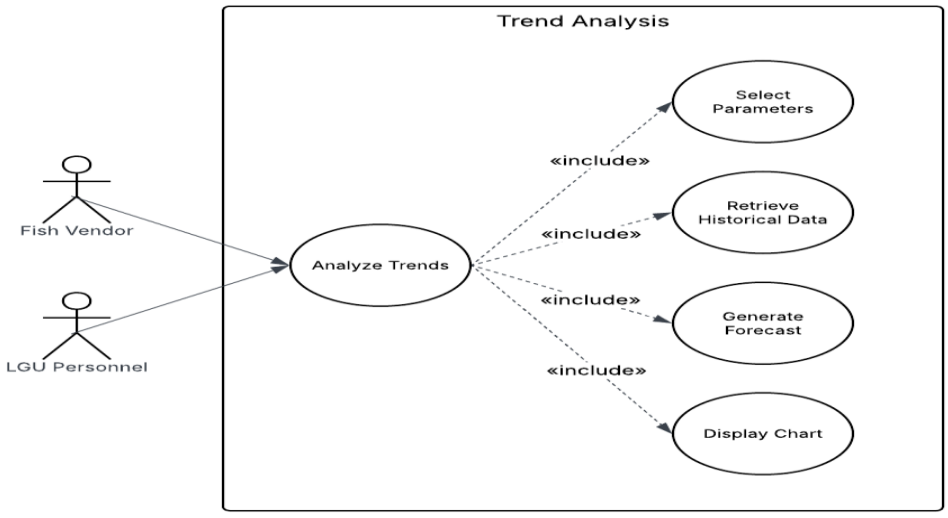


Figure 25

Price Trend Analysis and Forecasting Use Case Diagram

Figure 25 shows the Price Trend Analysis and Forecasting use case. In this flow, both Fish Vendor and LGU Personnel interact with the system to analyze market behavior. The process begins when the user selects parameters, then the system retrieves historical data to provide context. Next, the system generates a forecast based on the data, and finally displays the chart for visualization.

Table 12

Price Trend Analysis and Forecasting

Use Case Name: Price Trend Analysis and Forecasting	ID: UC-6	Priority: Medium
Actor: Fish Vendor, LGU Personnel		
Description: The actor analyzes historical price trends and views forecasted prices using predictive analytics. The actor selects parameters, the system retrieves historical data, analyzes trends, generates a forecast (e.g., moving average, seasonal trend), and displays a combined chart.		
Trigger: Actor selects the trend analysis feature from the dashboard. Type: ☑ External ☐ Temporal		
Preconditions:		

Actor is logged in. Sufficient historical data exists (minimum 4 weeks).

Normal Course:	Information for Steps:
1. Actor selects analysis parameters (species, date range). →	Analysis parameters
2. System retrieves historical price data. →	Historical price records
3. System analyzes trends (moving average, percent change, etc.). →	Trend metrics
4. System generates a forecast (e.g., next 2 weeks) using a model. →	Forecast values
5. System displays a combined chart (historical + forecast). ←	Chart visualization

Alternative Course:

2.1 Insufficient data: system displays warning "Not enough historical data for forecasting. Showing only available trends."

Postconditions:

Trends and forecast (if possible) displayed.

Exceptions:

E1. Forecast model error: system falls back to simple trend line, notifies "Forecast unavailable. Showing trend only."

Summary Inputs	Source	Outputs	Destination
Analysis parameters (species, date range)	Actor	Historical price data	Analysis engine
Forecast model	System Administrator	Forecast values	Chart display
Warning message	System	Alert display	Actor

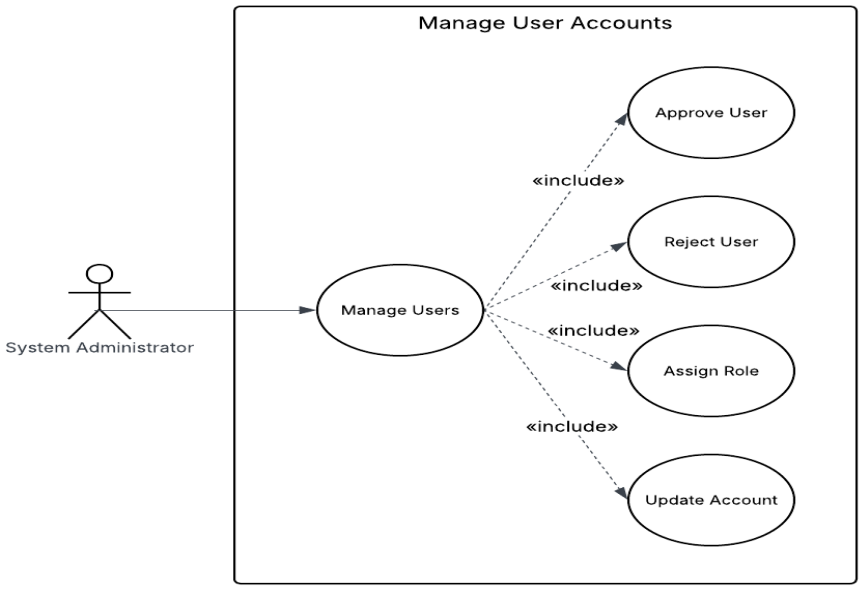


Figure 26

Manage User Accounts Use Case Diagram

Figure 26 shows the Manage User Accounts use case. In this flow, the System Administrator interacts with the system to oversee and control user access. The administrator can approve new users, reject accounts, assign roles, and update account details.

Table 13

Manage User Accounts

Use Case Name: Manage User Accounts	ID: UC-7	Priority: High
Actor: System Administrator		
Description: The System Administrator manages user registrations, roles, and access. The admin reviews pending registrations, approves or rejects users, assigns roles (Fish Vendor, LGU Personnel, System Administrator), and the system updates account status accordingly. Also allows editing, deleting, or deactivating existing accounts.		
Trigger: Administrator accesses the user management module. Type: ☑ External ☐ Temporal		

Preconditions: Administrator is logged in.	
Normal Course: 1. Admin opens the user management module. 2. System displays list of pending registrations (or search results). 3. Admin approves or rejects a user. 4. Admin assigns or modifies roles (Fish Vendor, LGU Personnel, System Administrator). 5. System updates account status (active/rejected) and role. 6. System displays a confirmation message.	Information for Steps: User management interface List of user accounts Decision (approve/reject) Role assignment Updated user record Success message
Alternative Course: 3.1 Missing requirements (branch at step 3 – approval): system prevents approval and displays "User missing required fields (e.g., no municipality assigned)."	
Postconditions: User accounts updated with correct roles and status. Audit log records admin's action	
Exceptions: E1. Role assignment conflict: system prevents contradictory roles (e.g., both Fisherfolk and LGU Personnel) and asks admin to choose one.	

Summary Inputs	Source	Outputs	Destination
User details	Pending user / DB	Account status update	User datastore
Role assignment	System Administrator	Role change confirmation	Audit log
Approval/rejection	System Administrator	Notification to user	User (email/in-app)

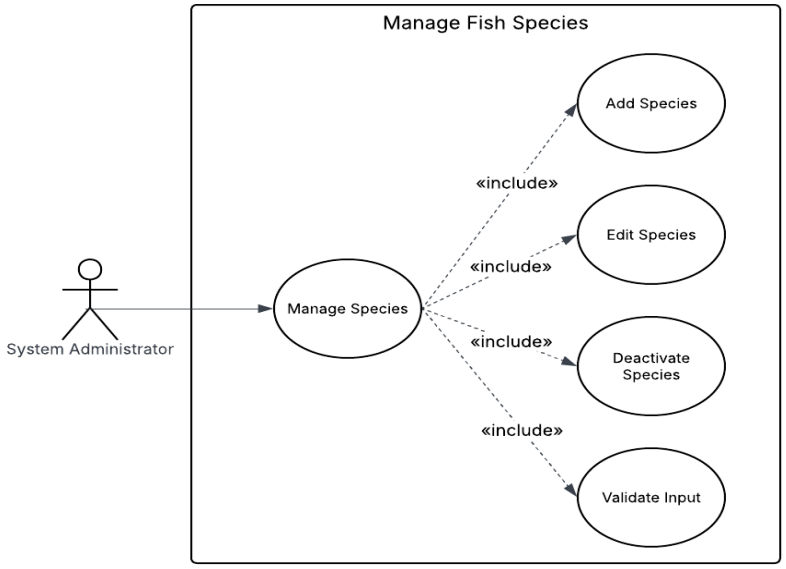


Figure 27

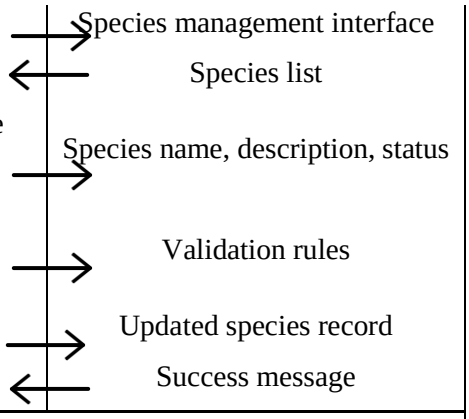
Manage Fish Species Use Case Diagram

Figure 27 shows the Manage Fish Species use case. In this flow, the System Administrator interacts with the system to maintain and control the list of fish species. The administrator can add new species, edit existing entries, deactivate species when necessary, and ensure accuracy by validating input.

Table 14

Manage Fish Species

Use Case Name: Manage Fish Species	ID: UC-8	Priority: Medium
Actor: System Administrator		
Description: The System Administrator adds, updates, or deactivates fish species. The admin enters species name, description, and status; the system validates input and saves changes.		
Trigger: Admin accesses the species management module. Type: ☑ External ☐ Temporal		
Preconditions: Administrator is logged in.		
Normal Course:		Information for Steps:

1. Admin opens the species management module. 2. System displays current list of species. 3. Admin adds a new species or edits an existing one (name, description, status). 4. System validates input (name not empty, no disallowed special characters). 5. System saves changes to the species datastore. 6. System displays a confirmation message.	

Alternative Course:

4.1 Duplicate entry (branch at step 4 for new species): system displays “Species name already exists.”

Postconditions:

Species data updated (add/edit/deactivate). Deactivated species hidden from price entry dropdowns.

Exceptions:

E1. Attempt to deactivate species with active price references: system warns “This species has price entries. Deactivate anyway?”

Summary Inputs	Source	Outputs	Destination
Species name, description	System Administrator	Validation result	Species datastore
Status change (active/inactive)	System Administrator	Updated species list	User interface
Duplicate error	System	Alert display	System Administrator

Sequence Diagram

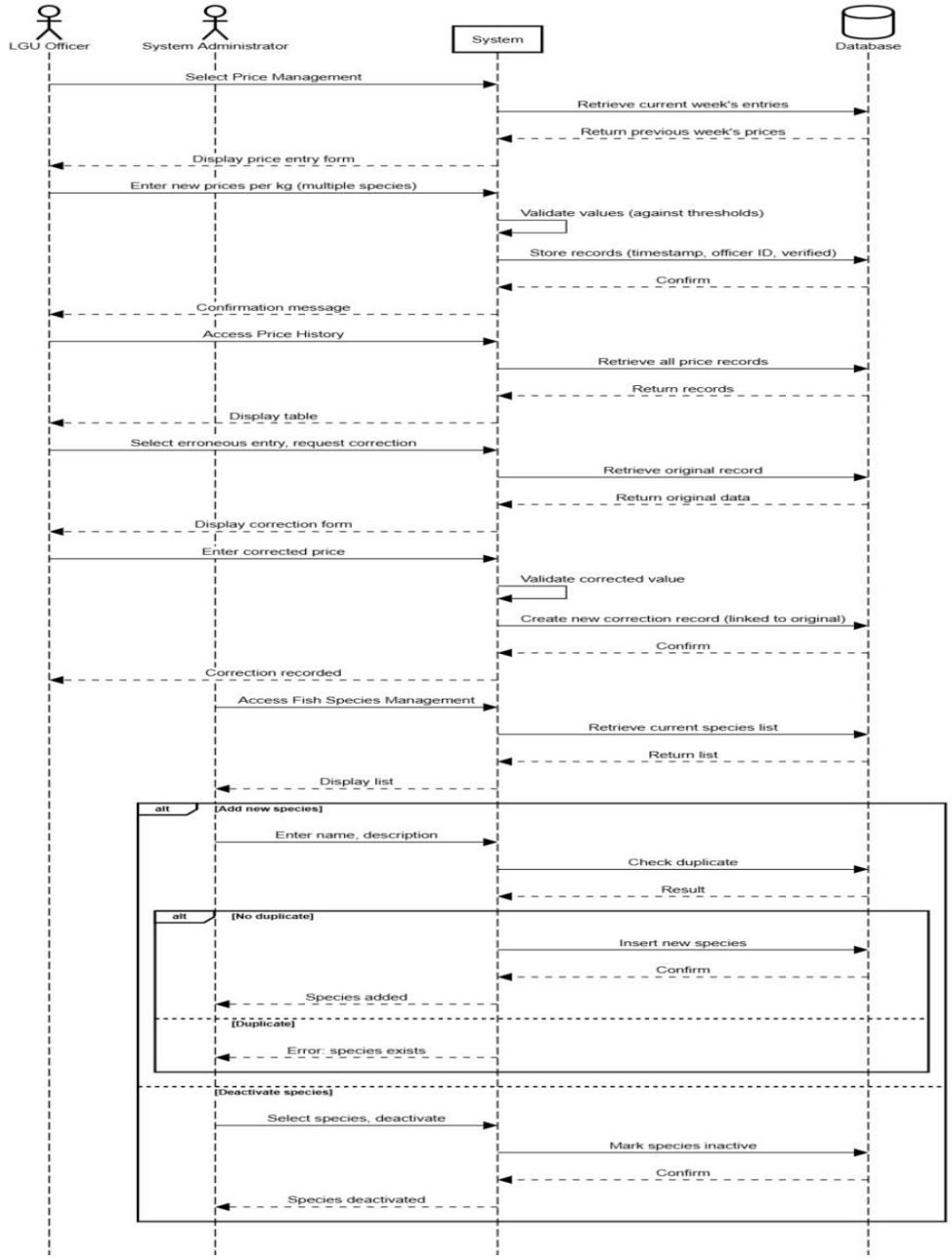


Figure 28

Sequence Diagram 1: Price Management & Administration

As shown in Figure 28, it illustrates the interaction among the LGU Officer, System Administrator, System, and Database during price data management and administrative tasks. The process begins when the LGU Officer accesses the price management module; the system retrieves the previous week's entries and displays a pre-filled form. The officer

enters new prices per kilogram for each fish species, and the system validates these values against configured thresholds. Each price record is stored with a timestamp, officer identity, and verification status, ensuring a complete audit trail. When an erroneous entry is discovered, the officer can request a correction; the system retrieves the original record, displays a correction form, and after validation creates a new corrected record linked to the original. Meanwhile, the System Administrator manages the fish species list by adding, editing, or deactivating species, with duplicate checks and confirmation messages. All actions are confirmed by the database and reported back to the user.

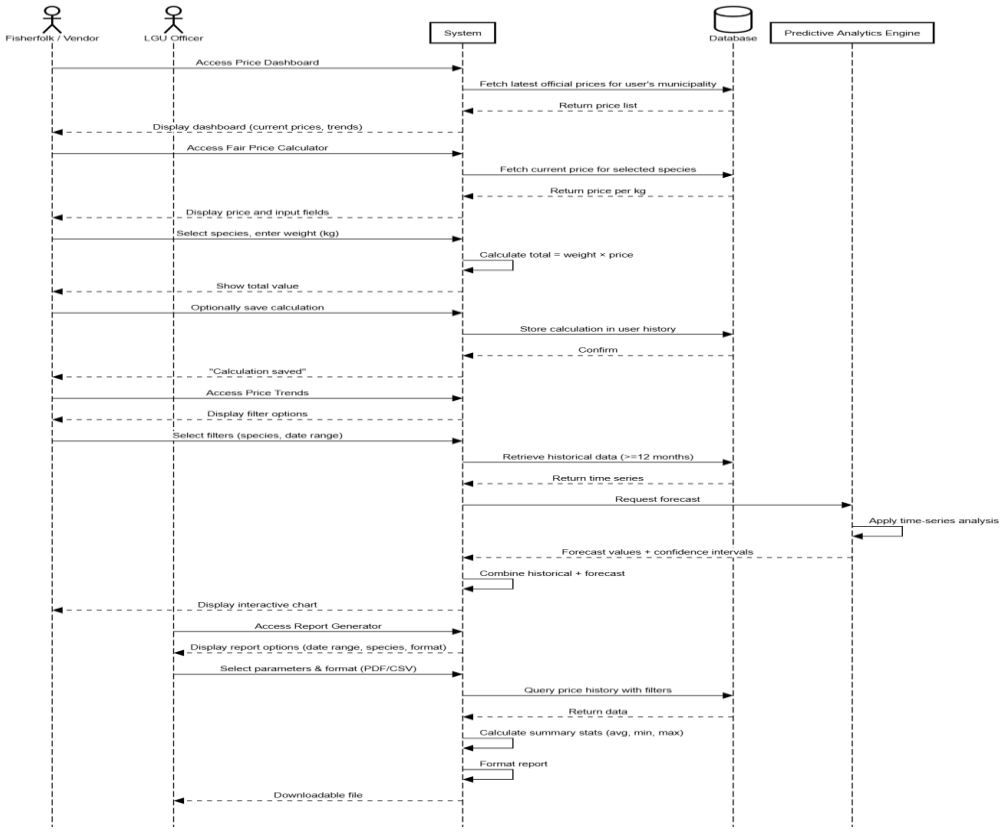


Figure 29

Sequence Diagram 2: Market Use & Analytics

As shown in Figure 29, it illustrates the interaction among the Vendor, LGU Officer, System, Database, and Predictive Analytics Engine during market information access and decision-support activities. The Vendor views current prices and trends, uses the fair price calculator (species, weight → total price), and saves calculations. For forecasts, the Vendor selects filters, the system retrieves historical data, and the analytics

engine returns confidence intervals displayed as an interactive chart. The LGU Officer generates market reports (PDF/CSV) by selecting parameters, and the system provides a downloadable file with summary statistics.

Activity Diagram

Activity diagrams model the workflow of a process in terms of sequential actions, parallel flows, decision points, and swimlane-based role assignments. They are particularly useful for showing the step-by-step control flow of complex, multi-actor processes within AquaTimbang. Three activity diagrams are presented covering the three primary operational workflows of the system.

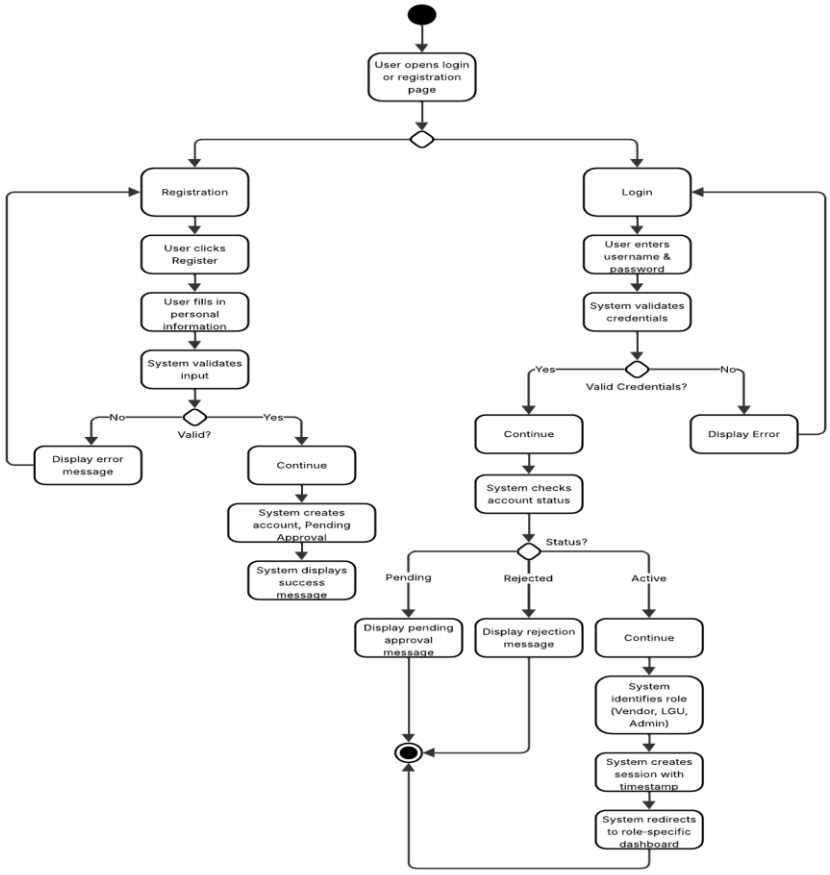


Figure 30

Activity Diagram 1: User Authentication, Registration and Role-Based Access

As shown in Figure 30, the diagram presents the authentication and registration process in AquaTimbang. It begins with the user opening the authentication page, where they can either register or log in. In registration, the user provides personal information which the system validates; invalid inputs trigger error messages, while valid inputs lead to account creation marked as pending approval. For login, the system checks the entered credentials and verifies account status, displaying appropriate messages if credentials are invalid, or if the account is pending or rejected. Once validated and active, the system identifies the user's role—such as Vendor, LGU personnel, or Administrator—creates a session with a timestamp, and redirects them to their respective dashboard.

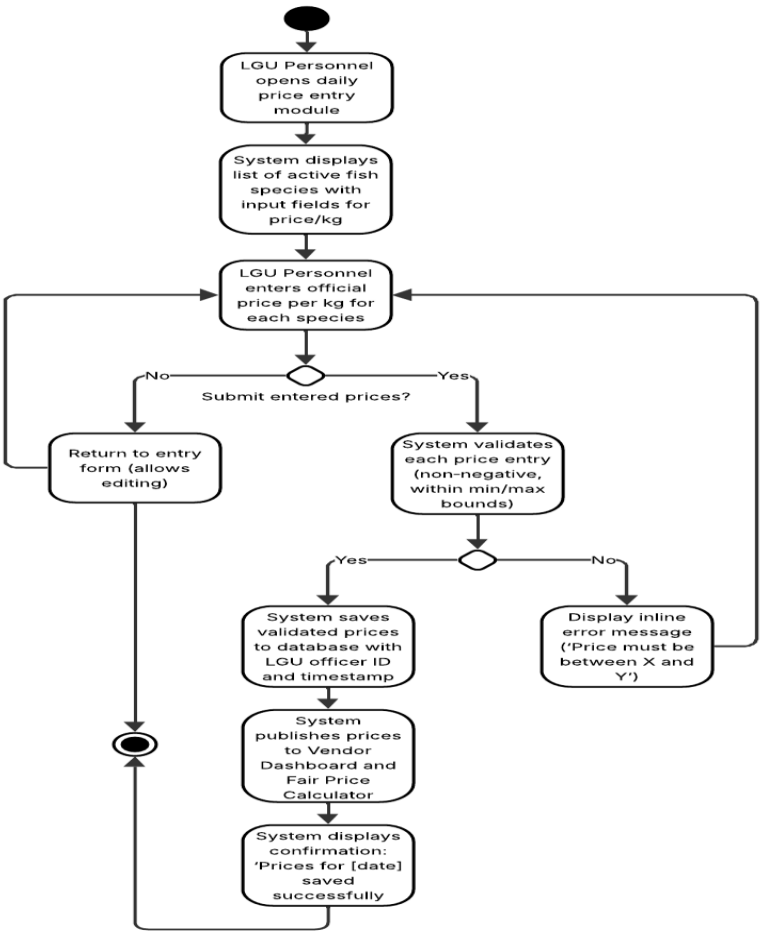


Figure 31

Activity Diagram 2: Price Participatory Entry and Validation

As shown in Figure 31, it shows the daily price entry and validation performed by LGU Personnel. It begins with the officer accessing the daily price entry module, where a list of active fish species is displayed with fields for entering the price per kilogram. After inputting the prices, the officer decides whether to submit them; if not, they can return to the form for editing. Once submitted, the system validates each entry against defined rules such as non-negative values and acceptable price ranges. Invalid entries trigger inline error messages, while successful validation results in the prices being saved to the database along with the officer's ID and timestamp. The validated data is then published to the Vendor Dashboard and Fair Price Calculator, ensuring transparency and accessibility. Finally, the system confirms successful saving of the day's prices.

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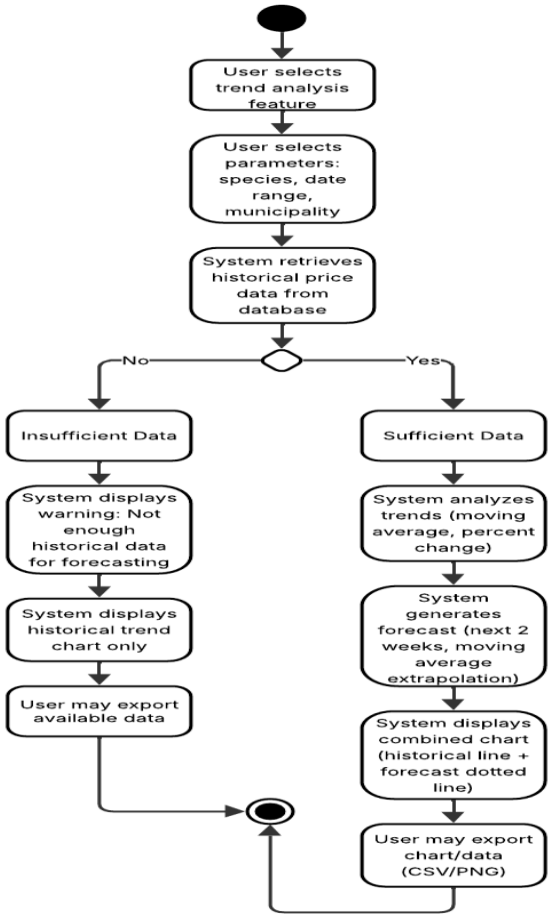


Figure 32

Activity Diagram 3: Price Trend Analysis and Forecasting

As shown in Figure 32, it illustrates the price trend analysis and forecasting of Aquatimbang. The process begins when a user selects the trend analysis feature and specifies parameters such as species, date range, and municipality. The system retrieves relevant historical data from the database and evaluates whether the dataset is sufficient for forecasting. If sufficient, the system applies methods like moving averages and percent changes to identify trends, then extrapolates forecasts for the upcoming two weeks. The output is a combined chart showing both historical data and forecasted values, which users can export in formats such as CSV or PNG.

DESIGN PHASE

The Design Phase is where the conceptual framework of AquaTimbang is transformed into a structured system blueprint. In this stage, the researchers developed the system architecture, database design, user interface layouts, and process flow diagrams to ensure that every feature aligns with the identified user requirements. This phase focused on creating an organized, user-friendly, and efficient design for Fish Vendors, LGU officers and administrators, providing a clear visual and technical guide for the system's actual development.

User Interface Design

The User Interface (UI) design phase establishes the visual and navigational architecture of the system, tailored specifically to the distinct operational environments of its users. For LGU personnel and system administrators, a comprehensive desktop interface is designed to maximize screen real estate, providing multi-pane dashboards for daily price entry and validation, fish species management, report generation, and forecast monitoring. Conversely, the vendor interface is designed as a mobile-responsive web application, prioritizing large touch targets, simplified navigation, and clear price displays to ensure fast and accurate access to the fair price calculator, current market prices, and trend forecasts on smartphone devices in the bustling market environment.

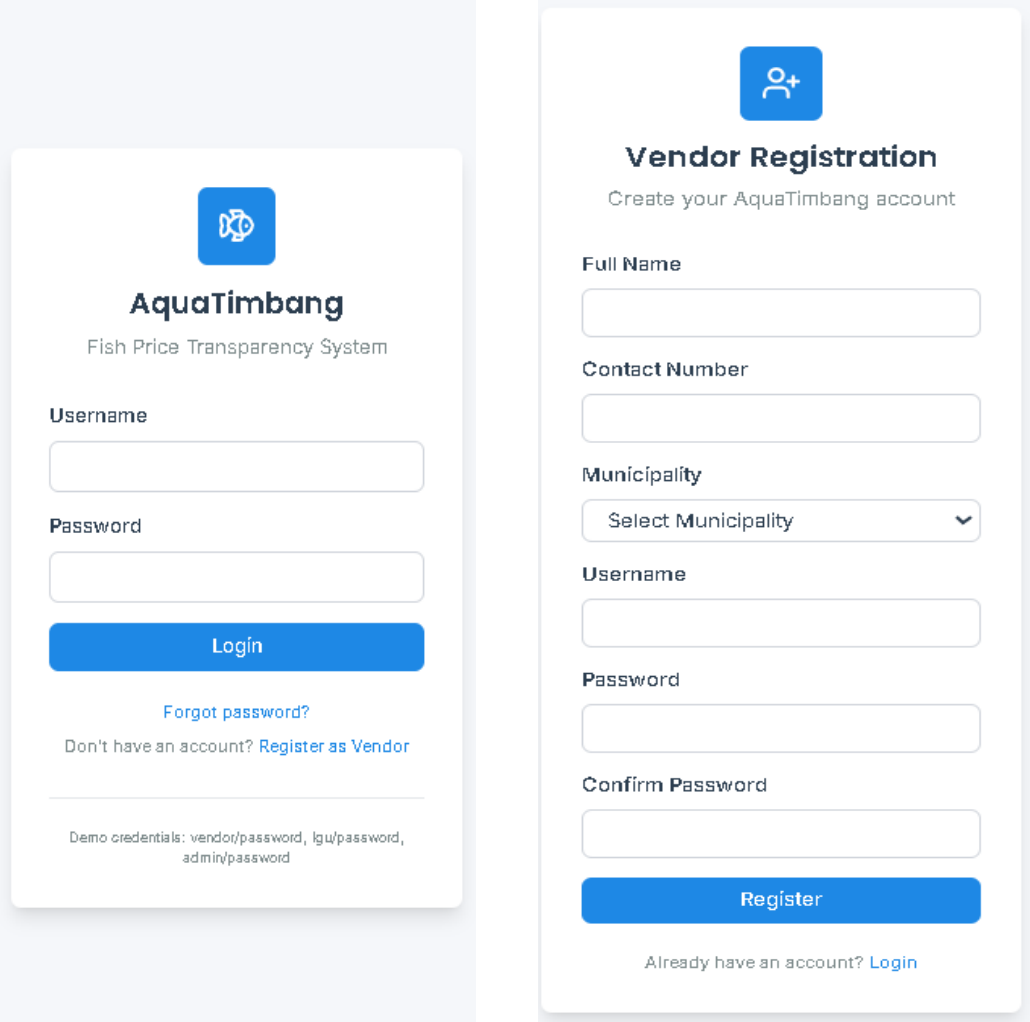


Figure 33

Aquatimbang Register/Login

As shown in Figure 33, the login page provides username and password fields, a “Forgot password?” link, and a “Register” button for vendors. After credential validation, the system checks account status (Active/Pending/Rejected), locks after 5 failed attempts, and redirects to the role-specific dashboard upon successful login. The registration page collects vendor details (name, contact, municipality, username, password). Upon submission, the system creates a “Pending Approval” account and displays a confirmation message. The vendor cannot log in until an LGU Personnel or Admin approves the account.

Fish Vendor

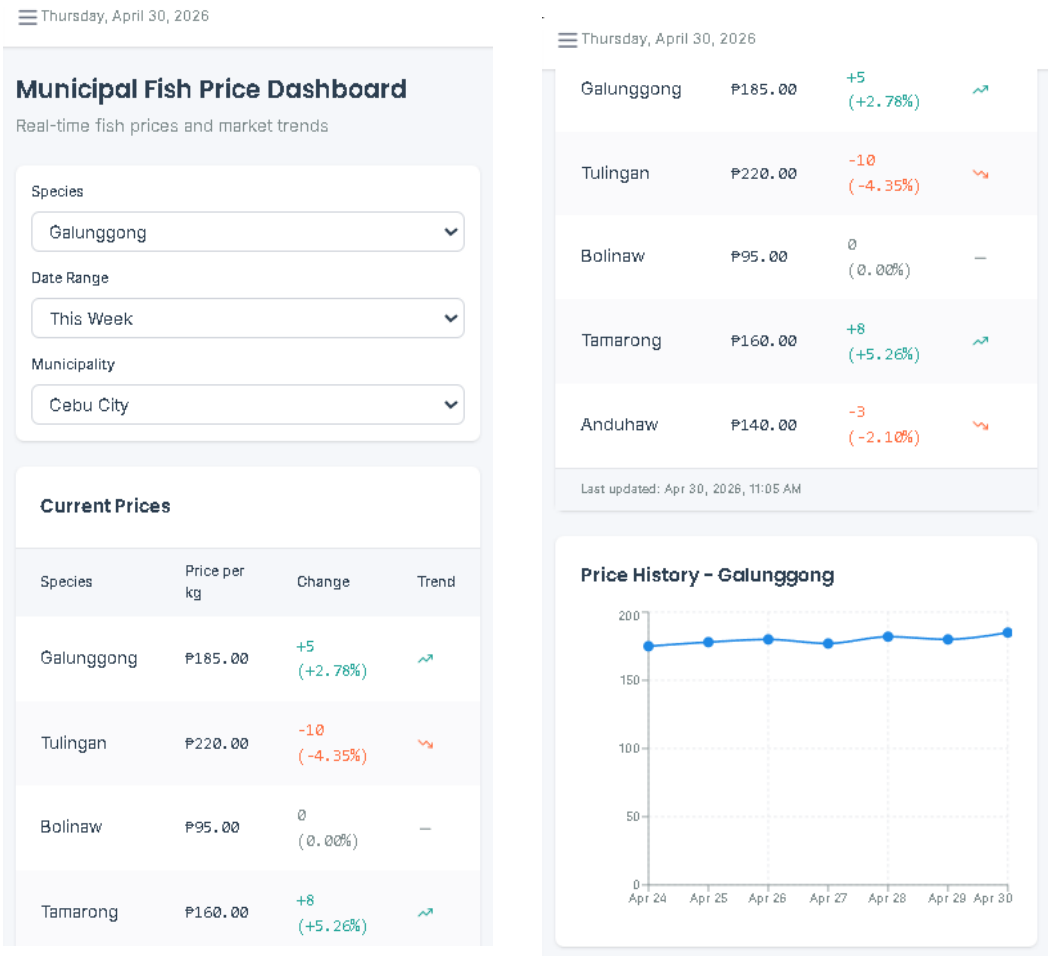


Figure 34

Fish Vendor Fish Price Dashboard

As shown in Figure 34, the Fish Price Dashboard displays current and historical prices for selected wet fish species (e.g., galunggong, tulingan, bolinaw, tamarong, anduhaw). Vendors can filter by species, date range, and municipality to view price trends through line charts and simple moving averages.

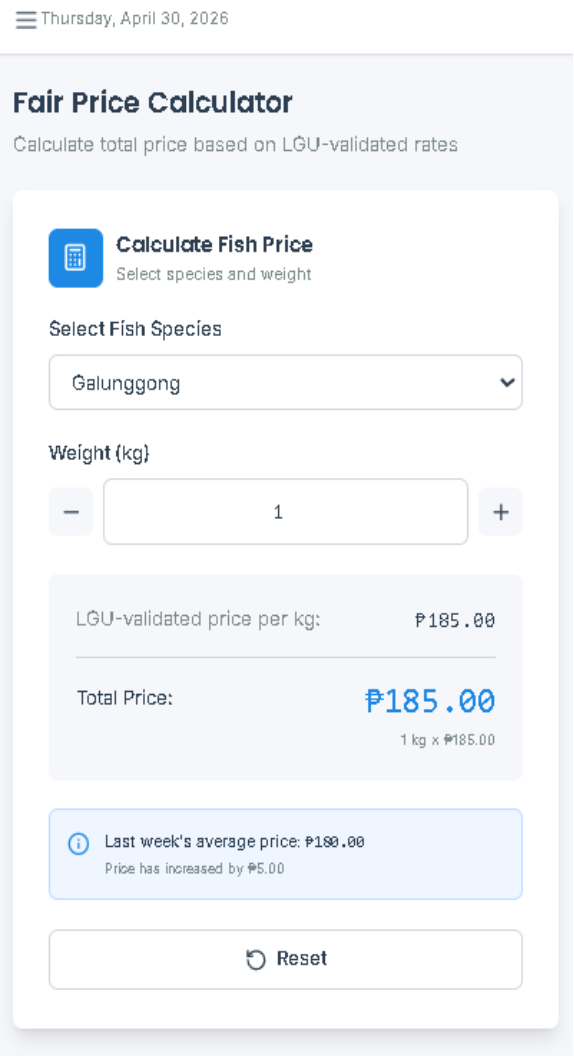


Figure 35

Fair Price Calculator

As shown in Figure 35, the Fair Price Calculator allows the vendor to select a fish species, enter weight in kilograms, and instantly compute the total price using the LGU-validated price per kg (Weight × Price per kg). The result is displayed in a large bold font.

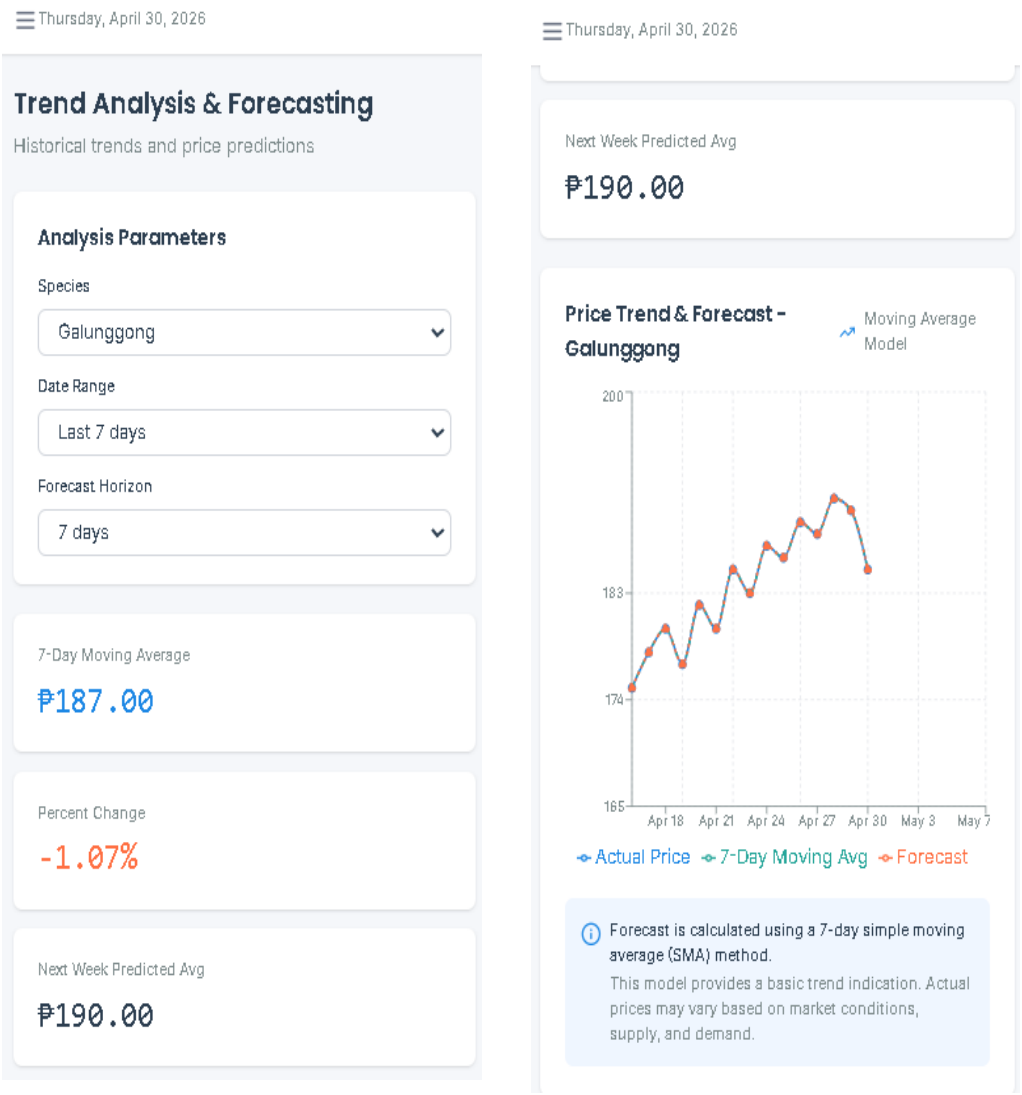


Figure 36

Trend Analysis and Forecasting Dasboard

As shown in Figure 36, the Trend Analysis and Forecasting dashboard enables vendors to view historical price trends and basic forecasts (e.g., next 2 weeks using moving average) with a combined chart. If insufficient data exists (less than 4 weeks), the system shows a warning and displays only the available historical trend. These features empower vendors to make informed selling decisions, secure fairer incomes, and reduce dependency.

LGU Officer

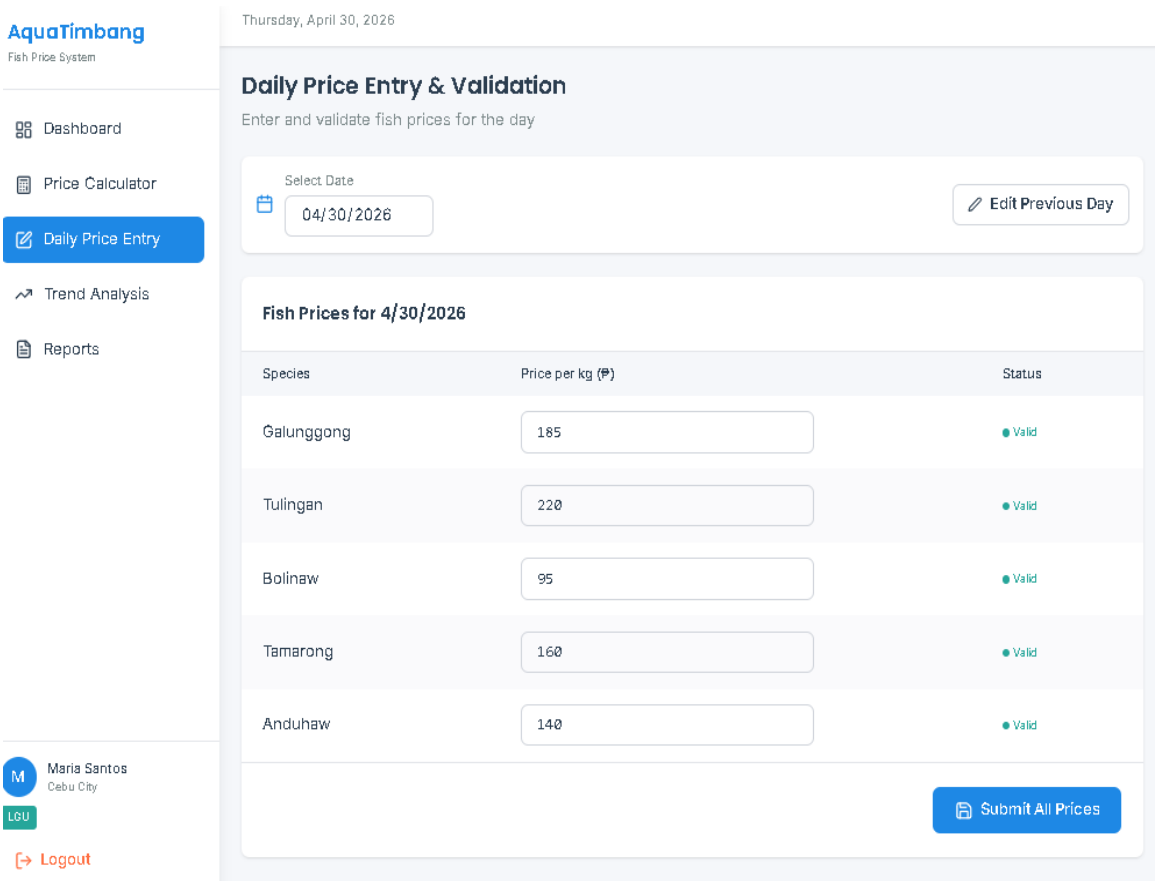


Figure 37

Daily Price Entry and Validation

As shown in Figure 37, the Daily Price Entry and Validation module provides a tabular form where LGU personnel select the current date, enter the official price per kg for each active fish species, and submit the entries. The system validates that the price is non-negative and within configurable bounds, shows inline errors for invalid entries, and displays a success toast after saving.

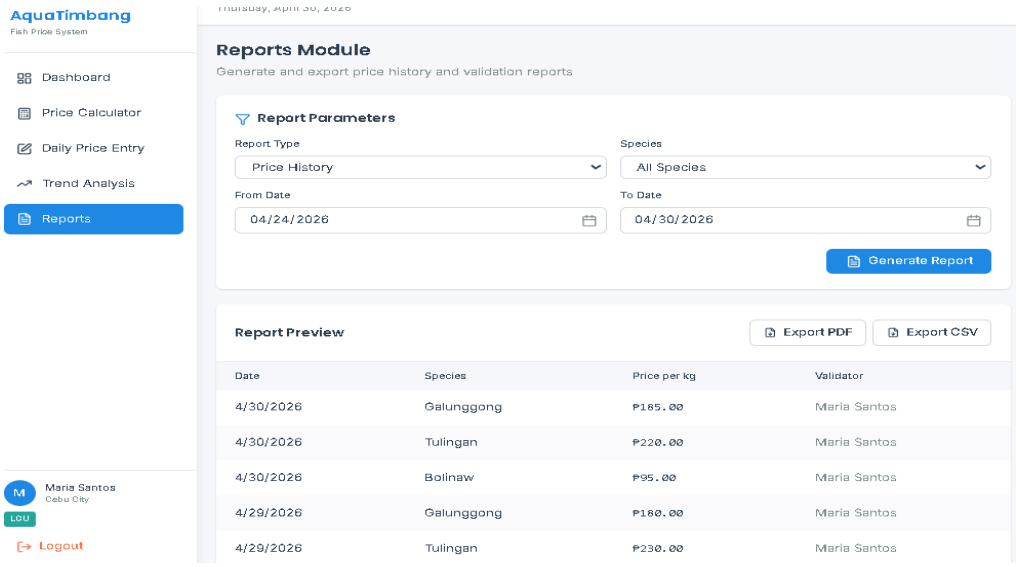


Figure 38

Generate Reports Dashboard

As shown in Figure 38, the Reports Module allows LGU personnel to generate and export price history reports (PDF or CSV) by selecting date range, species, and report type. A preview table shows the filtered data before export.

Administrator

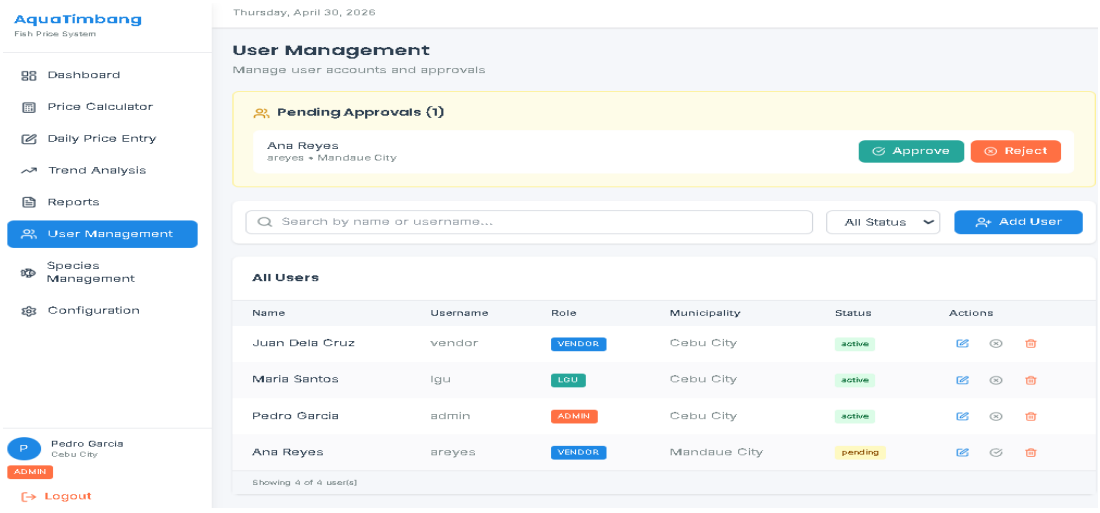


Figure 39

User Management Dashboard

As shown in Figure 39, User Management Module displays a list of pending vendor registrations with approve/reject buttons, and a searchable table of all users (Fish Vendors, LGU Personnel, other Administrators). The admin can edit user details (name, contact, municipality), assign or modify roles, delete or deactivate accounts, and manually add new LGU or admin accounts.

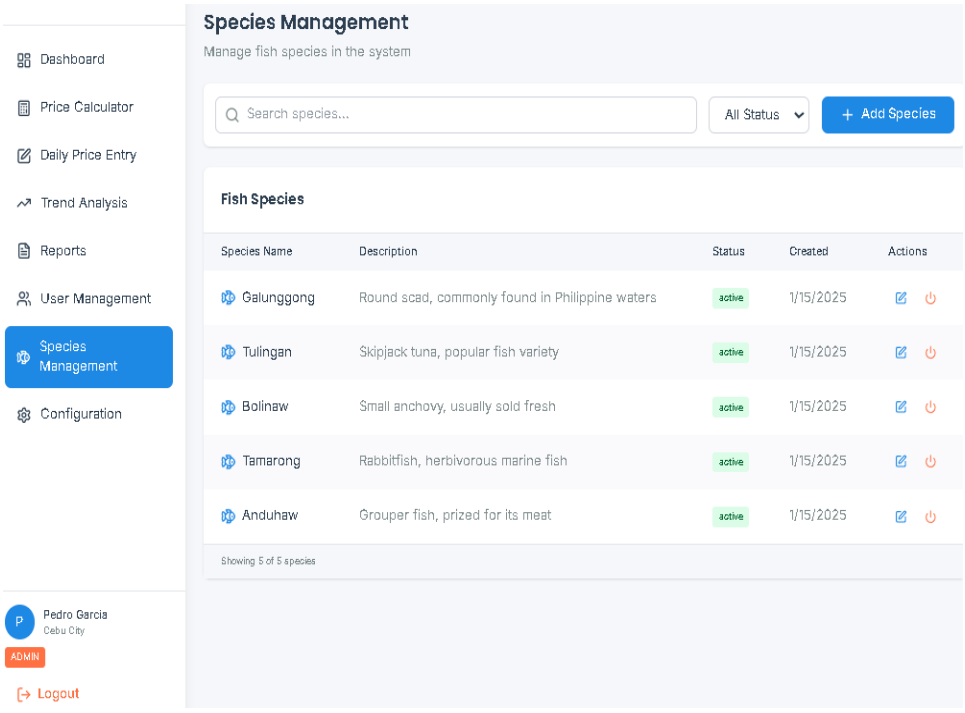


Figure 40

Fish Species Management Dashboard

As shown in Figure 40, the Fish Species Management Module provides a table of all fish species with fields for name, description, and status (active/inactive). The admin can add new species, edit existing ones, or deactivate species (with a warning confirmation if price entries exist). Deactivated species are hidden from dropdowns across the system.

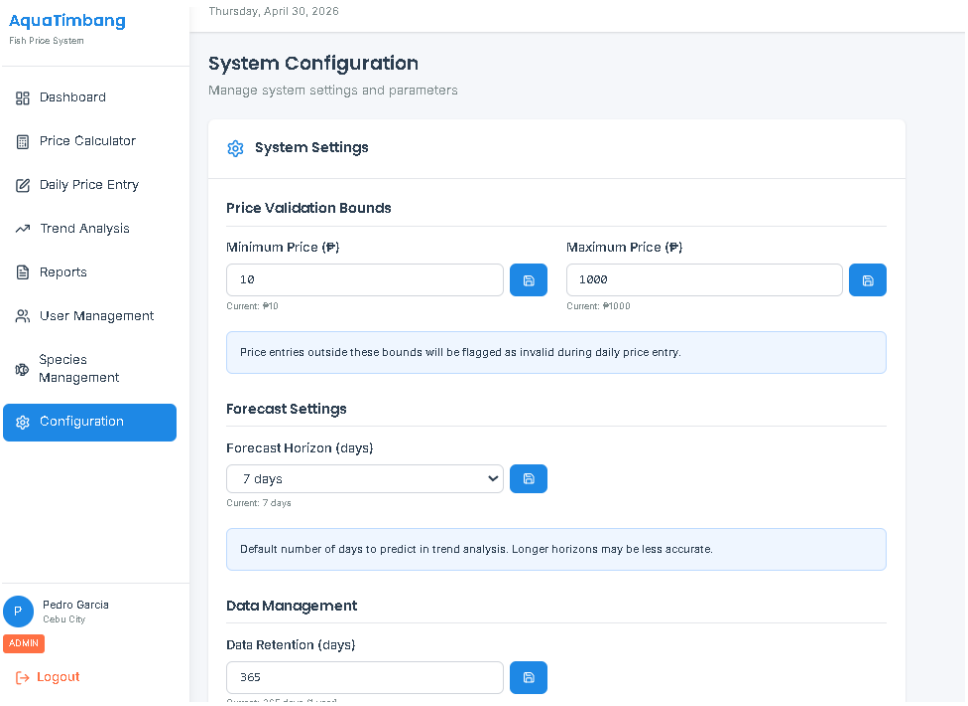


Figure 41

System Configuration Dashboard

As shown in Figure 41, the System Configuration Module allows the admin to set price validation bounds (minimum and maximum allowed price per kg), forecast horizon (e.g., 7-day moving average), and data retention policies. All settings are saved with immediate effect and logged in the audit trail.

Entity Relationship Diagram

The Entity-Relationship Diagram (ERD) presents the database structure of the AquaTimbang Fish Price Transparency System, which is designed to improve fairness, transparency, and data-driven decision-making in local fish markets. The system integrates various entities that support user management, fish species tracking, price recording, predictive analytics, reporting, and access monitoring. It provides a comprehensive representation of how different components interact to ensure efficient and reliable market operations.

Database Design

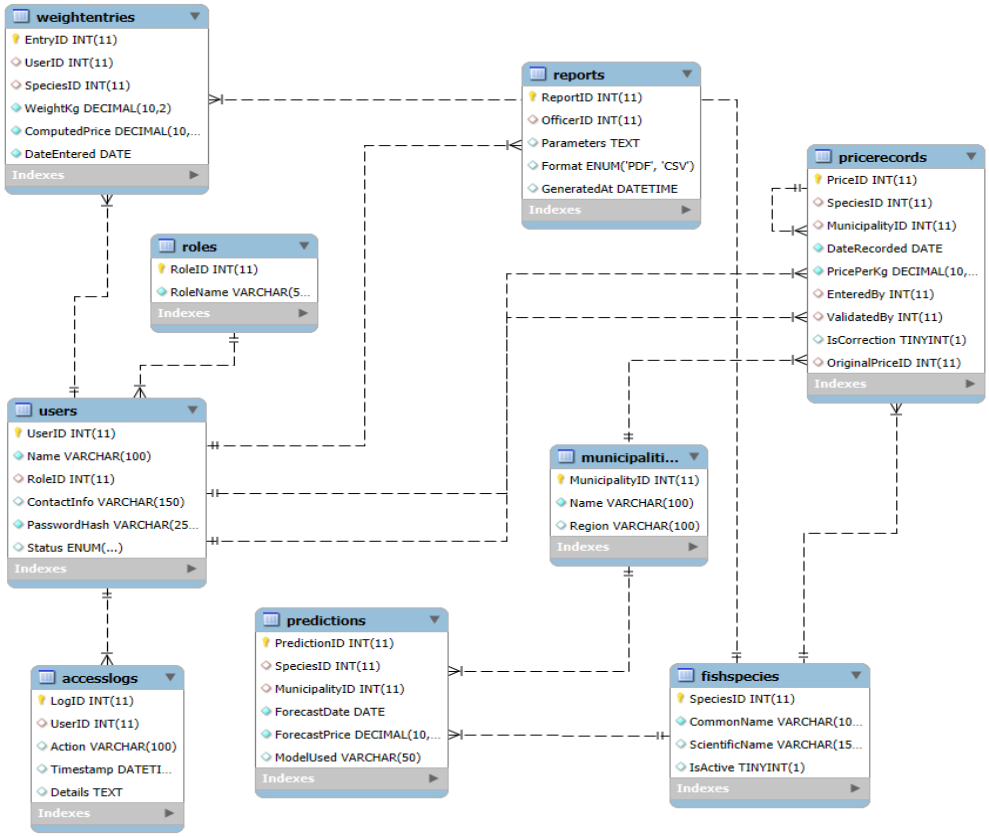


Figure 42

AquaTimbang ERD

As shown in Figure 42, it illustrates the structure of the system's database, including its entities, attributes, and relationships. The design follows a relational database model to ensure organized data storage, efficient retrieval, and strong data integrity. Central to the system is the Users entity, linked to Roles to enforce role-based access control. A status attribute manages account workflows. FishSpecies and Municipalities support market context by cataloging fish types and local markets. The PriceRecords entity stores weekly price data entered and validated by LGU officers, with references to the responsible users and audit fields for corrections. WeightEntries captures user-input fish weights and computes prices using verified rates, enabling fair and standardized transactions. For decision support, Predictions holds forecasted prices generated by basic predictive analytics, while Reports logs official outputs created by LGU officers. AccessLogs records system activities (logins, data entries, administrative actions) to ensure accountability.

Data Dictionary

The Data Dictionary of AquaTimbang serves as the centralized reference repository that explicitly defines the structural metadata of the AquaTimbang database, including table schemas, data types, field constraints, and foreign key relationships. It acts as an essential architectural blueprint for the backend, ensuring absolute data integrity and consistency across the system's core modules, such as fish species catalogs, historical price records, user accounts, and forecast results.

Table 15

User Entity Data Dictionary

USER				
Field Name	**Data Type**	**SIze**	**Constraints**	**Description**
UserID	**INT**	**11**	**PK, Auto Increment**	**Unique identifier for each user**
Name	**VARCHAR**	**100**	**NOT NULL**	**Full name of the user**
RoleID	**INT**	**11**	**FK → Roles.RoleID**	**Defines the user's role (Admin, LGU, Vendor)**
ContactInfo	**VARCHAR**	**150**	**NULL**	**Email or phone number**
PasswordHash	**VARCHAR**	**255**	**NOT NULL**	**Encrypted password for authentication**
Status	**ENUM**	**N/A**	**Default 'Pending'**	**Registration status (Pending, Active, Inactive, Rejected)**

As shown in Table 15, User Entity Data Dictionary table stores all user accounts in the AquaTimbang system, including fish vendors, LGU personnel, and the system administrator. Each user is assigned a unique UserID as the primary key. The Name field

records the full name of the user, while ContactInfo stores optional email or phone number for communication purposes. The PasswordHash field securely stores encrypted passwords to protect user credentials. The Status field, which defaults to 'Pending', controls account approval before system access, allowing administrators to review and activate or reject new registrations. The RoleID foreign key links each user to a specific role defined in the Roles table, enabling role-based access control throughout the system.

Table 16

Roles Entity Data Dictionary

ROLES				
Field Name	**Data Type**	**Size**	**Constraints**	**Description**
RoleID	**INT**	**11**	**PK, Auto Increment**	**Unique identifier for each role**
RoleName	**VARCHAR**	**50**	**NOT NULL**	**Role name (Admin, LGU Officer, Vendor)**

As shown in Table 16, Roles Entity Data Dictionary table defines the three distinct user roles within AquaTimbang: Administrator, LGU Officer, and Fish Vendor. The RoleID serves as the primary key and is referenced by the User table to assign appropriate permissions to each user account. The RoleName field stores the descriptive name of each role, which determines the functional features and data access levels available to users. This structure supports the system's role-based access control (RBAC) mechanism, ensuring that vendors can only view prices and use the calculator, LGU officers can enter and validate price data, and administrators have full system management capabilities.

Table 17

Municipalities Entity Data Dictionary

MUNICIPALITIES				
Field Name	**Data Type**	**SIze**	**Constraints**	**Description**
MunicipalityID	**INT**	**11**	**PK, Auto Increment**	**Unique identifier for municipality**
Name	**VARCHAR**	**100**	**NOT NULL**	**Municipality name**
Region	**VARCHAR**	**100**	**NULL**	**Region name**

As shown in Table 17, Municipalities Entity Data Dictionary table stores the list of municipalities where fish price monitoring is conducted within the AquaTimbang system. The MunicipalityID is the primary key that uniquely identifies each location. The Name field records the official municipality name, while the Region field provides optional regional categorization for broader geographic filtering.

Table 18

FishSpecies Entity Data Dictionary

FISHSPECIES				
Field Name	**Data Type**	**SIze**	**Constraints**	**Description**
SpeciesID	**INT**	**11**	**PK, Auto Increment**	**Unique identifier for fish species**
CommonName	**VARCHAR**	**100**	**NOT NULL**	**Local/common name of fish**
ScientificName	**VARCHAR**	**100**	**NULL**	**Scientific name of fish**
IsActive	**BOOLEAN**	**1**	**Default TRUE**	**Indicates if species is active**

As shown in Table 18, FishSpecies Entity Data Dictionary table catalogs all wet fish species available for price monitoring and forecasting within the AquaTimbang system. The SpeciesID serves as the unique primary key for each fish species entry. The CommonName field stores the local or common name of the fish (e.g., Bangus, Tilapia, Galunggong), which is displayed to users on the dashboard and calculator. The ScientificName field is optional and provides the formal taxonomic name for reference purposes. The IsActive boolean flag, which defaults to TRUE, allows administrators to deactivate species that are no longer sold in the market without permanently deleting historical price records, preserving data integrity for reporting and forecasting.

Table 19

PriceRecords Entity Data Dictionary

PriceRecords				
Field Name	**Data Type**	**SIze**	**Constraints**	**Description**
PriceID	**INT**	**11**	**PK, Auto Increment**	**Unique identifier for price record**
SpeciesID	**INT**	**11**	**FK → FishSpecies.SpeciesID**	**Fish species being priced**
MunicipalityID	**INT**	**11**	**FK → Municipalities. MunicipalityID**	**Municipality where price is recorded**
DateRecorded	**DATE**	**N/A**	**NOT NULL**	**Date of price entry**
PricePerKg	**DECIMAL**	**10,2**	**NOT NULL**	**Price per kilogram**
EnteredBy	**INT**	**11**	**FK → Users.UserID**	**User who entered the price**
ValidatedBy	**INT**	**11**	**FK → Users.UserID**	**User who validated the price**

IsCorrection	**BOOLEAN**	**1**	**Default FALSE**	**Marks if record is a correction**
OriginalPriceID	**INT**	**11**	**FK → PriceRecords.PriceID**	**Links corrected record to original**

As shown in Table 19, PriceRecords Entity Data Dictionary table stores all daily price entries made by LGU personnel as part of the system's core price monitoring function. The PriceID is the primary key that uniquely identifies each price record. Foreign keys link to SpeciesID (from FishSpecies) and MunicipalityID (from Municipalities) to specify which fish species is being priced and where. The DateRecorded field captures the date of price entry, and PricePerKg stores the validated price per kilogram as a decimal value. The EnteredBy and ValidatedBy fields reference the User table to track which LGU officer entered the price and which officer validated it, ensuring accountability. The IsCorrection flag and OriginalPriceID field enable price corrections while preserving audit history, allowing the system to maintain a complete trail of changes without overwriting original records.

Table 20

Weight Entries Entity Data Dictionary

WEIGHT ENTRIES				
Field Name	**Data Type**	**SIze**	**Constraints**	**Description**
EntryID	**INT**	**11**	**PK, Auto Increment**	**Unique identifier for weight entry**
UserID	**INT**	**11**	**FK → Users.UserID**	**User who made the entry**
SpeciesID	**INT**	**11**	**FK → FishSpecies.SpeciesID**	**Fish species being weighed**
WeightKg	**DECIMAL**	**10,2**	**NOT NULL**	**Weight of fish in kilograms**
ComputedPrice	**DECIMAL**	**10,2**	**NOT NULL**	**Computed fair**

				price (Weight × PricePerKg)
DateEntered	**DATE**	**N/A**	**NOT NULL**	**Date of entry**
Saved	**BOOLEAN**	**1**	**Default FALSE**	**Indicates if calculation was saved**

As shown in Table 20, WeightEntries Entity Data Dictionary table logs all fair price calculations performed by fish vendors using the system's price calculator feature. The EntryID serves as the unique primary key for each calculation record. The UserID and SpeciesID foreign keys identify which vendor performed the calculation and which fish species was involved. The WeightKg field stores the weight of fish entered by the vendor, while the ComputedPrice field automatically stores the result of multiplying WeightKg by the current LGU-validated PricePerKg. The DateEntered captures when the calculation was made, and the Saved boolean flag indicates whether the vendor chose to save the calculation to their personal history for future reference, allowing vendors to track their pricing decisions over time.

Table 21

Predictions Entity Data Dictionary

PREDICTIONS				
Field Name	**Data Type**	**SIze**	**Constraints**	**Description**
PredictionID	**INT**	**11**	**PK, Auto Increment**	**Unique identifier for prediction**
SpeciesID	**INT**	**11**	**FK → FishSpecies.SpeciesID**	**Fish species being forecasted**
MunicipalityID	**INT**	**11**	**FK → Municipalities. MunicipalityID**	**Municipality for forecast**
ForecastDate	**DATE**	**N/A**	**NOT NULL**	**Date of forecast**
ForecastPrice	**DECIMAL**	**10,2**	**NOT NULL**	**Forecasted**

				price
ModelUsed	**VARCHAR**	**50**	**NULL**	**Predictive model used (e.g., ARIMA)**

As shown in Table 21, Predictions Entity Data Dictionary table stores all forecast outputs generated by the AquaTimbang predictive analytics module, which uses the SARIMAX model to produce short-term price forecasts. The PredictionID is the primary key that uniquely identifies each forecast record. Foreign keys link to SpeciesID and MunicipalityID to specify which fish species and location the forecast applies to. The ForecastDate field indicates the future date for which the price is being predicted, while ForecastPrice stores the predicted price per kilogram. The ModelUsed field records which predictive model was employed allowing administrators to track model versions and performance.

Table 22

Reports Entity Data Dictionary

REPORTS				
Field Name	**Data Type**	**SIze**	**Constraints**	**Description**
ReportID	**INT**	**11**	**PK, Auto Increment**	**Unique identifier for report**
OfficerID	**INT**	**11**	**FK → Users.UserID**	**LGU officer who generated the report**
Parameters	**TEXT**	**N/A**	**NULL**	**Report parameters (species, date range, etc.)**
Format	**ENUM**	**N/A**	**Values: 'PDF', 'CSV'**	**Format of report**
GeneratedAt	**DATETIME**	**N/A**	**Default CURRENT_TIMESTAMP**	**Timestamp of report generation**

As shown in Table 22, Reports Entity Data Dictionary table logs all market reports generated by LGU officers within the AquaTimbang system. The ReportID serves as the primary key for each report record. The OfficerID foreign key references the User table to identify which LGU officer generated the report. The Parameters field stores the filter settings applied during report generation, such as date range, selected fish species, or municipality, as a text block. The Format field uses an ENUM data type to specify whether the report was exported as PDF or CSV. The GeneratedAt field automatically records the timestamp of report creation using the current date and time. This table provides an audit trail of all generated reports, helping LGUs track reporting activities and ensure transparency in market data dissemination.

Table 23

AccessLogs Entity Data Dictionary

ACCESSLOGS				
Field Name	**Data Type**	**SIze**	**Constraints**	**Description**
LogID	**INT**	**11**	**PK, Auto Increment**	**Unique identifier for log entry**
UserID	**INT**	**11**	**FK → Users.UserID**	**User who performed the action**
Action	**VARCHAR**	**100**	**NOT NULL**	**Action performed (Login, Price Entry, etc.)**
Timestamp	**DATETIME**	**N/A**	**Default CURRENT_TIMESTAMP**	**Date and time of action**
Details	**TEXT**	**N/A**	**NULL**	**Additional details about the action**

As shown in Table 23, AccessLogs Entity Data Dictionary table maintains a comprehensive audit trail of all user actions within the AquaTimbang system for security,

accountability, and troubleshooting purposes. The LogID is the primary key that uniquely identifies each log entry. The UserID foreign key references the User table to identify which user performed the action. The Action field records the type of activity performed, such as login, logout, price entry, price validation, report generation, or forecast viewing. The Timestamp field automatically records the exact date and time of the action using the current timestamp, providing a chronological history of system usage. The Details field is an optional text field that can store additional context about the action, such as which price record was modified or which report was generated. This table supports security auditing, helps detect unauthorized access attempts, and provides valuable data for system performance analysis and user behavior studies.

System Architecture Design

The architecture of AquaTimbang is built on a hybrid client-server model that integrates a cloud-hosted backend with local client devices. This design ensures secure, real-time data exchange between fish vendors, LGU officers, and administrators, supporting all core functionalities such as price entry, validation, fair price calculation, and predictive analytics

Network Design

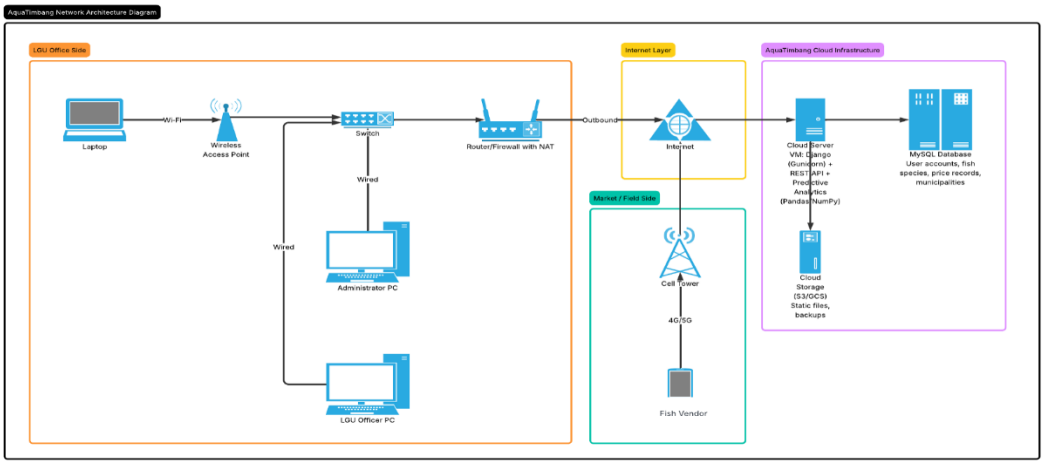


Figure 43

AquaTimbang Network Design

As shown in Figure 43, the network design of AquaTimbang employs a hybrid client-server model that accommodates both wireless and wired connections. Wireless access points connect laptops and mobile devices to a central router, which is protected by a firewall before reaching the internet. A switch handles wired devices—such as desktop computers in LGU offices—to ensure stable backend operations. This structure supports all user roles (fish vendors, LGU officers, administrators) by providing secure, reliable communication for tasks like price entry, validation, trend analysis, and report generation. The design prioritizes flexibility, scalability, and data protection to meet the needs of a multi-stakeholder environment.

Network Topology

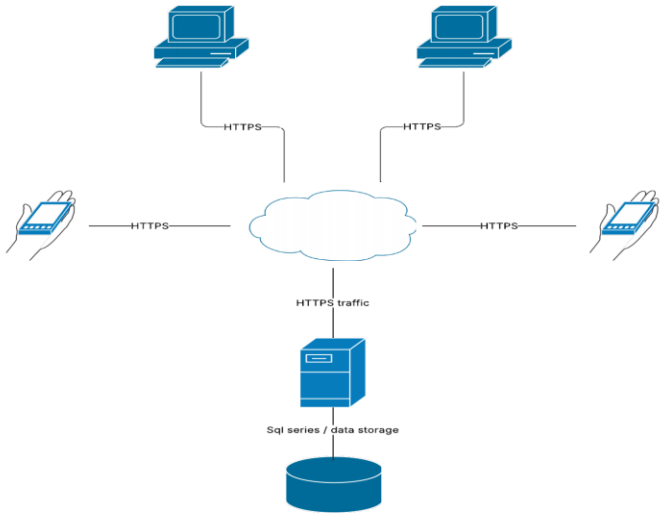


Figure 44

AquaTimbang Network Topology

As shown in Figure 44, AquaTimbang uses a cloud-based star topology. The central hub is the AquaTimbang Cloud Server, which hosts the Django backend, REST API, and predictive analytics engine. All client nodes—including LGU workstations, vendor smartphones, and administrator computers—connect directly to this hub via the internet. External services (e.g., future mapping or weather APIs) also communicate with the hub over secure HTTPS connections.

Security

Security is implemented across multiple layers to protect system data, integrity, and user privacy. Network security ensures that all communication between client devices and the cloud server is encrypted using HTTPS with TLS 1.3. User authentication is managed using Django's built-in framework, with passwords securely hashed. Role-Based Access Control (RBAC) is enforced to restrict system features based on user roles. Session management includes automatic expiration after inactivity and temporary account lockouts after multiple failed login attempts. API security requires authentication tokens for all endpoints, validates all incoming requests, and applies rate limiting to prevent abuse. Database security ensures that the MySQL database is accessed only through secure connections, with credentials stored as environment variables. Encryption at rest is enabled when supported by the hosting provider.

The system complies with the Data Privacy Act of 2012 by ensuring that personal data is collected with consent, used only for system purposes, and accessible only to authorized users. Client-side security ensures that no sensitive data is stored in plain text, authentication tokens are securely handled, and cross-origin requests are restricted to trusted sources.

Table 24

Security Design - Layer Summary

Layer	Implementation
Network Security	All communication encrypted via HTTPS with TLS 1.3. Firewalls at local and cloud levels prevent unauthorized access.
User Authentication	Managed with Django's built-in framework. Passwords securely hashed.
Access Control	Role-Based Access Control (RBAC) restricts features based on user roles.
Session Management	Automatic expiration after inactivity. Temporary account lockouts after multiple failed login attempts.

API Security	Authentication tokens required for all endpoints. Incoming requests validated. Rate limiting applied to prevent abuse.
Database Security	MySQL is accessed only through secure connections. Credentials stored as environment variables. Encryption at rest enabled when supported.
Legal Compliance	Adheres to Data Privacy Act of 2012: personal data collected with consent, used only for system purposes, and accessible only to authorized users.
Client-Side Security	No sensitive data stored in plain text. Authentication tokens securely handled. Cross-origin requests restricted to trusted sources.

Devices

AquaTimbang relies on a combination of physical and virtual devices to facilitate communication and data processing.

Client Devices

LGU officers use desktop or laptop computers with modern web browsers and internet access to perform price entry, validation, and monitoring. Vendors and fisherfolk use smartphones or tablets with internet connectivity to access price trends, calculate fair prices, and compare market data. Administrators use desktop or laptop computers to manage users, configure the system, and oversee data.

Server Infrastructure

The system is hosted on a cloud-based virtual machine running a Linux operating system. The server hosts the Django application using Gunicorn and serves the React frontend through Nginx. The predictive analytics component operates within the Django environment using data processing libraries. The database server uses MySQL to store all structured system data, including users, fish species, municipalities, and price records.

Networking Hardware (Client Side)

A router or modem provides internet connectivity to the local network. A wireless access point enables Wi-Fi connections for mobile devices and laptops. A firewall protects the network from unauthorized external access. A network switch supports wired connections for stable and high-bandwidth devices such as desktop computers.

DEVELOPMENT PHASE

The development phase of AquaTimbang focuses on translating the system requirements into a functional web-based and mobile-responsive application that supports fish price monitoring, data entry, and basic predictive analytics. This stage ensures that all technical components, tools, and environments are properly defined and integrated to deliver a reliable, secure, and user-friendly system for fish vendors, and local government units.

Programming Environment

The system is developed as a responsive web application that can be accessed through modern browsers on both desktop and mobile devices. Developers use local machines with Python virtual environments and Node.js for frontend development and build processes. The production environment is deployed on a cloud-based virtual server running Ubuntu Linux.

Table 25

Programming Environment

Component	Specification
Development	Windows 10 or higher
Code Editor	Visual Studio Code (latest version)
Local Dev Environment	Python 3.9+ (with Django, Pandas, NumPy, Statsmodels, Pmdarima) Node.js 18+ (for React.js) MySQL 8+

	Github
UI/Wireframe Design	React.js
Web Browsers for Testing	Google Chrome (latest), Microsoft Edge (latest)

Programming Language

The backend of the system is developed using Python. The frontend is developed using JavaScript with the React.js library. HTML5 and CSS3 are used for structuring and styling the web pages.

Development Tools / IDE

Visual Studio Code is used for coding, debugging, and managing the project. GitHub is used as the version control and collaboration platform.

Table 26

Development Tools / IDE

Tool	Purpose
Visual Studio Code	Primary code editor for PHP,
GitHub	Version control and collaboration platform for managing source code, tracking changes, and team coordination.
Google Chrome Dev/Tools	Front-end debugging, responsive layout testing , network inspection
MySQL Workbench	Database design, query development, and management of MySQL database (price records, user accounts, fish species).

Postman	API testing for Django REST Framework endpoints (price entry, calculator, forecast retrieval) before frontend integration.

Build and Dependency Management

Backend dependencies are managed using pip and are listed in a requirements.txt file. Frontend dependencies are managed using Node.js package managers such as npm or yarn.

Version Control

The system uses a Git repository with a structured branching strategy. The main branch contains stable and production-ready code. The develop branch is used for integrating completed features. Feature branches are used for developing individual functionalities.

Frameworks and Libraries

Django is used as the backend framework, and Django REST Framework is used to build RESTful APIs. React.js is used to develop the user interface using reusable components. Chart.js or Recharts is used for data visualization of price trends and forecast confidence intervals. Pandas and NumPy are used for data manipulation and numerical computing. Statsmodels provides the SARIMAX class for seasonal time-series forecasting with exogenous regressors, while pmdarima supplies auto_arima for automatic selection of non-seasonal and seasonal parameters (p, d, q, P, D, Q, s). Django ORM is used to manage database interactions with MySQL, including storage of price records, external factor data (fuel prices, weather conditions, holiday flags), and forecast outputs. Axios is used to handle HTTP requests between the React frontend and the Django backend.

List of Modules

AquaTimbang is composed into eight top-level functional modules. Each presents the complete list of system modules, their assigned programmers, and the corresponding functions accessible to Fish Vendors, LGU Officers, and Administrators.

Table 2

List of Modules

Programmer	Modules	FishVendor	LGU officer	Admin
User Authentication Module				
Ybañez, JoshKent U. Padillo, Christian Dave T. Sagaad, Rio Jen J.	Registration	*	*	*
	Login	*	*	*
	Logout	*	*	*
No. of Points:		1		
Municipal Fish Price Dashboard Module				
Ybañez, JoshKent U. Padillo, Christian Dave T. Sagaad, Rio Jen J.	Real-Time Price Display	*	*	*
	Historical Price Visualization	*	*	*
	Forecast Integration (SARIMAX)	*	*	*
	Interactive Charts & Graphs	*	*	*
	Data Export & Sharing		*	*
	Responsive Interface	*	*	*
No. of Points:		1		
Fair Price Calculator Module				
Ybañez, JoshKent U. Padillo, Christian	Weight Input Interface	*		
	Computation Logic	*		

Dave T. Sagaad, Rio Jen J.	Validation & Error Handling	*		
No. of Points:		1		
Price Entry and Validation Module				
Ybañez, JoshKent U. Padillo, Christian Dave T. Sagaad, Rio Jen J.	Daily Price Input Interface		*	
	Submission & Review Panel		*	
	Official Price Publication		*	
	Audit & Logging System		*	
	External Factor Integration		*	*
No. of Points:		1		
Predictive Analytics Module (SARIMAX-Based Forecasting)				
Ybañez, JoshKent U. Padillo, Christian Dave T. Sagaad, Rio Jen J	SARIMAX Forecasting Core	*	*	
	Visualization & Insights Generator	*	*	
	External Factor Integration	*	*	
No. of Points:		1		
Reports Module				
Ybañez, JoshKent U. Padillo, Christian Dave T. Sagaad, Rio Jen J	Price History Reports		*	*
	External Factor Summaries		*	*
	User Activity Logs			*

	Export Engine (CSV/PDF)		*	*
No. of Points:		1		
User Management Module				
	User Account Creation			*
Ybañez, JoshKent U. Padillo, Christian Dave T. Sagaad, Rio Jen J	Edit & Update User Information			*
	Deactivate/Delete Accounts			*
	Role Assignment & Management			*
	Password Reset & Recovery			*
	User Activity Logs			*
No. of Points:		1		
Fish Species Management Module				
Ybañez, JoshKent U. Padillo, Christian Dave T. Sagaad, Rio Jen J	Species List Maintenance		*	*
	Common & Scientific Name Registry		*	*
	Active/Inactive Status Control		*	*
No. of Points:		1		
Total Number of Modules		8		

Database Stack

MySQL is used as the relational database management system for AquaTimbang. The database stores user accounts, fish species catalogs, municipalities, and historical price

records with defined foreign key relationships to maintain referential integrity. The database schema is normalized to eliminate redundancy and ensure data consistency across all tables. Indexes are applied to frequently queried columns such as municipality, species, and date to optimize query performance, particularly for the price dashboard and forecast generation features.

Table 28

Database Stack

Component	Specification
Database System	MySQL 8+
Stored Data	User accounts, fish species catalogs, municipalities, historical price records, external factors (fuel prices, weather, holidays), forecast results
Key Design Principles	Normalized schema, foreign key constraints, indexes on (municipality, species, date)
Access Method	Django ORM with MySQL backend

Testing Tools (Developer Side)

Django's built-in test framework is used for unit and integration testing of models, views, and APIs to ensure backend reliability. Frontend components are tested manually using browser developer tools, with optional unit testing available through Jest. API endpoints are validated using the Django REST Framework browsable API and tools such as Postman. User Acceptance Testing (UAT) is conducted with selected fish vendors, LGU personnel, and the system administrator using the deployed system prototype to gather real-world feedback.

Table 29

Testing Tools (Developer Side)

Tool	Purpose
Django Built-in Test Framework	Unit and integration testing for models, views, and APIs
Browser Developer Tools (Chrome/Edge)	Manual frontend testing, responsive layout checks, console debugging
Jest (optional)	Unit testing for React.js components
Django REST Framework Browsable API	API endpoint validation and manual exploratory testing
Postman	Automated API request testing, collection runs, and environment configuration
Deployed System Prototype	User Acceptance Testing (UAT) with actual stakeholders

Configuration

Environment variables are used to store sensitive information such as database credentials, Django secret keys, and deployment settings. These variables are managed through .env files in local development and cloud environment settings in production. Separate Django settings modules are used to manage configurations for development and production environments, ensuring that debugging, logging, and security settings are appropriately scoped.

Table 30

Configuration

Configuration Item	Approach
Sensitive Data Storage	Environment variables (database credentials, secret keys, API keys)
Local Development	.env files (excluded from version control via .gitignore)
Production	Cloud environment variables (e.g., AWS Secrets Manager, GCP Secret Manager, or platform settings)
Django Settings Management	Separate settings modules (e.g., settings/base.py, settings/dev.py, settings/prod.py)
Environment Detection	DJANGO_SETTINGS_MODULE or environment-specific configuration files

Operating System

The development environment for AquaTimbang is operating system-agnostic, allowing the team to use Windows, macOS, or Linux based on individual preference, provided the system supports Python, Node.js, and MySQL. For production deployment, the system runs on a Linux-based server to ensure stability, security, and compatibility with cloud hosting platforms. Ubuntu 20.04 or 22.04 LTS is selected as the production operating system due to its long-term support, wide community adoption, and optimized performance for Django, MySQL, and SARIMAX forecasting workloads.

Table 31

Operating System

Environment	Specification
Development	Windows, macOS, or Linux (any OS supporting Python, Node.js, and MySQL)
Production	Linux-based server (Ubuntu 20.04 or 22.04 LTS) hosted on a cloud platform (e.g., AWS EC2, GCP Compute Engine)

Hardware Requirements

LGU officers require desktop or laptop computers with modern web browsers and stable internet connections to perform daily price entry, validation, and report generation. Administrators have similar hardware requirements, with optional higher processing capabilities for data management tasks such as user account oversight and system configuration. Vendors require smartphones or tablets with internet access to use the mobile-optimized interface for viewing prices, using the fair price calculator, and checking forecasts. The cloud server infrastructure requires a virtual machine with sufficient CPU and RAM to run the Django application and MySQL database, with optional use of managed database services for scalability.

Table 32

Hardware Requirements

Stakeholder / Component	Hardware Requirements
LGU Personnel	Desktop or laptop computer, modern web browser (Chrome/Edge), stable internet connection

System Administrator	Desktop or laptop computer (optionally higher processing power for data management), modern web browser, stable internet connection
Fish Vendors	Smartphone or tablet with internet access (mobile-optimized interface)
Cloud Server	Virtual machine (e.g., AWS EC2 or GCP Compute Engine) with sufficient CPU (e.g., 2 vCPUs) and RAM (e.g., 4-8 GB) to run Django + MySQL; optionally, managed database services (e.g., AWS RDS)

TESTING PHASE

The AquaTimbang system undergoes continuous testing to ensure reliability, accuracy, and usability. Testing is integrated with development using an iterative version control workflow, where feature-specific branches are merged into the develop branch for continuous integration before deployment to the main branch. Testing is conducted systematically across all core system modules.

Unit Testing

Unit testing verifies that individual modules and functions operate correctly before integration. The web dashboard components are tested using Jest and React Testing Library to ensure proper rendering, state updates, and correct API interactions. The mobile interface is tested by validating UI elements and JavaScript functions using browser developer tools. The backend components are tested using pytest-django to validate authentication logic, CRUD operations, predictive analytics functions, and API responses. Each REST API endpoint is manually tested using Postman to verify request and response structures, status codes, and authentication requirements.

Integration Testing

Integration testing evaluates how different system components interact within the development environment. The frontend and backend communication is tested to ensure that user actions correctly trigger API processes and update the interface. Database integration is tested to confirm that data is accurately inserted, updated, and retrieved using Django ORM, while maintaining data integrity. The predictive analytics module is tested to ensure it correctly processes historical data and generates accurate forecasts displayed on the dashboard. Authentication and Role-Based Access Control are tested to ensure that users can only access features permitted by their roles.

System Testing

System testing evaluates the complete and fully integrated system in a simulated environment. End-to-end testing is conducted using realistic scenarios to verify that all modules function together correctly. The system supports processes such as user registration and approval, price entry and validation, fair price computation, and predictive trend generation.Edge-case testing is performed to ensure the system handles invalid inputs, duplicate data, and missing fields properly. Performance testing is conducted using Google Lighthouse to measure system responsiveness, accessibility, and API performance.

Acceptance Testing

Acceptance testing determines whether the system meets business requirements and is ready for deployment. LGU officers and administrators test the web dashboard in a simulated environment to perform tasks such as data entry, validation, and monitoring. System usability is measured using the System Usability Scale, with a target score of at least 68. Vendors test the mobile interface to ensure usability, accurate price calculations, and timely data updates. This phase validates that the system is practical, intuitive, and effective for real-world use.

User Acceptance Testing Questionnaire

A Technology Acceptance Model-based questionnaire is used to evaluate system acceptance after user testing. The questionnaire uses a 5-point Likert scale to measure perceived usefulness, perceived ease of use, and user intention. For LGU officers and

administrators, the system is evaluated based on its ability to improve efficiency in recording and validating prices, monitor trends, and support fair market practices. The system is also assessed for ease of use, clarity of interface, and willingness of users to adopt it regularly. For fish vendors, the system is evaluated based on its ability to provide reliable pricing information, ensure fair pricing through calculation, and support decision-making using price trends. The mobile interface is assessed for simplicity, clarity, and usability during real market transactions. User intention to continue using the system is also measured.

Table 33

UAT Checklist

Module	Test Scenario	Expected Result	Status (Pass/Fail)
User Authentication	Vendor logs in with valid credentials	Dashboard displayed; vendor cannot access admin pages	Pass
Daily Price Entry	LGU personnel enters price for Bangus (₱180/kg) on current date	Price saved to database; appears on vendor dashboard	Pass
Fair Price Calculator	Vendor enters 2.5 kg of Bangus	Calculated price = ₱450 (2.5 × 180) displayed	Pass
Price Dashboard	Vendor views today's prices for all species	List shows validated prices with timestamps	Pass

Forecast Visualization	Vendor clicks "View Forecast" for Tulingan	Graph shows 7-day forecast with confidence intervals	Pass
Report Generation	LGU generates PDF report for last 7 days	PDF contains price table and date range	Pass
Role-Based Access	Vendor attempts to edit a price record	Access denied; error message shown	Pass

DEPLOYMENT PHASE

The AquaTimbang system is deployed as a cloud-based web application composed of a frontend user interface, a backend API with an analytics engine, and a relational database. The system uses a cloud-hosted architecture to ensure accessibility, scalability, and reliable performance for fish vendors, LGU officers, and administrators.

Hardware Requirements

The hardware requirements specify the minimum computational resources needed for both client devices and cloud infrastructure to ensure smooth system operation.

Table 34

Hardware Requirements

Stakeholder / Component	Hardware Requirements
LGU Officers & Administrators (Client)	Desktop or laptop with at least 4GB RAM, modern processor, screen resolution of 1366×768 or higher
Fish Vendors (Client)	Smartphone or tablet running Android 10 or iOS 13 and above (optional GPS capability)

All Users	Stable internet connection with minimum bandwidth of 5 Mbps
Cloud Server (Virtual Machine)	At least 1 vCPU and 2GB RAM to run Django and MySQL
Persistent Storage	SSD-backed storage with minimum capacity of 20GB
Database (Alternative)	Managed database service (e.g., AWS RDS, GCP Cloud SQL) may be used for dedicated database hosting

Operating System

The system is platform-independent on the client side and operates across multiple operating systems. Supported client operating systems include Windows, macOS, Android, and iOS to accommodate both LGU personnel (using desktops/laptops) and fish vendors (using smartphones/tablets). The backend services run on a Linux-based environment to ensure stability, security, and compatibility with Django, MySQL, and the SARIMAX forecasting stack.

Table 35

Operating System

Environment	Supported Operating Systems
Client (LGU Officers & Administrators)	Windows 10/11, macOS 11 or later
Client (Fish Vendors)	Android 10 or later, iOS 13 or later
Backend Server (Production)	Linux-based (Ubuntu 20.04 LTS or 22.04 LTS)

Runtime Requirements

The runtime requirements define the software environments and dependencies needed to execute the system components. The frontend is a client-side React.js application served via Nginx and runs on modern web browsers. The backend is a Django application running on Python with Gunicorn as the WSGI server. Predictive analytics for fish price forecasting is implemented using SARIMAX from statsmodels, with auto_arima from pmdarima for automatic parameter selection, while pandas and NumPy handle data manipulation and numerical computing.

Table 36

Runtime Requirements

Component	Requirements / Technologies
Frontend (Browser)	HTML5, CSS3, JavaScript (ES6+); compatible with latest versions of Chrome, Edge, Firefox, Safari
Frontend Framework	React.js (served via Nginx)
Backend Runtime	Python 3.10 or later
Backend Framework	Django 4.2 or higher, Django REST Framework
WSGI Server	Gunicorn
Reverse Proxy / Static Server	Nginx
Database Adapter	mysqlclient or pymysql
Data Manipulation	pandas, NumPy

Forecasting Libraries	statsmodels (SARIMAX class), pmdarima (auto_arima)
Utilities	django-cors-headers, python-dotenv

Network Requirements

The network requirements define the connectivity standards necessary for system communication and real-time updates. All client devices must have a stable internet connection to access the AquaTimbang platform. The system uses HTTPS protocol to ensure secure communication between clients and the server. Communication occurs through RESTful APIs between the React frontend, Django backend, and MySQL database, with all internal traffic secured within a virtual private cloud (VPC). Price data is updated based on user actions, and users can manually refresh the dashboard to view the latest data. The system does not require external APIs for its core functionality.

Table 37

Network Requirements

Requirement	Specification
Minimum Bandwidth	5 Mbps (client to server)
Protocol	HTTPS (TLS 1.2 or higher)
Communication Pattern	RESTful APIs (frontend ↔ backend ↔ database)
Internal Security	Virtual Private Cloud (VPC) for backend and database communication
Data Update Mechanism	User-triggered actions; manual dashboard refresh
External Dependencies	None (core functionality does not require external APIs)

Database Server Environment

The database server environment manages all persistent system data. The system uses MySQL Community Edition as the primary relational database. The database can be hosted on the same virtual machine as the application or on a managed cloud service for improved scalability and maintenance. The database stores user accounts, roles and permissions, fish species, municipalities, price records, weight entries, prediction results, reports, and logs. Security measures include restricted database access, encrypted connections, and regular automated backups to ensure data integrity and availability.

Table 38

Database Server Environment

Component	Specification
Database System	MySQL Community Edition (8+)
Hosting Options	Same VM as application OR managed cloud service (e.g., AWS RDS, GCP Cloud SQL)
Stored Data	User accounts, roles/permissions, fish species, municipalities, price records, weight entries, prediction results (forecasts), reports, system logs
Security Measures	Restricted database access (least privilege), encrypted connections (SSL/TLS), regular automated backups
Backup Frequency	Daily automated backups (e.g., using mysqldump or cloud provider snapshots)

biApplication Hosting or Deployment Environment

The system uses a cloud-based architecture that separates the frontend, backend, and database layers. The frontend is built as static files and served using Nginx on the same virtual machine as the backend. Alternatively, the frontend may be hosted on a static hosting platform for improved performance. The backend runs on a cloud virtual machine using Django and Gunicorn, with Nginx acting as a reverse proxy to handle requests and serve static files. The database is hosted either locally on the same virtual machine or on a managed cloud service. The system architecture consists of a UI layer (React), a logic layer (Django), and a data layer (MySQL). All communication is conducted through RESTful APIs using JSON over HTTPS.

Table 39

Application Hosting or Deployment Environment

Layer	Technology / Component	Hosting Approach
UI Layer (Frontend)	React.js (static files)	Served via Nginx on the same VM as backend, OR hosted on a static hosting platform (e.g., Vercel, Netlify)
Logic Layer (Backend)	Django + Gunicorn	Runs on a cloud virtual machine (e.g., AWS EC2, GCP Compute Engine)
Reverse Proxy & Static Server	Nginx	Runs on the same cloud VM, forwards API requests to Gunicorn and serves frontend static files
Data Layer (Database)	MySQL 8+	Hosted locally on the same VM OR managed cloud

		service (e.g., AWS RDS, GCP Cloud SQL)
Communication Protocol	RESTful APIs, JSON over HTTPS	Between all layers

User Environment Requirements

The user environment requirements define the necessary conditions for each stakeholder to use the system effectively. LGU officers and administrators require a desktop or laptop with a modern web browser and a minimum screen resolution of 1366×768. They must also have a stable internet connection for system operations. Vendors require a smartphone or tablet with internet access and a mobile web browser. The system does not require installation and can be accessed through a URL. A responsive design ensures usability on both desktop and mobile devices.

Table 40

User Environment Requirements

User Group	Environment Requirements
LGU Officers & Administrators	Desktop or laptop with modern web browser (Chrome/Edge/Firefox/Safari), minimum resolution 1366×768, stable internet connection (5 Mbps recommended)
Fish Vendors	Smartphone or tablet with mobile web browser, internet access, stable connection (5 Mbps recommended)

All Users	No software installation required; system accessed via URL; responsive design adapts to screen size

Security Requirements

The system implements security measures to protect user data, ensure system integrity, and secure communication. All users are required to log in using valid credentials, and passwords are securely hashed. Role-Based Access Control (RBAC) is enforced to manage user permissions. Session management includes timeout mechanisms, automatic logout, and secure cookies. All data transmission is secured using HTTPS, and sensitive data is protected through encryption and environment variables. The system complies with the Data Privacy Act of 2012 by ensuring proper data handling and access control. API endpoints require authentication, and input validation is applied to prevent security threats. CORS policies restrict access to trusted domains. Server security includes firewall configuration, regular updates, and secure backup storage.

Table 41

Security Requirements

Security Category	Implementation
Authentication	Valid credentials required for login; passwords securely hashed (e.g., using Django's PBKDF2)
Authorization	Role-Based Access Control (RBAC) enforced (Vendor, LGU, Admin)
Session Management	Timeout mechanisms, automatic logout, secure cookies (HttpOnly, Secure, SameSite)

Data Transmission	HTTPS (TLS 1.2 or higher) for all communication
Sensitive Data Protection	Environment variables for credentials, secrets, and API keys (no hardcoding)
Compliance	Data Privacy Act of 2012 (proper data handling, access control, user consent)
API Security	Authentication required for all endpoints; input validation to prevent injection attacks
CORS Policy	Restricted to trusted domains only
Server Security	Firewall configuration (e.g., UFW), regular OS and package updates, secure backup storage

CURRICULUM VITAE

CURRICULUM VITAE



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APPENDICES

Appendix A

Capstone Project 1
Oath of Confirmation

Date: February 20, 2026

OATH OF CONFIRMATION

This is to confirm that I, Rio Jen J. Sagaad, a BSIS-3 student, currently enrolled in the course of Capstone Project 1, have aptly received a copy; religiously read and understood its contents; and openly submit to the terms, rules, conditions and regulations stated in the Capstone Project Manual document implemented by the Subject Teacher.

This is also to affirm that said guidelines were judiciously discussed and explicitly elaborated in a forum conducted by the Subject Teacher.

CONFORME:

Rio Jen J. Sagaad / February 20, 2026

(Signature of Student over Printed name)/ Date

Date: February 20, 2026

OATH OF CONFIRMATION

This is to confirm that I, Josh Kent U. Ybañez, a BSIS-3 student, currently enrolled in the course of Capstone Project 1, have aptly received a copy; religiously read and understood its contents; and openly submit to the terms, rules, conditions and regulations stated in the Capstone Project Manual document implemented by the Subject Teacher.

This is also to affirm that said guidelines were judiciously discussed and explicitly elaborated in a forum conducted by the Subject Teacher.

CONFORME:

Josh Kent U. Ybañez / February 20, 2026

(Signature of Student over Printed name)/ Date

Date: February 20, 2026

OATH OF CONFIRMATION

This is to confirm that I, Christian Dave T. Padillo, a BSIS-3 student, currently enrolled in the course of Capstone Project 1, have aptly received a copy; religiously read and understood its contents; and openly submit to the terms, rules, conditions and regulations stated in the Capstone Project Manual document implemented by the Subject Teacher.

This is also to affirm that said guidelines were judiciously discussed and explicitly elaborated in a forum conducted by the Subject Teacher.

CONFORME:

Christian Dave T. Padillo / February 20, 2026

(Signature of Student over Printed name)/ Date

Appendix B

Capstone Project Team Composition Form

Name	Signature	Role Assignment	Email	Contact No.
Rio Jen J. Sagaad		Hustler/Project Manager	riojensagaad31@gmail.com	09977997355
Josh Kent U. Ybañez		Hacker	joshkentybanez@gmail.com	09692498929
Christian Dave T. Padillo		Hipster	padillochristiandave@gmail.com	09666224513

Appendix C

Acceptance of Capstone Adviser

College of Computer, Information and Communications Technology

Acceptance of Capstone Adviser

This is to certify that I am accepting the ***Cozy Trio Team*** to be their Capstone Project adviser in the degree of ***Bachelor of Science in Information Systems.***

_____
(Signature of Adviser over printed name)
Date:_____

Noted by:

_____
(Signature of Subject Teacher over printed name)
Date:_____

Appendix D

Capstone Project Business Feasibility Analysis

Technical Feasibility

This technical feasibility analysis evaluates the proposed AquaTimbang: A Predictive Analytics-Driven Fish Price Transparency System for Local Government Units (LGUs). It identifies potential risks in system development and adoption, and outlines mitigation strategies to ensure successful implementation.

Table 5
Technical Feasibility

Criteria	Description	Risk Level	Mitigation Strategy
Familiarity with the Application	LGU Officers have limited experience with digital platforms for fish price monitoring. Manual processes dominate, creating a learning curve for users.	Moderately High	Conduct training sessions, design a user-friendly interface, and provide helpdesk support to ensure smooth adoption.
Familiarity with the Technology	Required technologies (MySQL, Pandas/NumPy for analytics, AWS/GCP hosting, RBAC) are widely available and supported. Development teams are familiar with these tools.	Low	Use established frameworks, follow secure coding practices, and leverage cloud services for scalability and reliability.
Project Size	The system includes multiple modules: price entry/validation, fair price calculator, municipal dashboard, predictive analytics, and user management. Managing all within the capstone timeline may be challenging.	Medium	Prioritize core modules (price entry, fair price calculator, dashboard). Advanced features (trend forecasting, extended reporting) can be phased in later.
Compatibility	LGUs already use basic IT infrastructure (computers, internet, spreadsheets). AquaTimbang is compatible with existing workflows and requires only moderate upgrades.	Low	Ensure cross-device/browser testing, adopt lightweight system architecture, and provide offline fallback options

Economic Feasibility

Economic feasibility evaluates whether the expected benefits of the system outweigh the estimated costs of development, implementation, operation, and maintenance. It analyzes the financial viability of the system by assessing development expenses, operational costs, potential savings, and long-term benefits. Economic feasibility also determines whether the system can provide cost-effective improvements in efficiency, security, and overall operational performance while ensuring that available resources are used efficiently.

Economic Feasibility Table

		2026	2027	2028	2029	Total
Benefits						
	Increased Sales	₱30,000	₱31,800	₱33,708	₱35,730	₱131,238
	Reduced Pricing Errors	₱8,000	₱8,320	₱8,653	₱8,999	₱33,972
	Optimized LGU Staff Time	₱5,000	₱5,200	₱5,408	₱5,624	₱21,232
Total Benefits		₱43,000	₱45,320	₱47,769	₱50,353	₱186,442
Present Value of Total Benefits (10%)		₱39,091	₱37,455	₱35,891	₱34,396.00	₱146,833
Development Costs						
	Labor: Analysis & Design	₱10,000	₱0	₱0	₱0	₱10,000
	Labor: Implementation	₱15,000	₱0	₱0	₱0	₱15,000
	Development Training	₱2,000	₱0	₱0	₱0	₱2,000
	Office Space & Equipment	₱1,000	₱0	₱0	₱0	₱1,000
Total Development Costs		₱28,000	₱0	₱0	₱0	₱28,000
Operational Costs						
	Cloud Hosting & Domain	₱4,000	₱4,160	₱4,326	₱4,499	₱16,985
	Internet & Communication	₱2,000	₱2,080	₱2,163	₱2,250	₱8,493
	LGU Personnel Time	₱5,200	₱5,408	₱5,624	₱5,849	₱22,081
Total Operational Costs		₱11,200	₱11,648	₱12,113	₱12,598	₱47,559
Total Costs		₱39,200	₱11,648	₱12,113	₱12,598	₱47,559
Present Value of Total Costs (10%)		₱39,200	₱10,589	₱10,011	₱9,466	₱69,266
Total Benefits - Total Costs		₱3,800	₱33,672	₱35,656	₱37,755	₱110,883
Cumulative Net Cash Flow		₱3,800	₱37,472	₱73,128	₱110,883	
Present Value of Net Benefits		-₱109	₱26,866	₱25,880	₱24,930	₱77,567
Intangible Benefits						
	Enhanced public trust	Transparent fish pricing increases community confidence in LGU market governance.				
	Better decision-making	LGU officers gain access to real-time, verified price data for policy formulation.				
	Improved LGU image	Adoption of digital innovation positions the LGU as modern and responsive.				

The table presents the cost-benefit analysis of the Aquatimbang System by comparing the projected development and operational costs with the expected benefits over several years. It highlights the financial gains from reduced manual processing, improved security efficiency, and faster entry and exit verification. The analysis also shows the system's return on investment (ROI), break-even point, and net present value (NPV), indicating that the project is financially feasible and beneficial for long-term campus security operations.

Organizational Feasibility

Organizational feasibility examines whether the proposed system can be successfully adopted and sustained within the existing structure, culture, and resources of the organization. It evaluates the alignment of AquaTimbang with local government priorities, the roles and responsibilities of stakeholders, and the capacity of organizational management to support implementation. This analysis ensures that the system is not only technically viable but also organizationally practical, with clear leadership, stakeholder participation, and structured management processes to guarantee long-term success.

AquaTimbang is organizationally feasible because the LGU has a legal responsibility to monitor and validate fish prices, fisherfolk have a clear need for transparent pricing, and a designated system administrator can manage the backend. Potential challenges—such as low fisherfolk participation or limited technical literacy—can be addressed through community orientation, a simple mobile interface, and phased implementation.

AquaTimbang is strategically aligned with the Local Government Unit's (LGU) mandate to ensure fair trade, protect fisherfolk from exploitative middlemen, and modernize market governance. By digitizing daily price entry and validation, the system supports the Department of Veterinary Medicine and Fisheries (DVMF) in Cebu City, which oversees market-related activities and data monitoring. The project also complements national initiatives such as the Electronic Fisheries Price Monitoring System (eFPMS) and the Department of Information and Communications Technology's online price monitoring platforms by extending their focus from data collection to predictive analytics and fair, weight-based pricing. At a broader level, AquaTimbang contributes to the United Nations Sustainable Development Goals (SDGs), specifically SDG 1 (No Poverty) by enabling vendors to secure fairer and more stable incomes, SDG 2 (Zero Hunger) by promoting equitable fish distribution, SDG 8 (Decent Work and Economic Growth) by fostering transparent and trustworthy market environments, SDG 9 (Industry, Innovation, and Infrastructure) by applying accessible predictive analytics to traditional markets, and SDG 12 (Responsible Consumption and Production) by reducing inefficiencies caused by information asymmetry. By aligning with both local governance priorities and global development goals, AquaTimbang ensures organizational support and long-term relevance.

Table 7
Stakeholder Analysis

Stakeholder	Role in the System	Contribution to Feasibility
Stakeholder Champion	LGU Fisheries Officer or Market Supervisor with authority to mandate system use	Provide leadership, enforce adoption, allocate staff, and embed AquaTimbang into official operations
LGU Personnel	Market regulators responsible for inputting and validating daily fish price data	Guarantee transparency and accountability by generating reports and overseeing market practices
System Administrator	Technical manager overseeing accounts, species lists, and system configuration	Maintain data integrity, ensure system uptime, and safeguard security for reliable operations
Fish Vendors	End users and primary beneficiaries who use the fair price calculator and monitor trends	Ensure data accuracy and adoption through active participation; benefit from fair and transparent pricing
Organizational Management	LGU hierarchy including Market Office, Agriculture Office, and IT Office	Supervise operations, schedule training, enforce system use, and align processes with organizational goals

Appendix E

COMPARATIVE MATRIX

System	Type	Price Transparency	Weight-Based Calculator	Predictive Analytics	LGU Validation	Mobile Access	Local PH Focus	% Score
AquaTimbang (Proposed)	Integrated Platform	✓	✓	✓	✓	✓	✓	100 %
Sinay Fish Price App	International Commercial	✓	✗	✗	✗	✓	✗	33%
BFAR eFPMS	Government Monitoring	✓	✗	✗	✓	✗	✓	50%
FAO Fish Price Index	International Index	✗	✗	✗	✗	✗	✗	17%
ABALOBI	Catch Documentation	✗	✗	✗	✗	✓	✗	17%
TanFish Market	Digital MarketPlace	✗	✗	✗	✗	✓	✗	17%
UDSM Platform	Digital MarketPlace	✓	✗	✗	✗	✓	✗	33%
OurFish App (RARE)	Catch Documentation	v	✗	✗	✓	✓	✓	67%
Tateh AquaBiz	Aquaculture Support	✓	✗	✗	✗	✓	✓	50%
Mayani	E-Commerce	✗	✗	✗	✗	✓	✓	33%
Bantay Presyo (DA)	Price Monitoring	✓	✗	✗	✓	✗	✓	50%

Appendix F

Research Working Title Form

Name of Proponents	
1. Rio Jen J. Sagaad	
2. Josh Kent U. Ybañez	
3. Christian Dave T. Padillo	
Proposed Research Title:	
AquaTimbang: A Predictive Analytics–Driven Fish Price Transparency System for Informed Market Decision- Making.	
Date:	Date:
Submitted by:	Noted and Approved by:
(Signature of Project Leader over Printed Name)	(Signature of Subject Teacher over Printed Name)

Appendix G

Consultation Log Form

Research/Capstone Project Title

AquaTimbang: A Predictive Analytics–Driven Fish Price Transparency System for Informed Market Decision- Making.

Name of Proponents

Rio Jen J. Sagaad

Christian Dave T. Padillo

Josh Kent U. Ybanez

Name of Group

Cozy Trio

1st Consultation: CHAPTER 1 must be completely delivered for adviser's evaluation

Date	PMs Signature	Adviser's Signature

2nd Consultation: CHAPTER 1 and 2 must be completely delivered for adviser's evaluation

Date	PMs Signature	Adviser's Signature

3rd Consultation: CHAPTER 1,2 and 3 must be completely delivered for adviser's evaluation

Date	PMs Signature	Adviser's Signature

4th Consultation: CHAPTER 1, 2, 3 and including the initial and finals pages must be completely delivered for adviser's evaluation

Date	**PMs Signature**	**Adviser's Signature**

This is to certify that I have been regularly consulted by my advisees; have reviewed their manuscript of the above-stated study. As their adviser, I therefore submit them ready for Proposal Hearing as the chapters 1,2, and 3 and the pertinent parts of their manuscript are complete.

Dr. Emelie C. Villaceran, Ph. D.
(Signature of Adviser over printed name)

Appendix H

List of Modules

Programmer	Modules	FishVendor	LGU officer	Admin
User Authentication Module				
Ybañez, JoshKent U. Padillo, Christian Dave T. Sagaad, Rio Jen J.	Registration	*	*	*
	Login	*	*	*
	Logout	*	*	*
No. of Points:		1		
Municipal Fish Price Dashboard Module				
Ybañez, JoshKent U. Padillo, Christian Dave T. Sagaad, Rio Jen J.	Real-Time Price Display	*	*	*
	Historical Price Visualization	*	*	*
	Forecast Integration (SARIMAX)	*	*	*
	Interactive Charts & Graphs	*	*	*
	Data Export & Sharing		*	*
	Responsive Interface	*	*	*
No. of Points:		1		
Fair Price Calculator Module				
	Weight Input Interface	*		

Ybañez, JoshKent U. Padillo, Christian Dave T. Sagaad, Rio Jen J.	Computation Logic	*		
	Validation & Error Handling	*		
No. of Points:		1		
Price Entry and Validation Module				
Ybañez, JoshKent U. Padillo, Christian Dave T. Sagaad, Rio Jen J.	Daily Price Input Interface		*	
	Submission & Review Panel		*	
	Official Price Publication		*	
	Audit & Logging System		*	
	External Factor Integration		*	*
No. of Points:		1		
Predictive Analytics Module (SARIMAX-Based Forecasting)				
Ybañez, JoshKent U. Padillo, Christian Dave T. Sagaad, Rio Jen J	SARIMAX Forecasting Core	*	*	
	Visualization & Insights Generator	*	*	
	External Factor Integration	*	*	
No. of Points:		1		
Reports Module				
Ybañez, JoshKent U.	Price History Reports		*	*
	External Factor Summaries		*	*

Padillo, Christian Dave T. Sagaad, Rio Jen J	User Activity Logs			*
	Export Engine (CSV/PDF)		*	*
No. of Points:		1		
User Management Module				
	User Account Creation			*
Ybañez, JoshKent U. Padillo, Christian Dave T. Sagaad, Rio Jen J	Edit & Update User Information			*
	Deactivate/Delete Accounts			*
	Role Assignment & Management			*
	Password Reset & Recovery			*
	User Activity Logs			*
No. of Points:		1		
Fish Species Management Module				
Ybañez, JoshKent U. Padillo, Christian Dave T. Sagaad, Rio Jen J	Species List Maintenance		*	*
	Common & Scientific Name Registry		*	*
	Active/Inactive Status Control		*	*
No. of Points:		1		
Total NUmber of Modules		8		

Appendix I

User Acceptance Questionnaire

Dear Respondent,

Good day! We are currently conducting a study on the development of AquaTimbang: A Predictive Analytics–Driven Fish Price Transparency System for Informed Market Decision-Making. This system aims to improve fish price transparency, ensure fair weight-based pricing, and provide reliable market data for fisherfolk and Local Government Units (LGUs).

In line with this, you are kindly requested to answer the following questionnaire. Your responses will be used solely for academic research purposes and will be treated with strict confidentiality in accordance with the Data Privacy Act of 2012. There are no right or wrong answers. Please answer each item honestly based on your perception and experience.

Thank you for your time and cooperation.

Respectfully,

The Researchers

PART I – DEMOGRAPHIC PROFILE

Directions: Please provide the following information by writing your answer or checking the appropriate box.

Age: _____

Sex: ☐ Male ☐ Female ☐ Prefer not to say

User Category: ☐ Fish Vendor ☐ LGU Personnel ☐ System Administrator

Municipality/Barangay: _____

Years of Experience in Fish Market / Work: ☐ Less than 1 year ☐ 1-3 years ☐ 4-6 years ☐ More than 6 years

Have you used any digital system for price monitoring or data recording before? ☐ Yes ☐ No.

PART II – PERCEIVED NEED

This section measures the respondents' perception of current fish pricing practices and the need for a digital price transparency system.

Statement	5	4	3	2	1
Current fish pricing practices in local markets are inconsistent.					
Manual pricing methods lead to unfair price determination.					
Lack of accessible fish price information affects decision-making.					
Dependence on middlemen leads to unequal profit distribution.					
Manual recording of fish prices results in errors and inaccuracies.					
There is a lack of centralized fish price data in local markets.					
A digital system is needed to improve fish price transparency.					

PART III – ANTICIPATED FUNCTIONALITY AND EFFECTIVENESS

This section evaluates the perceived effectiveness of AquaTimbang's core system features.

Statement	5	4	3	2	1
Fish Price Monitoring					
The system will provide accurate and updated fish price information.					
Viewing historical price data will help users make better decisions.					
Fair Price Computation					
The fair price calculator will ensure accurate pricing based on weight.					
Weight-based pricing will promote fairness in transactions.					
Participatory Pricing					
Allowing fisherfolk to suggest prices will improve pricing fairness.					
The averaging of suggested prices will produce a reliable reference price.					
LGU validation of prices will ensure credibility and accuracy.					
Rescue Resource Management					
The dashboard will effectively display price trends and comparisons.					
The system will help LGU personnel monitor fish market prices efficiently					

PART IV – SSTEM USABILITY AND PERFORMANCE (ISO/IEC 25010)

This section evaluates the system based on software quality characteristics.

Statement	5	4	3	2	1
The system is easy to use and understand.					
The interface is clear and user-friendly.					
The system is easy to learn even for first-time users.					
The system performs all required functions correctly.					
The features meet the needs of users.					
The system operates consistently without errors					
The system performs well during usage.					
The system responds quickly to user inputs.					
The system processes data efficiently.					
The system works well on different devices (mobile/web).					
The system can be easily updated or improved when needed.					

PART V – ANTICIPATED IMPACT AND USER ACCEPTANCE

This section measures the expected impact of AquaTimbang on transparency, fairness, and decision-making.

Statement	5	4	3	2	1
The system will improve transparency in fish pricing.					
The system will help fisherfolk earn fair and stable income.					
The system will support LGUs in better price monitoring and regulation.					
The system will reduce price inconsistencies across markets.					
The system will improve trust among fish market stakeholders.					
The system will enhance data-driven decision-making.					
The system will be a valuable improvement over current manual methods.					
I am willing to use AquaTimbang in my daily activities.					

INTERVIEW QUESTIONS:

Question 1: "Are you willing to allow the researchers to collect and use LGU fish price data (historical and current) for the development of the AquaTimbang system?

☐ Yes or ☐ No.

If No, what are your concerns?" _____

_____

Question 2: "Would the LGU be open to having the AquaTimbang system deployed in your local fish market as a pilot testing site? Why or why not?" _____

_____

Question 3: "What specific conditions or requirements would the LGU require before sharing price data with us? (e.g., data sharing agreement, memorandum of understanding, approval from higher office)"_____

_____

Question 4: "Is there an existing data privacy or data sharing policy in your LGU that we need to comply with? Please describe." _____

_____

Question 5: "Would you be willing to assign LGU personnel to act as validators and publishers of fish prices in the system? What would be needed to make this possible?"

_____

_____

PART B: Current Fish Price Monitoring Practices

These questions gather baseline information on how the LGU currently handles fish pricing.

Question 1: "How does the LGU currently monitor fish prices in the market? Is it done manually (paper, whiteboard) or digitally (spreadsheet, app)?"_____

_____

Question 2: "How often are fish prices recorded and updated? Daily, weekly, or monthly?"

_____

_____

Question 3: "Who is responsible for determining the official reference price for fish? What is the process – do you consult fish vendors, or use a formula?" _____

_____

Question 4: "Is the current fish price information shared publicly? If yes, how – bulletin board, social media, text message, or not at all?" _____

_____

Question 5: "How are fish weighed for pricing – is there a standard weighing scale at the market? Are the scales calibrated regularly?" _____

_____

PART C: Challenges and Problems in Current System

These questions identify pain points that AquaTimbang aims to solve.

Question 1: "What are the biggest challenges you face in monitoring and enforcing fair fish prices?" _____

_____

Question 2: "Have you received complaints from fish vendors or consumers about unfair pricing? How often?" _____

_____

Question 3: "Is price data sometimes lost, delayed, or inaccurate under your current system? Please give an example." _____

_____

Question 4: "Do you have difficulty tracking price trends over time – for example, comparing prices from last month to this month? Why?" _____

_____

Question 5: "What is the most time-consuming part of your current price monitoring process? _____

_____

PART D: Desired Features for a Digital Price Transparency System

These questions help validate that AquaTimbang features match LGU needs.

Question 1: "Would a digital dashboard showing real-time fish prices help your LGU? What information would you want to see on it – current prices, historical trends, forecasts?"

_____

_____

Question 3: "Would predictive analytics – forecasting future fish prices based on past data – be valuable for LGU planning? For what purpose?" _____

_____

Question 4: "What other features would you want in a fish price monitoring system? (e.g., mobile access, report generation, price alerts, integration with other government systems)"

_____

_____

Question 5: "Would you prefer a web-based system (accessible via computer) or a mobile-friendly system (accessible via smartphone)? Or both?" _____

_____

PART E: Training and Adoption Readiness

These questions assess organizational readiness for implementation.

Question 1: "Do LGU personnel have basic computer and internet access at their workstations? If not, is there a shared computer or mobile device available?" _____

_____

Question 2: "What level of training would your staff need to use a digital fish price system – basic computer literacy, system-specific training, or ongoing technical support?"

_____

_____

Question 3: "Would the LGU be willing to allocate staff time for price validation, user approval, and report generation using the system?"_____

_____

Question 4: "Is there an LGU champion or point person who could lead the adoption of AquaTimbang? Who would that be?"_____

_____

Question 5: "What barriers do you foresee that might prevent successful adoption of this system – budget, staff resistance, internet connectivity, or something else?" _____

_____

PART F: Data Specifics

Question 1: "What fish species are most commonly sold in your market? Please list at least 5–10." _____

_____

Question 2: "Do you have historical price records? For how many months or years? In what format (paper logs, spreadsheets, database)?" _____

_____

Question 3: "How often is price data collected? Daily, weekly, or monthly? Can we access the last 6–12 months of data?" _____

_____

Question 4: "Is price data separated by species and municipality? Is it tagged with dates and the name of the LGU officer who recorded it?" _____

_____

Question 5: "Would you allow us to take photos of existing price boards, weighing scales, or market areas for documentation purposes? (All identifying marks will be blurred if needed.)" _____

_____

Thank you for your participation!

Appendix J

POST-PROPOSAL HEARING COMPLIANCE CHECKLIST

A. General Information

Group Name	Cozy Trio
Group Members:	Rio Jen J. Sagaad Josh Kent U. Ybañez Christian Dave T. Padillo
Title	AQUATIMBANG: A PREDICTIVE ANALYTICS-DRIVEN FISH PRICE TRANSPARENCY SYSTEM FOR INFORMED MARKET DECISION-MAKING
Proposal Hearing Date:	May 05, 2026
Adviser:	Dr. Emelie C. Villaceran, Ph.D.
Chairman: **Panel 1:** **Panel 2:**	Dr. Angelbert P. Maghanoy, Ph.D. Prof. John Rey E. Furots, MBA Prof. Jayvee T. Cabije, MIT

B. Proposal Hearing Outcome

[ ] Approved with Minor Revisions
[/] Approved with Major Revisions
[ ] Disapproved (Re-proposal Required)

C. Compliance Checklist

Area	Requirements	Complied (Y/N)	Remarks/ Action
Initial Pages	Title Page Approval Sheet Abstract (not yet required) Acknowledgement Dedication Table of Contents List of Tables List of Figures		
Chapter 1 Project Context and Research Design	Project Context Introduction/Background of the Study Purpose and Description		

	Objectives Scope and Limitations Significance Research Design Methods Flow of the Study Environment Respondents Instruments Treatment of Data Scoring Procedures Definition of Terms		
Chapter 2 Related Literature and Technical Background	Review of Related Literature Related Studies Related Systems Technical Background Introduction to the Technology Theoretical Concepts Related Technologies		
Chapter 3 Methodology	Planning Phase Requirement Specifications Customer Validation Board System Context Diagram User Flow Diagram Functional Decomposition Diagram Work Breakdown Structure Gannt Chart Analysis Phase Use Case Diagram Sequence Diagram Activity Diagram Design Phase User Interface Design Database Design Data Dictionary System Architecture Development Phase Programming Environment Programming Language Development Tools/DE Build and Dependency Management Version Control Frameworks and Libraries List of Modules Database Stack		

	Testing Tools (Developer Side) Configuration Operating System Hardware Requirements Testing Phase Unit Testing Integration Testing System Testing Deployment Phase Operational Environment Hardware Requirements Operating System Runtime Requirements Network Requirements Database Server Environment Application Hosting or Deployment Environment User Environment Requirements Security Requirements		
Final Pages	Curriculum Vitae Bibliography Appendices A. Oath of Confirmation B. Capstone Project Team Composition C. Acceptance of Capstone Adviser D. Capstone Project Model E. Capstone Project Comparative Matrix F. Research Working Title G. Capstone Project Consultation Logs H. List of Modules I. Questionnaire J. Post-Proposal Hearing Compliance Checklist		

D. Panel-Specific Required Revisions
List mandatory changes identified by the panel:

Chairman: Dr. Angelbert P. Maghanoy

Mandatory Requirements	Action or Changes Made	Reference Pages
1. Revise Scope and Limitations	Changed the Scope and Limitations	Page 5

Panel 1: Prof. John Rey E. Furtos

Mandatory Requirements	Action or Changes Made	Reference Pages
1. Leaving of Big Blank Spaces.	Compressed each statements, leaving no big blank spaces.	x
2. Tables must not overlap within the required margin.	Aligned each tables, and fixed the required margin.	Pages 47 – 48, 59 – 60, 68-92

Panel 2: Prof. Jayvee T. Cabije

Mandatory Requirements	Action or Changes Made	Reference Pages
1. Revise Scope and Limitations	Changed the Scope and Limitations	Page 5

E. Adviser's Review

[ ] All required revisions have been complied with
[ ] Revisions are satisfactory but require minor edits
[ ] Further revisions required (specify)

Advisers Comments:	
1.	Revise the Scope and Limitations
2.	Reduce the Users of the System
3.	Change the Title
4.	Indentions must be applied
5.	Spacings must be followed
6.	Change the first paragraph in Rationale
7.	Insert SDGs before the addressed gaps in Rationale
8.	Remove Business Model Canvas and change it into Feasibility Study Analysis
9.	There should be a description after each figure.

F. Certification

This is to CERTIFY that the proponents have complied with the required revisions after the proposal hearing.

Name		Signature	Date
ADVISER	Dr. Emelie C. Villaceran, Ph.D.		
CHAIRMAN	Dr. Angelbert P. Maghanoy		
PANEL 1	Prof. John Rey E. Furtos		
PANEL 2	Prof. Jayvee T. Cabije		